Chapter 12: Introduction to Queueing Theory

12.1 & 12.2 The Elements of a Queueing Network and Little's Formula

12.1

~~9.1~~ a) M/M/1 Poisson arrivals, exponential service time, single server, no limit on number of customers

M/D/1/K Poisson arrivals, constant service time, single server, at most K customers allowed in system

M/G/3 Poisson arrivals, iid general service time, 3 servers, no limit on number of customers

D/M/2 Constant interarrival times, exponential service times, two servers, no limit on number of customers

G/D/1 Arrivals according to a general process, fixed constant service times, single server, no limit on number of customers

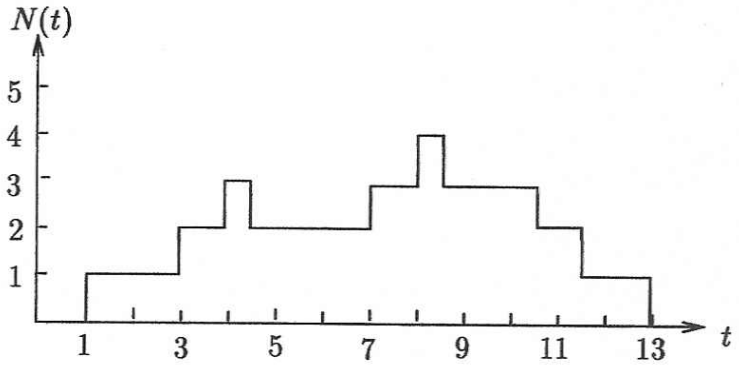
D/D/2 Constant interarrival times, constant service times, two servers, no limit on number of customers in the system

12.2 $\{S_i\} = \{1, 3, 4, 7, 8, 15\}$
$\{\tau_i\} = \{3.5, 4, 2, 1, 1.5, 4\}$

a) FCFS

i	S_i	τ_i	D_i	W_i	T_i
1	1	3.5	4.5	0	3.5
2	3	4	8.5	1.5	5.5
3	4	2	10.5	4.5	6.5
4	7	1	11.5	3.5	4.5
5	8	1.5	13.0	3.5	5.0
6	15	4			

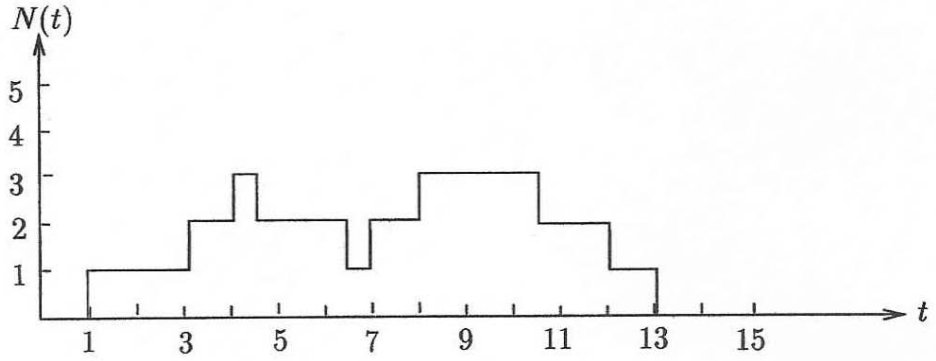
where $W_i = D_{i-1} - S_i = T_i - \tau_i$ and $T_i = D_i - S_i = W_i + \tau_i$



$$< N >_{13} = \frac{1}{13}\sum_{i=1}^{A_{13}} T_i = \frac{25}{13}$$
$$< \lambda >_{13} = \frac{A_{13}}{13} = \frac{5}{13}$$
$$< T >_{13} = \frac{1}{A_{13}}\sum_{i=1}^{A_{13}} T_i = \frac{25}{5}$$
$$< N >_{13} = \frac{25}{13} = < \lambda >_{13} < T >_{13} = \frac{5}{13}\frac{25}{5} \quad \surd$$

b) LCFS

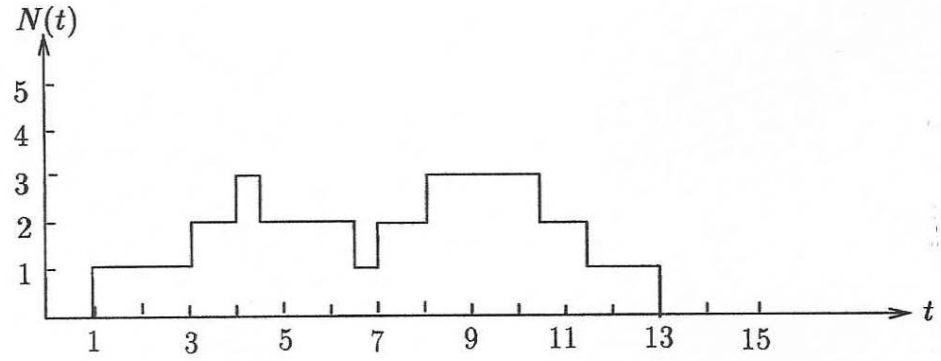
i	S_i	τ_i	D_i	$W_i = T_i - \tau_i$	$T_i = D_i - S_i$
1	1	3.5	4.5	0	3.5
2	3	4	10.5	3.5	7.5
3	4	2	6.5	0.5	2.5
4	7	1	13.0	5.0	6.0
5	8	1.5	12.0	2.5	4.0



$$< N >_{13} = \frac{23.5}{13} \qquad < \lambda >_{13} = \frac{5}{13} \qquad < T >_{13} = \frac{23.5}{5}$$
$$< N >_{13} = < \lambda >_{13} < T >_{13}$$

c) Shortest Job First:

i	S_i	τ_i	D_i	$W_i = T_i - \tau_i$	$T_i = D_i - S_i$
1	1	3.5	4.5	0	3.5
2	3	4	10.5	3.5	7.5
3	4	2	6.5	0.5	2.5
4	7	1	11.5	3.5	4.5
5	8	1.5	13.0	3.5	5.0



$$< N >_{13} = \frac{23}{13} \qquad < \lambda >_{13} = \frac{5}{13} \qquad < T >_{13} = \frac{23}{5}$$
$$< N >_{13} = < \lambda >_{13} < T >_{13}$$

12.3

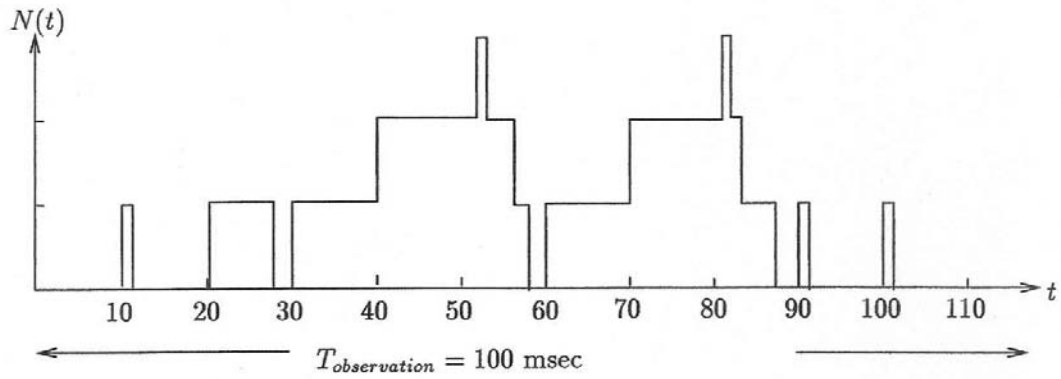
1) Interarrivals are constant with interarrival times = 10 μsec

2) Service time

if 0 error 1 μsec
1 error 1+5 μsec
> 1 error 1+20 μsec

arrival time	10	20	30	40	50	60	70	80	90	100
errors	0	1	3	1	0	4	0	1	0	0
service time	1	6	21	6	1	21	1	6	1	1
dep. time	11	26	51	57	58	81	82	88	91	101

a)



b) $$\frac{1}{100}\int_{10}^{110} N(t)dt$$
$$= \frac{1}{100}[1+6+10+20+3+12+1+10+20+3+2+6+1+1]$$
$$= 0.96$$

c) Server is working during 65 msec = Σ service times

$$\text{proportion idle time} = 1 - \frac{65}{100} = 0.35$$

12.4

9.4 a) One customer ⇒ no waiting

$$\Rightarrow \mathcal{E}[T] = m_1 + m_2 + m_3 \Rightarrow \lambda = \frac{1}{m_1 + m_2 + m_3}$$

Little's formula ⇒

$$\mathcal{E}[N_i] = \lambda m_i = \frac{m_i}{m_1 + m_2 + m_3} = \%\text{ time customer in queue } i$$

$$\sum_{i=1}^{3} \mathcal{E}[N_i] = \sum_{i=1}^{3} \frac{m_i}{m_1 + m_2 + m_3} = 1 \Rightarrow \text{one customer in system}$$

b) Let $\mathcal{E}[T]$ = mean cycle time per customer, then

$$\begin{matrix}\text{total \#} \\ \text{in system}\end{matrix} = N = \lambda \mathcal{E}[T] \quad \text{by Little's formula}$$

12.5

9.5 a) $\lambda T = 5$

b) $\lambda m = 2$

c) $T = \frac{5}{\lambda} = 5\left(\frac{m}{2}\right) = \frac{5}{2}m$

12.6

9.6 Let τ be the service time, then

$$P[\tau = 1] = p_0 \quad [\tau = 1 + 5] = p_1 \quad P[\tau = 1 + 20] = p_2$$

$$\mathcal{E}[\tau] = 1 \cdot p_0 + 6p_1 + 21p_2 \ \ 10^{-6} \text{ sec}$$

$$\lambda = 1 \text{ arrival every } 10\ \mu\text{s} = \frac{1}{10^{-5}} = 10^5$$

$$\mathcal{E}[N_d] = \lambda \mathcal{E}[\tau] = \frac{p_o + 6p_1 + 21p_2}{10}$$

12.7

a) $\mathcal{E}[T_i] = \frac{1}{\lambda_i}\mathcal{E}[N_i]$

b) $\mathcal{E}[T] = \frac{1}{\lambda}\mathcal{E}[N] = \frac{1}{\lambda}\sum_i \mathcal{E}[N_i]$

c) $\mathcal{E}[T] = \frac{1}{\lambda}\sum_i \lambda_i \mathcal{E}[T_i] = \sum_i \frac{\lambda_i}{\lambda}\mathcal{E}[T_i]$

Let

$$\begin{aligned} A_i(t) &= \#\text{ type } i \text{ arrivals during } [0,t] \\ A(t) &= \sum_i A_i(t) = \text{total } \# \text{ arrivals} \\ < T > &= \frac{1}{A(t)}\sum_i^{A(t)} T_i \quad \text{average time in system} \\ &= \frac{1}{A(t)}\left[\sum_{i_1}^{A_1(t)} T_{i_1} + \sum_{i_2}^{A_2(t)} T_{i_2} + \ldots + \sum_{i_n}^{A_n(t)} T_{i_n}\right] \\ &= \frac{1}{A(t)}\left[\frac{A_1(t)}{A_1(t)}\sum_{i_1}^{A_1(t)} T_{i_1} + \ldots + \frac{A_n(t)}{A_n(t)}\sum_{i_n}^{A_n(t)} T_{i_n}\right] \\ &= \underbrace{\frac{A_1(t)}{A(t)}}_{\frac{\lambda_1}{\lambda}}\left(\underbrace{\frac{1}{A_1(t)}\sum_{i_1}^{A_1(t)} T_{i_1}}_{\mathcal{E}[T_1]}\right) + \ldots + \underbrace{\frac{A_n(t)}{A(t)}}_{\frac{\lambda_n}{\lambda}}\left(\underbrace{\frac{1}{A_n(t)}\sum_{i_n}^{A_n(t)} T_{i_n}}_{\mathcal{E}[T_n]}\right) \end{aligned}$$

as $t \to \infty$

$\Rightarrow$ same result as above.

12.3 The M/M/1 Queue

12.8

a) $P[N \geq n] = (1-\rho)\sum_{j=n}^{\infty}\rho^j = (1-\rho)\frac{\rho^n}{1-\rho} = \rho^n$

b) $P[N \geq 10] = \rho^{10} = 10^{-3} \Rightarrow \rho = 10^{-0.3} \approx \frac{1}{2}$

$$\Rightarrow \lambda \approx \frac{1}{2}\mu$$

12.9

12.9 a) Machine 1 μ transactions/hr B \$/hr operating cost
Machine 2 2μ transactions/hr $2B$ \$/hr operating cost

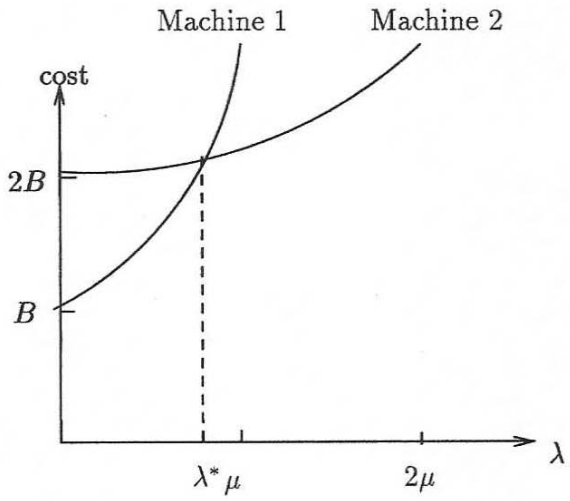
We assume "operating" cost is incurred regardless of whether machine is idle.

$$\text{Cost for \#1} = B + \lambda \frac{\text{cost}}{\text{hr}} \cdot \bar{W} \frac{\text{waiting hrs.}}{\text{cust}} \cdot A \frac{\$}{\text{hr}}$$

$$= B + \lambda \left(\frac{\rho}{1-\rho} \frac{1}{\mu} \right) A \quad \text{where } \rho = \frac{\lambda}{\mu}$$

$$= B + A \frac{\rho^2}{1-\rho}$$

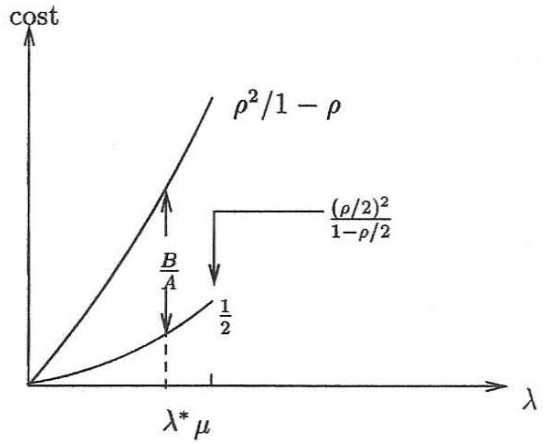
$$\text{Cost for \#2} = 2B + \lambda \left(\frac{\frac{\rho}{2}}{1-\frac{\rho}{2}} \right) \left(\frac{1}{2\mu} \right) A = 2B + A \frac{\left(\frac{\rho}{2}\right)^2}{1-\frac{\rho}{2}}$$



for $\lambda < \lambda^*$ machine 1 is less costly. Let $\rho^* = \lambda^*/\mu$

$$B + A \frac{\rho^{*2}}{1-\rho^{*2}} = 2B + A \frac{\left(\frac{\rho^*}{2}\right)^2}{1-\frac{\rho^*}{2}}$$

$$\Leftrightarrow \frac{\rho^{*2}}{1-\rho^*} - \frac{\left(\frac{\rho^*}{2}\right)^2}{1-\frac{\rho^*}{2}} = \frac{B}{A}$$



This requires finally the root of a cubic polynomial but we can estimate the root from the figure.

For $\frac{B}{A} >> 1$

$$\frac{\left(\frac{\rho^*}{2}\right)^2}{1-\frac{\rho^*}{2}} \approx \frac{1}{2}$$

so we solve the quadratic equation associated with

$$\frac{\rho^{*2}}{1-\rho^*} - \frac{1}{2} = \frac{B}{A}$$

In particular if $\frac{B}{A} = 10$ then $\rho^* \approx 0.91$.

For $\frac{B}{A} << 1$

$$\frac{\left(\frac{\rho^*}{2}\right)^2}{1-\frac{\rho^*}{2}} \approx 0$$

so we solve

$$\frac{\rho^{*2}}{1-\rho^*} = \frac{B}{A}$$

In particular if $\frac{B}{A} = \frac{1}{10}$ then $\rho^* \approx 0.27$.

NOTE: If "operating" cost is incurred only when a machine is in use then:

$$\text{Cost of Machine 1} = \underbrace{B\rho}_{\substack{\text{prop. of time} \\ \text{machine 1 in use}}} + \frac{\rho^2}{1-\rho}$$

$$\text{Cost of Machine 2} = 2B\left(\frac{\rho}{2}\right) + \frac{\left(\frac{\rho}{2}\right)^2}{1-\frac{\rho}{2}} = B\rho + \frac{\left(\frac{\rho}{2}\right)^2}{1-\frac{\rho}{2}}$$

In this case machine 2 is less costly than machine 1 for all ρ.

12.10

9.10 A net profit is made if

$$5 > \mathcal{E}[T] = \frac{\frac{1}{\mu}}{1-\rho}$$

$$\Rightarrow \quad 1 - \frac{\lambda}{\mu} > \frac{1}{5\mu} \Rightarrow 5\mu - 5\lambda > 1$$

$$\Rightarrow \qquad 0 < \lambda < \mu - \frac{1}{5}$$

12.11

9.11 a) $\mathcal{E}[N_q] = \dfrac{\rho^2}{1-\rho} = 5 \Rightarrow \rho^2 - 5\rho - 5 = 0$

$$\Rightarrow \rho = \frac{-5 + \sqrt{25 - 4(-5)}}{2} = \frac{\sqrt{45} - 5}{2} = 0.854$$

b)
$$\begin{aligned} P[N_q = j | N_q > 0] &= \frac{P[N_1 = j]}{P[N_q > 0]} \qquad j > 0 \\ &= \frac{(1-\rho)\rho^j}{\rho} = (1-\rho)\rho^{j-1} \\ \mathcal{E}[N_q | N_q > 0] &= \sum_{j=1}^{\infty} j(1-\rho)\rho^{j-1} = \frac{1}{1-\rho} = 5 \\ &\Rightarrow \rho = \frac{4}{5} = 0.8 \end{aligned}$$

c) It depends on whether one is concerned with average queue over all time (Part a) or on the average queue when one forms (Part b).

12.12

$$\begin{aligned} p &= P[W \le x] = \int_0^x ((1-\rho)\delta(t') + \lambda(1-\rho)e^{-\mu(1-\rho)t'})dt' \\ &= (1-\rho) + \left| -\frac{\lambda}{\mu} e^{-\mu(1-\rho)t'} \right|_0^x \\ &= 1 - \rho e^{-\mu(1-\rho)x} \\ \Rightarrow \frac{1-p}{\rho} &= e^{-\mu(1-\rho)x} \\ \Rightarrow x &= \frac{1}{\mu(1-\rho)} \ln \frac{\rho}{1-p} \qquad \surd \end{aligned}$$

12.13 $\frac{1}{\mu} = \frac{1}{2}$

a) From Example 12.5

$$\begin{aligned} x &= \frac{1}{\mu-\lambda}\ln\frac{1}{1-p} \\ \Rightarrow \mu-\lambda &= \frac{1}{x}\ln\frac{1}{1-p} \\ \lambda &= \mu - \frac{1}{x}\ln\frac{1}{1-p} = 2 - \frac{1}{3}\ln\frac{1}{1-.9} = 1.232 \end{aligned}$$

b) From Problem 12.12

$$\begin{aligned} x &= \frac{\frac{1}{\mu}}{1-\rho}\ln\frac{\rho}{1-\rho} = \frac{1}{\mu-\lambda}\ln\frac{\lambda}{\mu(1-p)} \\ 2 &= \frac{1}{2-\lambda}\ln 5\lambda \quad \Rightarrow \quad \lambda = 2 - \frac{1}{2}\ln 5\lambda \\ &\quad \Rightarrow \quad \lambda = 1.13 \end{aligned}$$

12.14

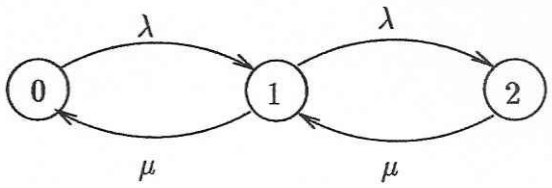
~~9.14~~ From Example 11.40 we have

$$\begin{aligned} P_j &= \frac{\lambda}{\mu}P_{j-1} = \left(\frac{\lambda}{\mu}\right)^j P_0 \qquad 1 \le j \le K \\ 1 &= \sum_{j=0}^{K}\left(\frac{\lambda}{\mu}\right)^j P_0 \Rightarrow P_0 = \frac{1}{\sum_{j=0}^{K}\left(\frac{\lambda}{\mu}\right)^j} = \frac{1-\rho}{1-\rho^{K+1}} \end{aligned}$$

where $\rho = \frac{\lambda}{\mu}$

$$\therefore P_j = \frac{(1-\rho)}{1-\rho^{K+1}}\rho^j \qquad 0 \le j \le K$$

12.15



$$P_0 = \frac{1-\rho}{1-\rho^3} = \frac{1}{1+\rho+\rho^2}$$

$$P_1 = \frac{\rho}{1+\rho+\rho^2} \qquad P_2 = \frac{\rho^2}{1+\rho+\rho^2}$$

In a very long time interval of length T

$$\begin{aligned}
\text{Profit} &= \#\text{ accepted into system} \times \$5 - \#\text{ blocked} \times 1 \\
&= \lambda T \times (1-P_B) \times 5 - \lambda T \times P_B \times 1 \\
\text{Profit} &= 0 \quad \text{if} \quad 5\lambda T(1-P_B) = \lambda T P_B
\end{aligned}$$

$$\begin{aligned}
&\Leftrightarrow \quad 5 = 6P_B \\
&\Leftrightarrow \quad P_B = \frac{\rho^2}{1+\rho+\rho^2} = \frac{5}{6} \\
&\Leftrightarrow \quad \rho^2 - 5\rho - 5 = 6 \\
&\Rightarrow \quad \rho = \frac{5+\sqrt{25+20}}{2} = 5.854 \\
&\Rightarrow \quad \lambda = 5.854\mu
\end{aligned}$$

12.16 $$P[N=k|N<K] = \frac{P[N=k, N<K]}{P[N<K]} = \begin{cases} 0 & k \geq K \\ \frac{P[N=k]}{P[N<K]} & 0 \leq k < K \end{cases}$$

$\therefore$ for $0 \leq k < K$

$$P[N=k|N<K] = \frac{P[N=k]}{P[N<K]} = \frac{P[N=k]}{1-P[N=K]}$$

Arriving customers are allowed into the system only when $N < K$.

$\therefore P[N=k|N<K]$ represents the proportion of time when there are k in the system and arriving customers are allowed in. Since Poisson arrivals pick their arrival times at random, then $P[N=k|N<K]$ is the proportion of customers that see k in system upon being admitted in.

12.17 a) M/M/1/5 ⇒ 6 states

(i)

$$\Gamma = \begin{pmatrix} -\lambda & \lambda & 0 & 0 & 0 & 0 \\ \mu & -(\mu+\lambda) & \lambda & 0 & 0 & 0 \\ 0 & \mu & -(\mu+\lambda) & \lambda & 0 & 0 \\ 0 & 0 & \mu & -(\mu+\lambda) & \lambda & 0 \\ 0 & 0 & 0 & \mu & -(\mu+\lambda) & \lambda \\ 0 & 0 & 0 & 0 & \mu & -\mu \end{pmatrix} = \begin{pmatrix} -0.5 & 0.5 & 0 & 0 & 0 & 0 \\ 1 & -1.5 & 0.5 & 0 & 0 & 0 \\ 0 & 1 & -1.5 & 0.5 & 0 & 0 \\ 0 & 0 & 1 & -1.5 & 0.5 & 0 \\ 0 & 0 & 0 & 1 & -1.5 & 0.5 \\ 0 & 0 & 0 & 0 & 1 & -1 \end{pmatrix}$$

$$P(t) = e^{\Gamma t} = \bar{E}[e^{\lambda_i t}]\bar{E}^{-1}$$

$$\bar{E} = \begin{pmatrix} -0.2644 & -0.4487 & 0.6172 & 0.8661 & -0.7487 & -0.8318 \\ 0.5882 & 0.7661 & -0.6172 & 0.4331 & 0.2193 & -0.1876 \\ -0.5882 & -0.3173 & -0.3086 & 0.2165 & 0.5294 & 0.1870 \\ 0.4263 & -0.1587 & 0.3086 & 0.1083 & 0.2647 & 0.3225 \\ -0.2280 & 0.2708 & 0.1543 & 0.0541 & -0.0775 & 0.3014 \\ 0.0661 & -0.1122 & -0.1543 & 0.0271 & -0.1872 & 0.2080 \end{pmatrix}$$

$$[e^{\lambda_i t}] = \begin{pmatrix} e^{-2.72t} & 0 & 0 & 0 & 0 & 0 \\ 0 & e^{-2.21t} & 0 & 0 & 0 & 0 \\ 0 & 0 & e^{-1.50t} & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & e^{-0.79t} & 0 \\ 0 & 0 & 0 & 0 & 0 & e^{-0.27t} \end{pmatrix}$$

$$E^{-1} = \begin{pmatrix} -0.0578 & 0.2573 & -0.5147 & 0.7460 & -0.7980 & 0.4627 \\ -0.1262 & 0.4309 & -0.3570 & -0.3570 & 1.2188 & -1.0097 \\ 0.1800 & -0.3600 & -0.3600 & -0.7201 & 0.7201 & -1.4402 \\ 0.5864 & 0.5864 & 0.5864 & 0.5864 & 0.5864 & 0.5864 \\ -0.2106 & 0.1233 & 0.5956 & 0.5956 & -0.3489 & -1.6846 \\ -0.1820 & -0.0818 & 0.5643 & 0.5643 & 1.0651 & 1.4558 \end{pmatrix}$$

$$E\left[e^{\Lambda t}\right] E^{-1} = X \quad \Rightarrow \quad P(t) = P(0) X$$

$$N(0) = 0 \Rightarrow P(0) = (1,0,0,0,0,0) \Rightarrow P(t) = \text{first row of } X$$

$$\Rightarrow P(t) = \begin{pmatrix} 0.0153e^{-2.72t} + 0.0556e^{-2.21t} + 0.1111e^{-1.5t} + 0.5091 + 0.1577e^{-.79t} + 0.1514e^{-0.27t} \\ -0.3040e^{-2.72t} + (-0.0967)e^{-2.21t} - 0.1111e^{-1.5t} + 0.2340 + (-0.0462)e^{-.79t} + 0.0340e^{-0.27t} \\ 0.3040e^{-2.72t} + 0.04e^{-2.21t} - 0.0556e^{-1.5t} + 0.1270 - 0.1115e^{-0.79t} - 0.030e^{-0.27t} \\ -0.0246e^{-2.72t} + 0.02e^{-2.21t} + 0.0556e^{-1.5t} + 0.0635 - 0.0557e^{-0.79t} - 0.0587e^{-0.27t} \\ 0.0132e^{-2.72t} - 0.0342e^{-2.21t} + 0.0278e^{-1.5t} + 0.0317 + 0.0163e^{-0.79t} - 0.0549e^{-0.27t} \\ -0.0582e^{-2.72t} + 0.0142e^{-2.21t} - 0.0278e^{-1.5t} + 0.0159 + 0.0394e^{-0.79t} - 0.0379e^{-0.27t} \end{pmatrix}$$

for other initial conditions also we can obtain $P(t)$ in the same way:

$N(0) = 2 \Rightarrow P(0) = (0,0,1,0,0,0)$, $P(t) = P(0)\, E\left[e^{\Lambda t}\right] E^{-1} \Rightarrow$ third row of X

$N(0) = 5 \Rightarrow P(0) = (0,0,0,0,0,1)$, $P(t) = P(0)\, E\left[e^{\Lambda t}\right] E^{-1} \Rightarrow$ last row of X

(ii) if $\rho_{21} \Rightarrow$

$$\Gamma_2 \begin{pmatrix} -1 & 1 & 0 & 0 & 0 & 0 \\ +1 & -2 & 1 & 0 & 0 & 0 \\ 0 & +1 & -2 & 1 & 0 & 0 \\ 0 & 0 & +1 & -2 & 1 & 0 \\ 0 & 0 & 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & 0 & 1 & -1 \end{pmatrix}$$

```
% Problem 12.17
% (i)
lambda=0.5;
mu=1;

v=-(lambda+mu);
a=lambda;
b=mu;
L=[-a a 0 0 0 0
   b v a 0 0 0
   0 b v a 0 0
   0 0 b v a 0
   0 0 0 b v a
   0 0 0 0 b -b];

[E D]=eig(L);
t=sym('t');
N0=1;
p0=zeros(1,6);
p0(N0)=1;
p=p0*E*expm(t*D)*inv(E);
f=inline('E*expm(t*D)*inv(E)');
ff=p0*f(E,t,D); % Example: p0*f(E,2,D) will compute the amount of p(2)
%Plot symbolic function
ezplot(mean(ff));

N0=3;
p0=zeros(1,6);
p0(N0)=1;
p=p0*E*expm(t*D)*inv(E);
f=inline('E*expm(t*D)*inv(E)');
ff=p0*f(E,t,D);
%Plot symbolic function
ezplot(mean(ff));
```

```
N0=6;
p0=zeros(1,6);
p0(N0)=1;
p=p0*E*exp(t*D)*inv(E);
f=inline('E*expm(t*D)*inv(E)');
ff=p0*f(E,t,D);
%Plot symbolic function
ezplot(mean(ff));

% for part (ii) we repeat the same process with lambda=mu=1
```

```
% Problem 12.17
% (i)
lambda=1.0;
mu=1;

v=-(lambda+mu);
a=lambda;
b=mu;
L=[-a a 0 0 0 0
   b v a 0 0 0
   0 b v a 0 0
   0 0 b v a 0
   0 0 0 b v a
   0 0 0 0 b -b];

[E D]=eig(L);
t=sym('t');
N0=6;
p0=zeros(1,6);
p0(N0)=1;
p=p0*E*expm(t*D)*inv(E);
f=inline('E*expm(t*D)*inv(E)');
ff=p0*f(D,E,t); % Example: p0*f(E,2,D) will compute the amount of p(2)
%Plot symbolic function
ezplot(mean(ff));
```

```
% Problem 12.17
% (i)

lambda=0.5;
mu=1;

v=-(lambda+mu);
a=lambda;
b=mu;
L=[-a a 0 0 0 0
  b v a 0 0 0
  0 b v a 0 0
  0 0 b v a 0
  0 0 0 b v a
  0 0 0 0 b -b];

[E D]=eig(L);
t=sym('t');
N0=1;
p0=zeros(1,6);
p0(N0)=1;
p=p0*E*expm(t*D)*inv(E);
f=inline('E*expm(t*D)*inv(E)');
ff=p0*f(D,E,t); % Example: p0*f(E,2,D) will compute the amount of p(2)
%Plot symbolic function
ezplot(mean(ff));
```

```
L =

  -0.5000   0.5000        0        0        0        0
   1.0000  -1.5000   0.5000        0        0        0
        0   1.0000  -1.5000   0.5000        0        0
        0        0   1.0000  -1.5000   0.5000        0
        0        0        0   1.0000  -1.5000   0.5000
        0        0        0        0   1.0000  -1.0000
```

```
>> E

E =

   0.0438   0.0733  -0.0976  -0.1091  -0.4082  -0.0957
  -0.1950  -0.2504   0.1952   0.0639  -0.4082  -0.0430
   0.3900   0.2074   0.1952   0.3084  -0.4082   0.0860
  -0.5652   0.2074  -0.3904   0.3084  -0.4082   0.2967
   0.6046  -0.7082  -0.3904  -0.1807  -0.4082   0.5547
  -0.3506   0.5867   0.7807  -0.8724  -0.4082   0.7654

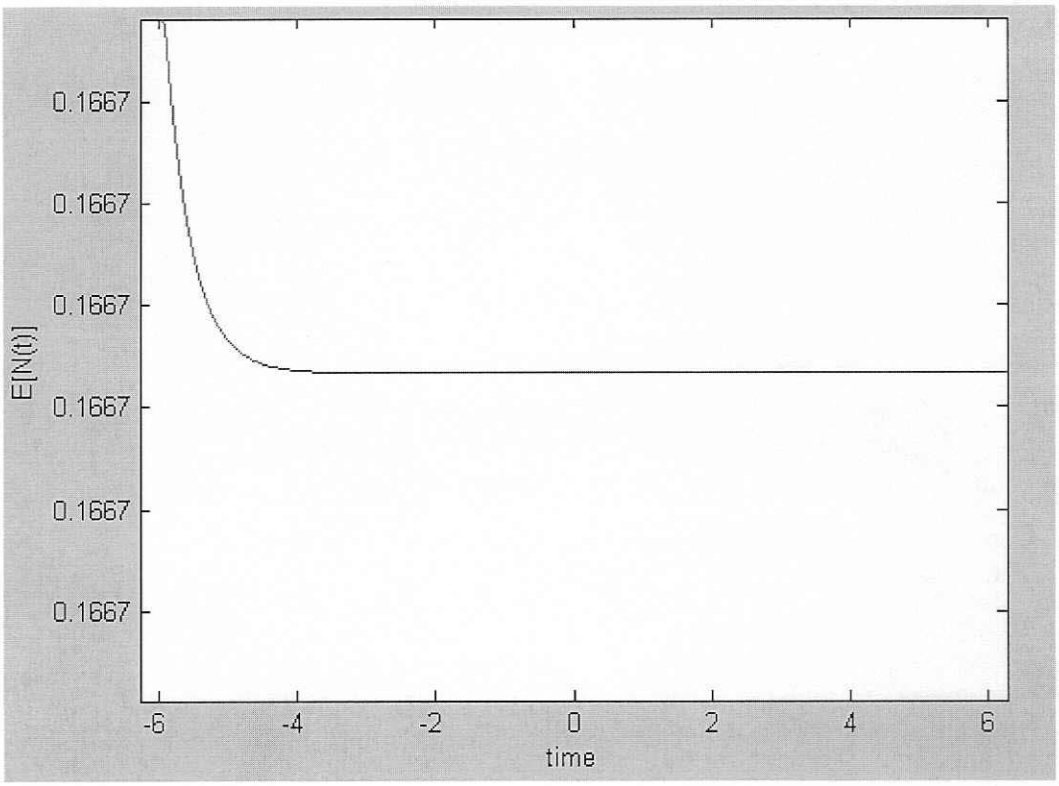

D =

  -2.7247        0        0        0        0        0
        0  -2.2071        0        0        0        0
        0        0  -1.5000        0        0        0
        0        0        0  -0.7929        0        0
        0        0        0        0  -0.0000        0
        0        0        0        0        0  -0.2753
```

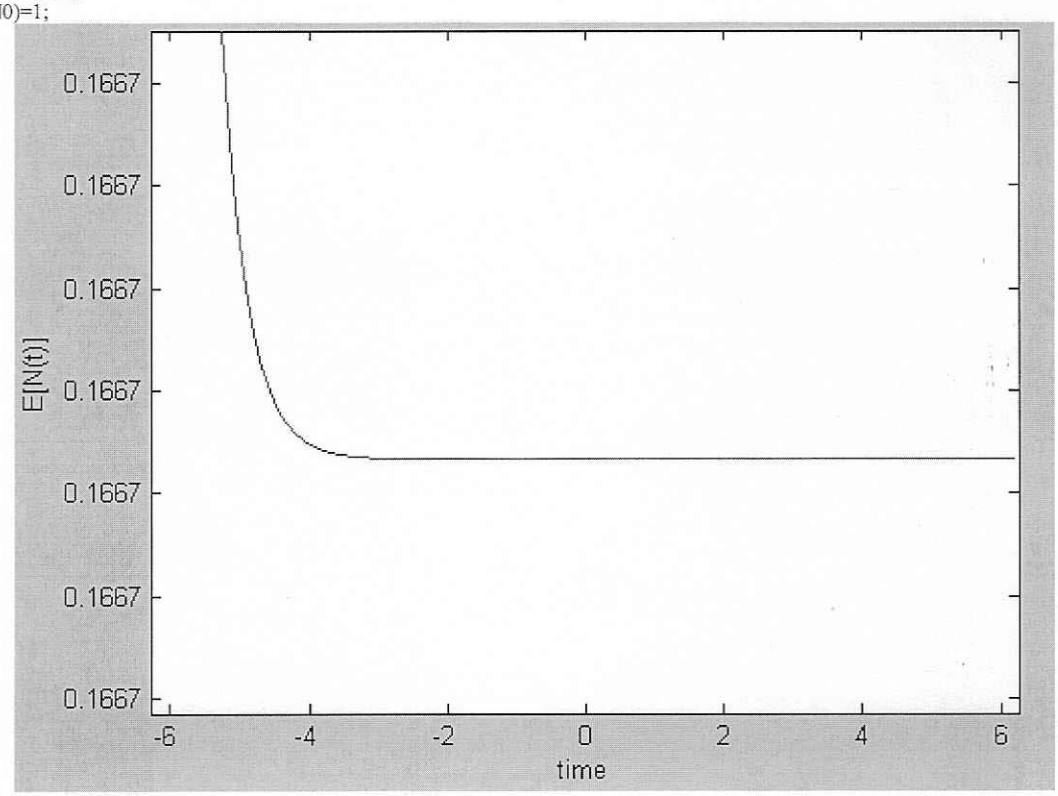
```
>> inv(E)

ans =

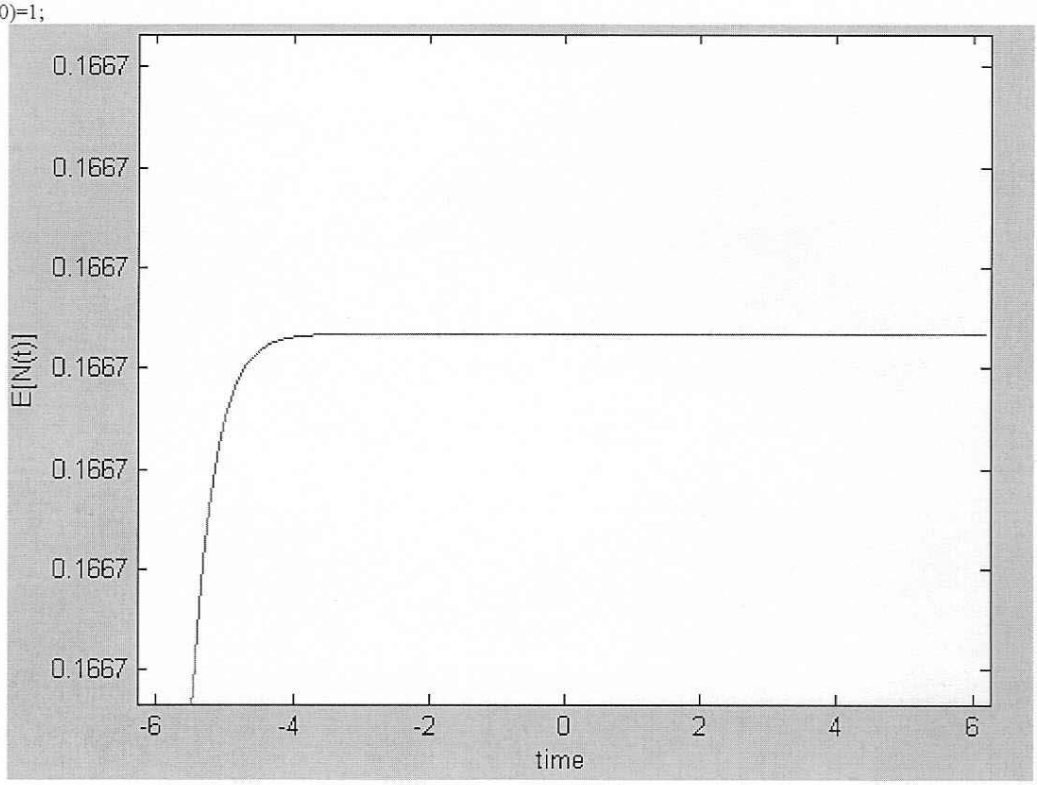
   0.3490  -0.7764   0.7764  -0.5627   0.3009  -0.0872
   0.7722  -1.3183   0.5460   0.2730  -0.4661   0.1931
  -1.1386   1.1386   0.5693  -0.5693  -0.2846   0.2846
  -1.4456   0.4234   1.0222   0.5111  -0.1497  -0.3614
  -1.2442  -0.6221  -0.3110  -0.1555  -0.0778  -0.0389
  -1.5822  -0.3556   0.3556   0.6133   0.5733   0.3955
```



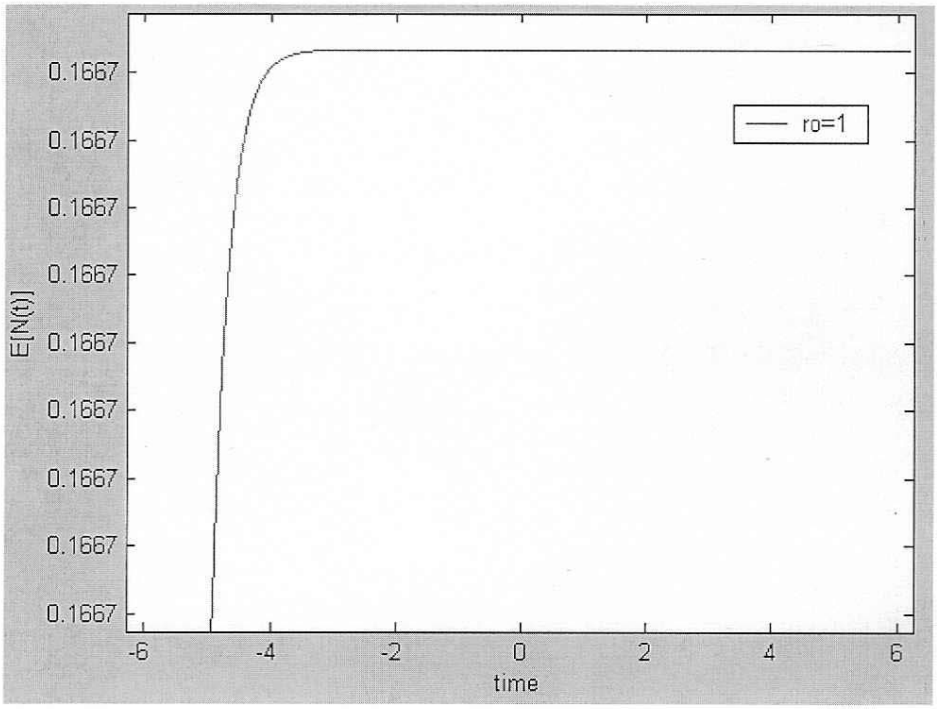

```
N0=3;
p0=zeros(1,6);
p0(N0)=1;
```



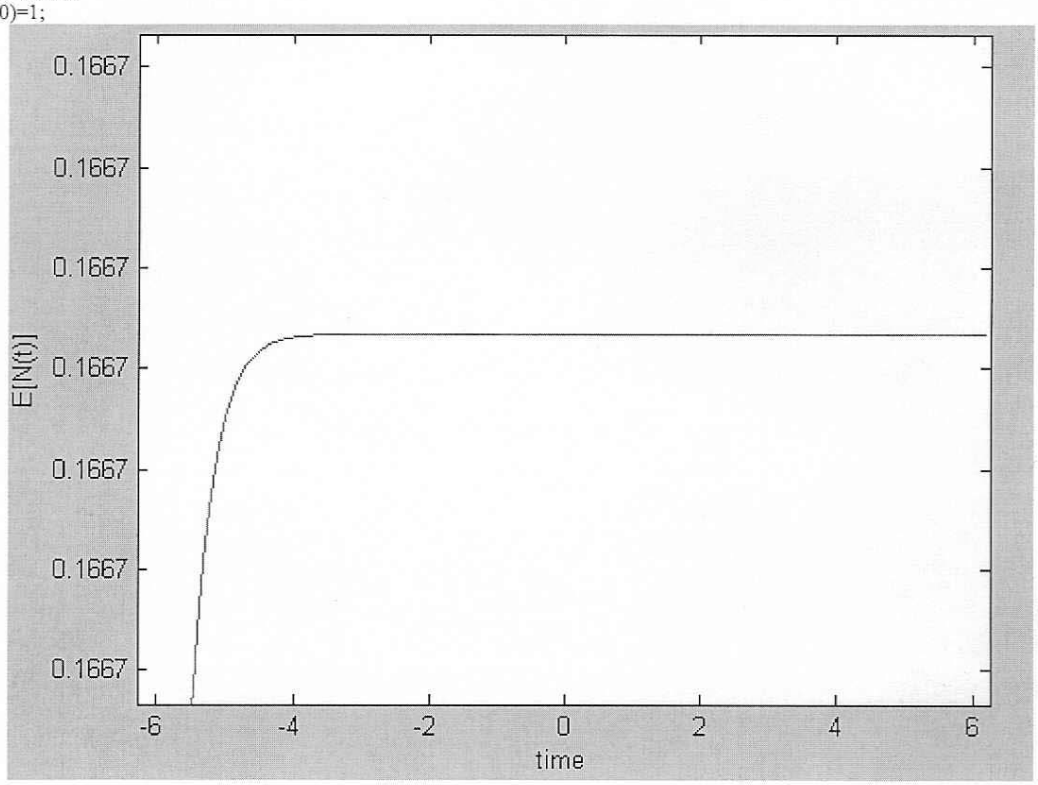
```
N0=6;
p0=zeros(1,6);
p0(N0)=1;
```



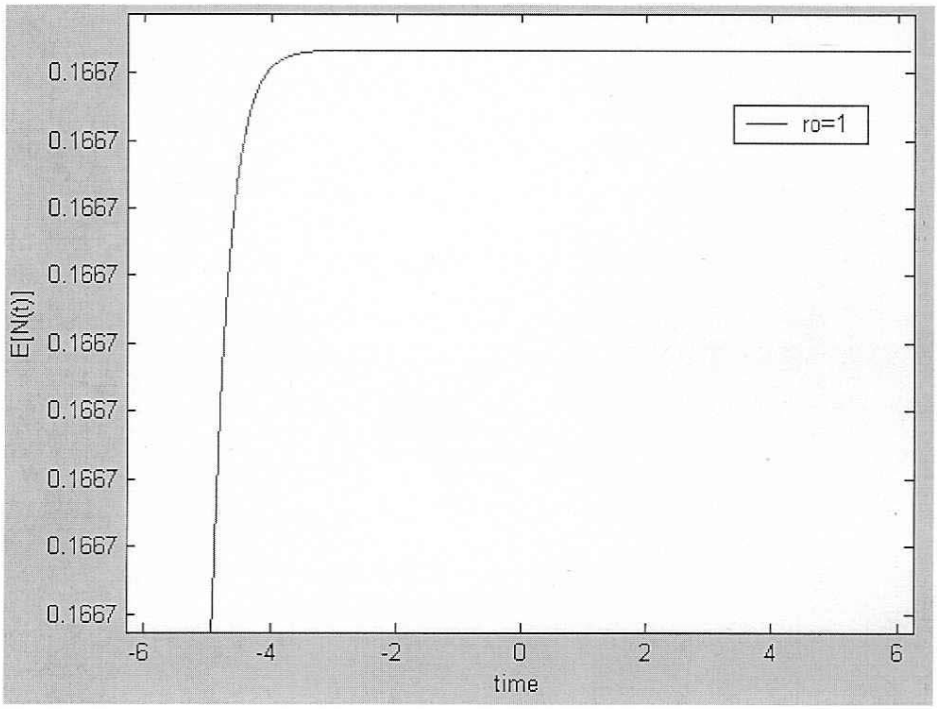

```
Part (ii)
Lambda=1;
N0=1;
p0=zeros(1,6);
p0(N0)=1;
```



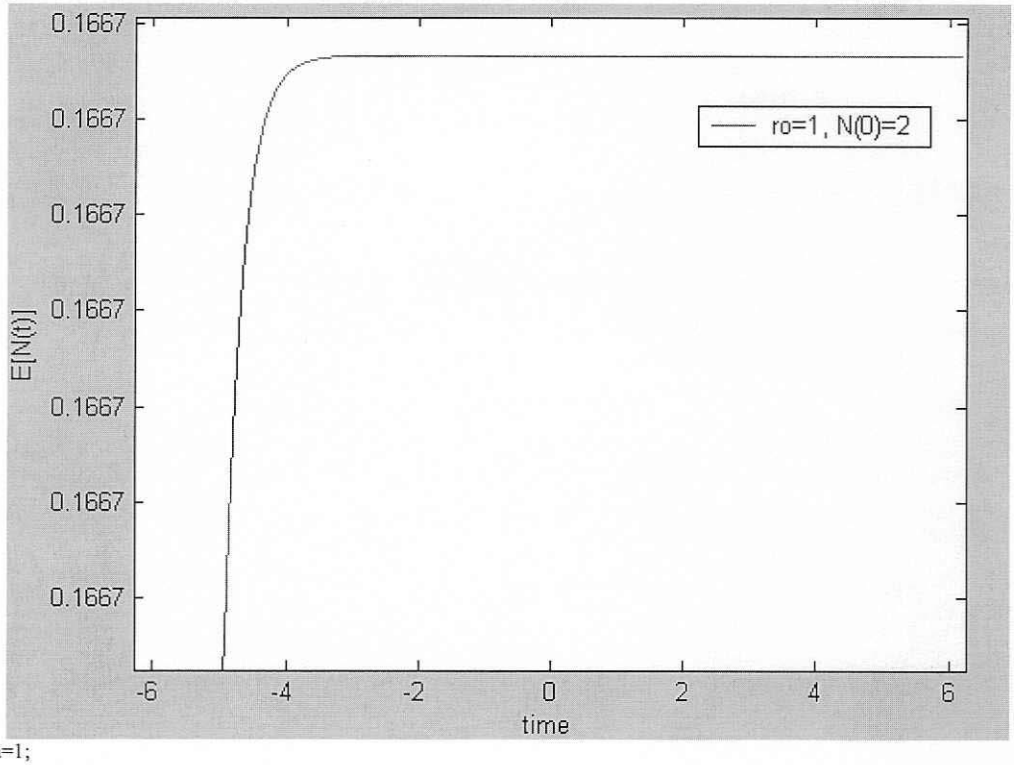
```
N0=6;
p0=zeros(1,6);
p0(N0)=1;
```



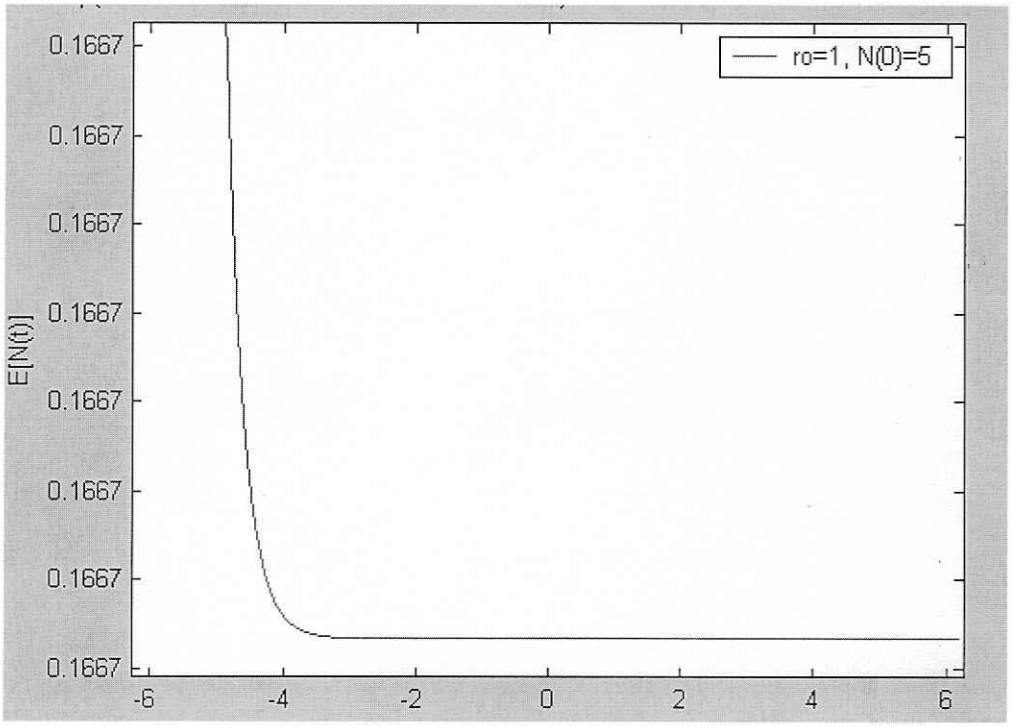

```
Part (ii)
Lambda=1;
N0=1;
p0=zeros(1,6);
p0(N0)=1;
```



```
Lambda=1;
N0=1;
p0=zeros(1,6);
p0(N0)=3
```

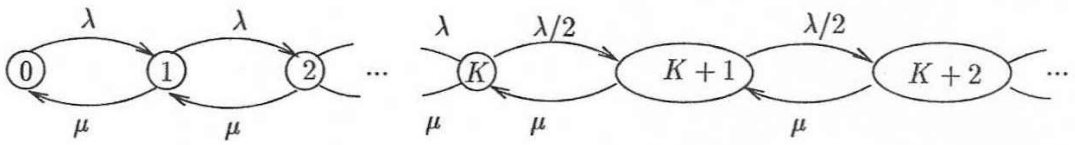



```
Lambda=1;
N0=1;
p0=zeros(1,6);
p0(N0)=6
```



12.18

9.17 If $N < K$ arrival rate is λ
If $N \geq K$ arrival rate is reduced to $\frac{\lambda}{2}$



For $0 \leq j \leq K$

$$P_j = \frac{\lambda}{\mu} P_{j-1} = \left(\frac{\lambda}{\mu}\right)^j P_0$$

For $K < j$

$$P_j = \frac{\lambda}{2\mu} P_{j-1} = \left(\frac{\lambda}{2\mu}\right)^{j-K} P_K = \left(\frac{\lambda}{2\mu}\right)^{j-K} \left(\frac{\lambda}{\mu}\right)^K P_0$$

$$1 = \sum_{j=0}^{\infty} P_j = \underbrace{P_0 \sum_{j=0}^{K-1} \left(\frac{\lambda}{\mu}\right)^j}_{\frac{1-\left(\frac{\lambda}{\mu}\right)^K}{1-\frac{\lambda}{\mu}}} + \underbrace{P_0 \sum_{j=K}^{\infty} \left(\frac{\lambda}{2\mu}\right)^{j-K} \left(\frac{\lambda}{\mu}\right)^K}_{\frac{\left(\frac{\lambda}{\mu}\right)^K}{1-\frac{\lambda}{2\mu}}}$$

$$P_0 = \left[\frac{1-\left(\frac{\lambda}{\mu}\right)^K}{1-\frac{\lambda}{\mu}} + \frac{\left(\frac{\lambda}{\mu}\right)^K}{1-\frac{\lambda}{2\mu}}\right]^{-1}$$

12.4 Multi-Server Systems: M/M/*c*, M/M/*c*/*c*, and M/M/∞

12.20

$$P[N \geq c+k] = \sum_{j=c+k}^{\infty} \frac{\rho^{j-c}}{c!} a^c P_0 = \frac{a^c}{c!} P_0 \rho^k \sum_{j'=0}^{\infty} \rho^j$$
$$= \frac{\frac{a^c}{c!} P_0 \rho^k}{1-\rho} = \frac{p_c \rho^k}{1-\rho}$$

12.21

$\lambda = 12 \quad \frac{1}{\mu} = \frac{5}{60} \quad c = 2$

$$\Rightarrow a = \frac{\lambda}{\mu} = 1 \qquad \rho = \frac{a}{2} = \frac{1}{2}$$

a)
$$P[N \geq c] = \frac{\frac{a^c}{c!} P_0}{1-\rho} - C(c,a)$$
$$P_0 = \left\{ \sum_{j=0}^{1} \frac{a^j}{j!} + \frac{a^2}{2!} \sum_{j=0}^{\infty} \rho^j \right\}^{-1} = \left\{ 1 + 1 + \frac{1}{2} \frac{1}{1-\frac{1}{2}} \right\}^{-1} = \frac{1}{3}$$
$$\Rightarrow P[N \geq 2] = \frac{\frac{1}{2!}\frac{1}{3}}{1-\frac{1}{2}} = \frac{1}{3} = C(2,1)$$

b)
$$\mathcal{E}[N] = \mathcal{E}[N_q] + a = \frac{\rho}{1-\rho} C(c,a) + a = \frac{\frac{1}{2}}{1-\frac{1}{2}} \frac{1}{3} + 1 = \frac{4}{3}$$
$$\mathcal{E}[T] = \frac{1}{\lambda} \mathcal{E}[N] = \frac{1}{9}$$

c) $P[N > 4] = P[N_q > 2] = \sum_{j=3}^{\infty} \rho^{j-2} P_2 = \frac{P_2 \rho}{1-\rho} = \frac{1}{6}$

12.22

~~9.20~~

$$\mathcal{E}[N_s] = \sum_{j=0}^{c} jP_j + \sum_{j=c+1}^{\infty} cP_j$$

$$= \sum_{j=0}^{c} j\frac{a^j}{j!}P_0 + c\sum_{j=c+1}^{\infty} \rho^{j-c}\frac{a^c}{c!}P_0$$

$$= \left[a\sum_{j=1}^{c}\frac{a^{j-1}}{(j-1)!} + \frac{ca^c}{c!}\frac{\rho}{1-\rho}\right]P_0 \quad \text{but } c\rho = a$$

$$= a\underbrace{\left[\sum_{j=0}^{c-1}\frac{a^j}{j!} + \frac{\frac{a^c}{c!}}{1-\rho}\right]}_{P_0^{-1}}P_0 = a = \lambda\mathcal{E}[\tau] \quad \surd$$

12.23

~~9.21~~ $\lambda = 10 \qquad \frac{1}{\mu} = \frac{1}{2} \qquad a = \frac{\lambda}{\mu} = 5$

a) $\mathcal{E}[T] = \mathcal{E}[W] + \frac{1}{\mu} \leq 4 \Rightarrow \mathcal{E}[W] \leq 4 - \frac{1}{\mu} = 2$

$$\mathcal{E}[W] = \frac{\frac{1}{\mu}}{c(1-\rho)}C(c,a) \leq 2 \Rightarrow C(c,a) \leq c(1-\rho)$$

$$C(c,a) \leq c - c\rho = c - a = c - 5$$

$\Rightarrow c \geq 5 + C(c,a)$

$\Rightarrow$ try $c = 6$ and check waiting time requirement

$$C(6,5) = \frac{p_0}{1-\rho} = 0.5875 \text{ where } p_0 = 0.004512$$

$$\therefore P[W \leq 8] = 1 - c(6,5)e^{-6\left(\frac{1}{2}\right)\left(1-\frac{5}{6}\right)8} = 0.9892$$

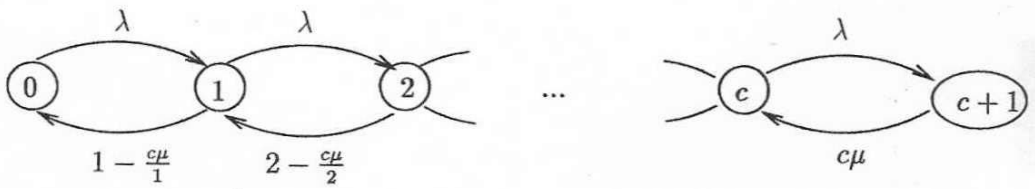
$\Rightarrow \quad c = 6 \quad$ OK

$$p_0 = \left\{\sum_{j=0}^{5}\frac{5^j}{j!} + \frac{\frac{5^6}{6!}}{1-\frac{5}{6}}\right\}^{-1} = 0.004512$$

$$P[\text{all servers busy}] = P[N \geq c] = C(c,a) = 0.5875$$

12.24

a)



b) $$P_j = \frac{\lambda}{c\mu}P_{j-1} \quad j \geq 1$$

$$= \left(\frac{\lambda}{c\mu}\right)^j P_0$$

$$\Rightarrow P_j = (1-\rho)\rho^j \quad j \geq 0, \quad \rho = \frac{\lambda}{c\mu}$$

c) $$\mathcal{E}[N] = \frac{\rho}{1-\rho} \qquad \mathcal{E}[T] = \frac{1}{\lambda}\mathcal{E}[N] = \frac{\frac{1}{c\mu}}{1-\rho}$$

$$\mathcal{E}[N_q] = \sum_{k=c}^{\infty}(k-c)P_k = \sum_{k=c}^{\infty}(k-c)(1-\rho)\rho^k$$

$$= \rho^c(1-\rho)\sum_{k'=0}^{\infty}k'\rho^{k'} = \frac{\rho^{c+1}}{1-\rho}$$

$$\mathcal{E}[W] = \frac{1}{\lambda}\mathcal{E}[N_q] = \frac{\rho^c}{1-\rho}\frac{1}{c\mu}$$

d) M/M/1

$$\mathcal{E}[T] = \frac{\frac{1}{c\mu}}{1-\rho} \quad \text{same as above system}$$

$$\mathcal{E}[W] = \frac{\rho}{1-\rho}\frac{1}{c\mu}$$

M/M/2

$$C(2,a) = \frac{a^2/2}{1-\rho}\left[1 + a + \frac{a^2/a}{1-\rho}\right]^{-1} = \frac{2\rho^2}{1+\rho} \quad \text{since } a = 2\rho$$

$$\mathcal{E}[W] = \frac{\frac{1}{\mu}}{c(1-\rho)}C(c,a) = \frac{2\rho^2(1+\rho)}{2\mu(1-\rho)}$$

$$\mathcal{E}[T] = \mathcal{E}[W] + \frac{1}{\mu} = \frac{2/(1+\rho)}{2\mu(1-\rho)}$$

Comparison:

	M/M/1	M/M/2	New System
$\mathcal{E}[T]$	$\frac{1}{2\mu(1-\rho)}$	$\frac{2/(1+\rho)}{2\mu(1-\rho)}$	$\frac{1}{2\mu(1-\rho)}$
$\mathcal{E}[W]$	$\frac{\rho}{2\mu(1-\rho)}$	$\frac{2\rho^2/(1+\rho)}{2\mu(1-\rho)}$	$\frac{1}{2\mu(1-\rho)}$

for $\mathcal{E}[T]$ New = M/M/1 < M/M/2

for $\mathcal{E}[W]$ New < M/M/2 < M/M/1

12.25

For the queue of P. 12.24 we have:

$$P_i = \frac{\lambda}{c\mu} P_{i-1} \Rightarrow P_i = \left(\frac{\lambda}{c\mu}\right)^i P_0 = (1-\rho)\rho^i$$

$$\sum_{j=0}^{\infty} P_j = 1 \Rightarrow P_0 + \sum_{j=1}^{\infty} P_j = 1 \Rightarrow \frac{1}{1-\frac{\lambda}{c\mu}} P_0 = 1 \Rightarrow P_0 = 1 - \frac{\lambda}{c\mu} = 1-\rho$$

$$E[N] = \frac{\rho}{1-\rho}, \quad E[T] = \frac{E[N]}{\lambda} = \frac{\frac{1}{c\mu}}{1-\frac{\lambda}{c\mu}} = \boxed{\frac{1}{c\mu-\lambda}}$$

$$E[N_q] = \sum_{i=c}^{\infty} (i-c) P_i = \sum_{i=c}^{\infty} (i-c)(1-\rho)\rho^i = \rho^c(1-\rho) \sum_{j=0}^{\infty} j\rho^j$$

$$= \rho^c(1-\rho) \times \frac{\rho}{(1-\rho)^2} = \boxed{\frac{\rho^{c+1}}{1-\rho}}$$

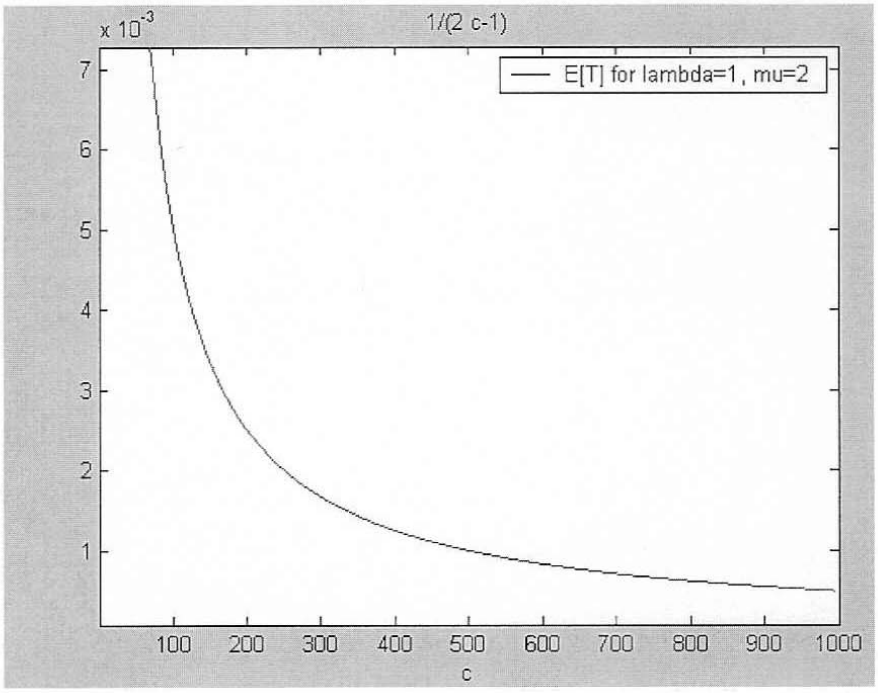
We write a simple code to plot $E[T]$ & $E(N_q)$

```
c=sym('c');
>> f=inline('1/(c*mu-lambda)');
>> f

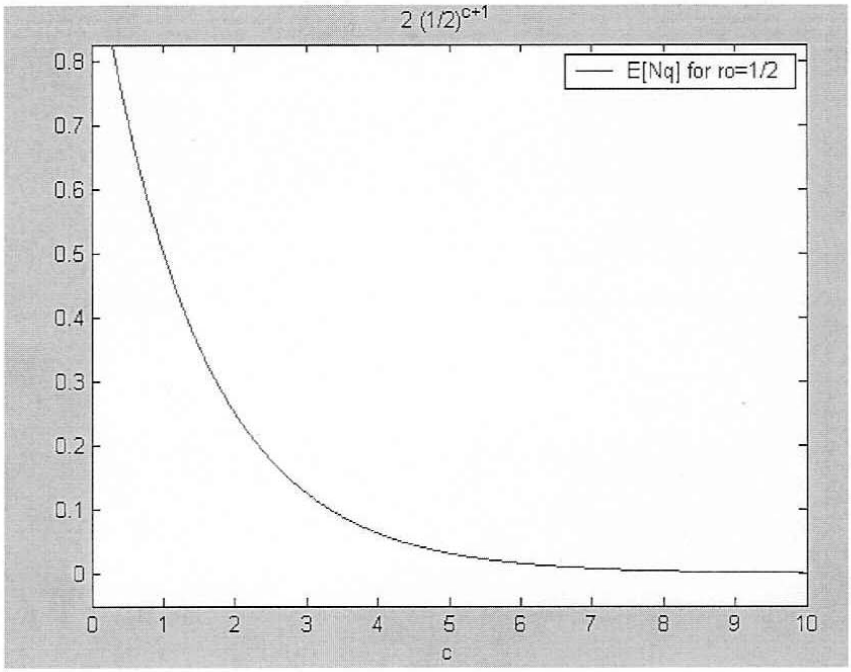
f =

    Inline function:
    f(c,lambda,mu) = 1/(c*mu-lambda)

>> ezplot(f(c,1,1))
```




```
f=inline('ro^(c+1)/(1-ro)');
ezplot(f(c,1/2))
```



12.26

$$\begin{aligned} B(c,a) &= \frac{\frac{a^c}{c!}}{\sum_{j=0}^{c}\frac{a^j}{j!}} = \frac{\frac{a^c}{c!}}{\frac{a^c}{c!}+\sum_{j=0}^{c-1}\frac{a^j}{j!}} = \frac{\frac{a^c}{c!}/\sum_{j=0}^{c-1}\frac{a^j}{j!}}{1+\frac{a^c}{c!}/\sum_{j=0}^{c-1}\frac{a^j}{j!}} \\ &= \frac{\frac{a}{c}B(c-1,a)}{1+\frac{a}{c}B(c-1,a)} = \frac{aB(c-1,a)}{c+aB(c-1,a)} \end{aligned}$$

12.27

~~9.24~~ $\lambda = 10 \qquad \frac{1}{\mu} = \frac{1}{2} \qquad \frac{\lambda}{\mu} = 5 = a$

$$\begin{aligned} B(0,5) &= 1 \\ B(1,5) &= \frac{5\cdot 1}{1+5\cdot 1} = \frac{5}{6} \\ B(2,5) &= \frac{5\left(\frac{5}{6}\right)}{2+5\left(\frac{5}{6}\right)} = \frac{25}{37} \\ B(3,5) &= \frac{5\left(\frac{25}{37}\right)}{3+5\left(\frac{25}{37}\right)} = \frac{125}{236} \\ B(4,5) &= \frac{5\left(\frac{125}{236}\right)}{4+5\left(\frac{125}{236}\right)} = \frac{625}{1569} \\ \rightarrow B(5,5) &= \frac{5\left(\frac{625}{1569}\right)}{5+5\left(\frac{625}{1569}\right)} = \frac{625}{2194} \approx 28.5\% \\ &\vdots \\ B(8,5) &= 0.070 \quad \text{need 3 more servers} \end{aligned}$$

12.28

~~9.25~~ **a)** $\lambda = \frac{1}{2} \qquad \frac{1}{\mu} = 2 \qquad a = 1$

$$B(4,1) = \frac{1}{65} = 1.54\%$$

b) $\mathcal{E}[N] = a(1 - B(4,1)) = \frac{64}{65} = 0.985$

c) $B(3,1) = \frac{1}{16} = 6.25\%$ an increase of 4.7%

12.29

9.26 For $a < c$

$$\textbf{a)} \quad C(c,a) = \frac{\frac{a^c}{c!}\frac{1}{1-\rho}}{\sum_{j=0}^{c-1}\frac{a^j}{j!} + \frac{a^c}{c!}\left(\frac{1}{1-\rho}\right)} = \frac{\frac{a^c}{c!}}{(1-\rho)\sum_{j=0}^{c-1}\frac{a^j}{j!} + \frac{a^c}{c!}} \quad \text{but } \rho = \frac{a}{c}$$

$$= \frac{\frac{a^c}{c!}}{\left(1-\frac{a}{c}\right)\sum_{j=0}^{c-1}\frac{a^j}{j!} + \frac{a^c}{c!}} = \frac{a\frac{a^{c-1}}{(c-1)!}}{(c-a)\sum_{j=0}^{c-1}\frac{a^j}{j!} + a\frac{a^{c-1}}{(c-1)!}}$$

$$= \frac{aB(c-1,a)}{(c-a)+1B(c-1,a)} \quad \text{since} \quad \frac{\frac{a^{c-1}}{(c-1)!}}{\sum_{j=0}^{c-1}\frac{a^j}{j!}} = B(c-1,a)$$

Problem 12.26 $\Rightarrow aB(c-1,a) = \dfrac{cB(c,a)}{1-B(c,a)}$

$$\therefore C(c,a) = \frac{cB(c,a)}{(c-a)(1-B(c,a)) + cB(c,a)} = \frac{cB(c,a)}{c-a(1-B(c,a))}$$

b) $C(c,a) = \dfrac{B(c,a)}{1-\frac{a}{c}(1-B(c,a))} > B(c,a)$ since

$$\frac{1}{1-\frac{a}{c}\underbrace{(1-B(c,a))}_{<1}} > \frac{1}{1-\underbrace{\frac{a}{c}}_{<1}} > 1$$

12.30

9.27 $\lambda = 1 \qquad \dfrac{1}{\mu} = 2 \qquad a = \dfrac{\lambda}{\mu} = 2$

a) $P[\text{redirected}] = B(3,2) = \dfrac{4}{19} = 21.1\%$

b) $P[\text{redirected}] = B(6,4) = \dfrac{256}{2185} = 11.7\%$

$\lambda' = 2 \qquad \dfrac{1}{\mu} = 2$

12.31

M/M/10

λ cust/sec

$\frac{\lambda}{2}$ Type 1 $\quad \overline{X}_1 = 1$

$\frac{\lambda}{2}$ Type 2 $\quad \overline{X}_2 = 3$

$\overline{X} = \frac{1}{2}\overline{X}_1 + \frac{1}{2}\overline{X}_2 = \frac{1}{2}(1+3) = 2$

$P_B = B(10, a) \qquad a = \lambda \overline{X} = 2\lambda$

alternative: 2 M/M/5 systems

	$\lambda=1$	$\lambda=2$	$\lambda=3$
$P_{B_1} = B(5, \frac{\lambda}{2})$	0%	0%	1%
$P_{B_2} = B(5, \frac{3\lambda}{2})$	1%	11%	24%
$P_B = B(10, 2\lambda)$	0%	1%	4%

Combined system does better for overall blocking; short service customers affected only at higher load

---

M/M/100

	$\lambda=46$	$\lambda=48$	$\lambda=50$
$P_{B1} = B(50, \frac{\lambda}{2})$	0%	0%	0%
$P_{B2} = B(50, \frac{3\lambda}{2})$	31%	33%	35%
$P_B = B(100, 2\lambda)$	3%	5%	8%

Combined system highly beneficial to customers with longer service time.

12.32

$$P[N = c] = \frac{a^c}{c!}e^{-a}$$

$$B(c,a) = \frac{\frac{a^c}{c!}}{\sum_{j=0}^{c}\frac{a^j}{j!}} < \frac{\frac{a^c}{c!}}{\sum_{j=0}^{\infty}\frac{a^j}{j!}} = \frac{a^c}{c!}e^{-a}$$

$\Rightarrow P[N = c]$ estimate is conservative

12.33

This system can be modeled with an M/M/∞ queue. $\lambda = 10$ $\frac{1}{\mu} = 3600 \Rightarrow \mu = \frac{1}{3600}$

$a = \frac{\lambda}{\mu} = 36{,}000$

$\Rightarrow P_j = \frac{a^j}{j!} e^{-a}$ a poisson dist.

when a is large enough, the poisson dist. can be approximated with a Gaussian, which is the case in this problem

12.5 Finite-Source Queueing Systems

12.34

~~9.29~~ a)

$$\rho = 1 - p_0 = 1 - \frac{1}{\sum_{k=0}^{K} \frac{K!}{(K-k)!}\left(\frac{\alpha}{\mu}\right)^k} \quad \text{let } j = K - k$$

$$= 1 - \frac{\left(\frac{\mu}{\alpha}\right)^K / K!}{\sum_{j=0}^{K} \frac{(\mu\alpha)^j}{j!}}$$

$$= 1 - B\left(K, \frac{\mu}{\alpha}\right)$$

Erlang B

$$K = 15 \quad \frac{1}{\mu} = 2 \quad \frac{1}{\alpha} = 30 \quad \frac{\mu}{\alpha} = 15$$

$$B(15, 15) = 0.18$$

$$\rho = 1 - B\left(K, \frac{\mu}{\alpha}\right) = 11 - 0.18 = 0.82$$

$$\lambda = \mu\rho = \frac{1}{2}0.82 = 0.41$$

$$\mathcal{E}[T] = \frac{K}{\lambda} - \frac{1}{\alpha} = \frac{15}{0.41} - 30 = 6.6$$

b) $K^* = \frac{\frac{1}{\mu} + \frac{1}{\alpha}}{\frac{1}{\mu}} = \frac{32}{2} = 16$

c) If we add 5 users we exceed K^* so

$$\mathcal{E}[T] \approx \frac{K}{\mu} - \frac{1}{\alpha} = 20(2) - 30 = 10$$

$$\lambda \approx \mu = 2$$

12.35

(a) $\frac{1}{\mu} = 0.01$, $\frac{1}{\alpha} = 5$ $\qquad \frac{\alpha}{\mu} = \frac{1/5}{\frac{1}{0.01}} = 20$

$$K^* \simeq \frac{\frac{1}{\mu} + \frac{1}{\alpha}}{\frac{1}{\mu}} = \frac{0.01 + 5}{0.01} = 501$$

(b) The pmf of the number of request can be obtained using eq 12.84

$$P[N_a = k] = \frac{\left[(K-1)!/(K-k-1)!\right](\alpha/\mu)^k}{\sum_{j=0}^{K-1}\left[(K-1)!/(K-j-1)!\right](\alpha/\mu)^j}.$$

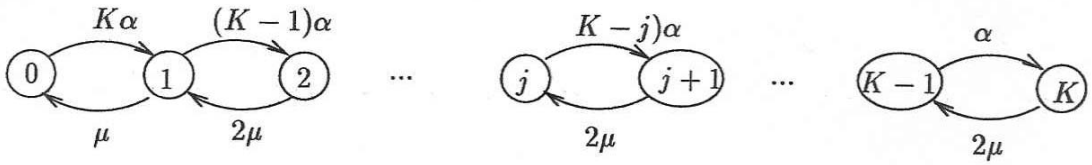
$$\Rightarrow P[N_a = k] = \frac{\frac{500!}{(500-k)!}(20)^k}{\sum_{j=0}^{500}\frac{500!}{(500-j)!}(20)^j}$$

```
% Problem 12.35

K=501;
alpha=1/5;
mu=100;
j=sym('j');
f=inline('factorial(500)/factorial(500-v)*(20)^v');
sum=0;
for j=0:(K-1)
    sum=sum+f(j);
end

sum
```

12.36



$$
\begin{aligned}
p_1 &= \frac{K\alpha}{\mu} p_0 \\
p_{j+1} &= \frac{(K-j)\alpha}{2\mu} p_j \qquad 1 < j \leq K-1 \\
\Rightarrow p_j &= \frac{K(K-1)\ldots(K-j+1)}{2^{j-1}} \left(\frac{\alpha}{\mu}\right)^j p_0 = 2\frac{K!}{(K-j)!}\left(\frac{\alpha}{2\mu}\right)^j p_0 \\
p_0 &= \left[1 + 2\sum_{j=1}^{K} \frac{K!}{(K-j)!}\left(\frac{\alpha}{2\mu}\right)^j\right]^{-1}
\end{aligned}
$$

12.37

$$P[N_a = k] = \frac{\frac{(K-1)!(\alpha/\mu)^k}{(K-1-k)!}}{\sum_{k'=0}^{K-1} \frac{(K-1)!(\alpha/\mu)^{k'}}{(K-1-k')!}} = \frac{\frac{(\alpha/\mu)^k}{(K-1-k)!}}{\sum_{k'=0}^{K-1} \frac{(\alpha/\mu)^{k'}}{(K-1-k')!}}$$

$$\begin{aligned}
\mathcal{E}[T] &= \frac{1}{\mu}\sum_{k=0}^{K-1}(k+1)P[N_a = k] \\
&= \frac{1}{\mu}\sum_{k=0}^{K-1}(k+1)\frac{\frac{(\alpha/\mu)^k}{(K-1-k)!}}{\sum_{k'=0}^{K-1}\frac{(\alpha/\mu)^{k'}}{(K-1-k')!}} \qquad \text{Let } j = K-1-k,\ j' = K-1-k' \\
&= \frac{1}{\mu}\sum_{j=0}^{K-1}(K-j)\frac{\frac{(\mu/\alpha)^j}{j!}}{\underbrace{\sum_{j'=0}^{K-1}\frac{(\mu/\alpha)^{j'}}{j'!}}_{\text{probs of M/M/K-1/K-1}}} \\
&= \frac{1}{\mu}\left[K - \underbrace{\frac{\mu}{\alpha}\left(1 - B(K-1, \frac{\mu}{\alpha}\right)}_{\text{mean \# in M/M/K-1/K-1}}\right] \\
&= \frac{K}{\mu} - \frac{1}{\alpha}\left(1 - B\left(K-1, \frac{\mu}{\alpha}\right)\right)
\end{aligned}$$

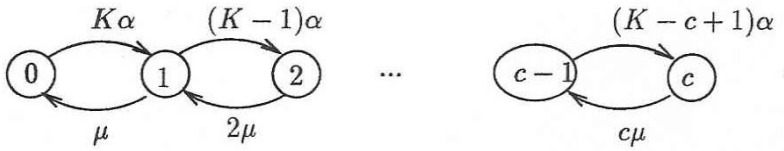
From Problem 12.26

$$\begin{aligned}
B\left(K-1, \frac{\mu}{\alpha}\right) &= \frac{\frac{\alpha K}{\mu}B\left(K, \frac{\mu}{\alpha}\right)}{1 - B\left(K, \frac{\mu}{\alpha}\right)} \\
\mathcal{E}[T] &= \frac{K}{\mu} - \frac{1}{\alpha} + \frac{\frac{K}{\mu}B\left(K, \frac{\mu}{\alpha}\right)}{1 - B\left(K, \frac{\mu}{\alpha}\right)} \\
\mathcal{E}[T] &= \frac{K}{\mu}\left[1 + \frac{B\left(K, \frac{\mu}{\alpha}\right)}{1 - B\left(K, \frac{\mu}{\alpha}\right)}\right] - \frac{1}{\alpha} \\
&= \frac{K}{\mu}\frac{1}{1 - B\left(K, \frac{\mu}{\alpha}\right)} - \frac{1}{\alpha}
\end{aligned}$$

But for Problem 12.34 solution

$$\begin{aligned}
\rho &= \frac{\lambda}{\mu} = 1 - B\left(K, \frac{\mu}{\alpha}\right) \\
\Rightarrow \mathcal{E}[T] &= \frac{K}{\lambda} - \frac{1}{\alpha} \qquad \text{as desired} \qquad \surd
\end{aligned}$$

12.38



where $c \leq K$

$$K\alpha P_0 = \mu P_1 \quad \Rightarrow \quad P_1 = \frac{K\alpha}{\mu} P_0$$

$$(K-j+1)\alpha P_{j-1} + j\mu P_j \quad \Rightarrow \quad P_j = \frac{(K-j+1)\alpha}{j\mu} P_{j-1}$$

$$\Rightarrow P_j = \frac{K...(K-j+1)}{j!}\left(\frac{\alpha}{\mu}\right)^j P_0 = \underbrace{\frac{K!}{j!(K-j)!}}_{\binom{K}{j}}\left(\frac{\alpha}{\mu}\right)^j P_0$$

$$\Rightarrow P_0 = \left[\sum_{j=0}^{c}\binom{K}{j}\left(\frac{\alpha}{\mu}\right)^j\right]^{-1}$$

$$\therefore P_j = \frac{\binom{K}{j}\left(\frac{\alpha}{\mu}\right)^j}{\sum_{j'=0}^{c}\binom{K}{j'}\left(\frac{\alpha}{\mu}\right)^{j'}} = \frac{\binom{K}{j}\left(\frac{\alpha}{\alpha+\mu}\right)^j\left(\frac{\mu}{\alpha+\mu}\right)^{K-j}}{\sum_{j'=0}^{c}\binom{K}{j'}\left(\frac{\alpha}{\alpha+\mu}\right)^j\left(\frac{\mu}{\alpha+\mu}\right)^{K-j}}$$

$$= \frac{\binom{K}{j}p^j(1-p)^{K-j}}{\sum_{j'=0}^{c}\binom{K}{j'}p^{j'}(1-p)^{K-j}}$$

b) $P[\text{all servers busy}] = P_c = \dfrac{\binom{K}{c}p^c(1-p)^{K-c}}{\sum_{j'=0}^{c}\binom{K}{j'}p^{j'}(1-p)^{K-j'}}$

c) Since an arriving customer sees the steady state of the system with one fewer server:

$P[\text{an arriving customer sees } c \text{ customers in system}]$

$$= P_{K-1}[N=c]$$

$$= \frac{\binom{K-1}{c}p^c(1-p)^{K-1-c}}{\sum_{j'=0}^{c}\binom{K-1}{j'}p^{j'}(1-p)^{K-1-j'}}$$

12.39

The probability that all the servers are busy in Engset (P. 12.38) is:

$$p_{busy} = \frac{\binom{K}{c} p^c (1-p)^{K-c}}{\sum_{i=0}^{c} \binom{K}{i} p^i (1-p)^{K-i}} \quad \text{where } p = \frac{\alpha}{\alpha+\mu}$$

$\mu = \frac{1}{0.1} = 10 \Rightarrow p = \frac{1}{11} \qquad c = 10$

$\alpha = 1$

we want $p_{busy} = 10\% = 0.1$

$$\Rightarrow \frac{\binom{K}{10} \left(\frac{1}{11}\right)^{10} \left(\frac{10}{11}\right)^{K-10}}{\sum_{i=0}^{10} \binom{K}{i} \left(\frac{1}{11}\right)^{i} \left(\frac{10}{11}\right)^{K-i}} = 0.1$$

12.6 M/G/1 Queueing Systems

12.40

9.33 A k-Erlang RV X with parameter k and λ has

$$\mathcal{E}[X] = \frac{k}{\lambda} \qquad VAR[X] = \frac{k}{\lambda^2}$$

Since $\mathcal{E}[X] = \frac{1}{\mu}$ we have that $\lambda = k\mu$ and

$$\begin{aligned} VAR[X] &= \frac{k}{k^2\mu^2} = \frac{1}{k\mu^2} \\ \Rightarrow C_X^2 &= \frac{VAR[X]}{\mathcal{E}[X]^2} = \frac{1}{k} \\ \therefore \mathcal{E}[W]_{M/E_k/1} &= \frac{\rho(1+C_X^2)}{2(1-\rho)}\mathcal{E}[\tau] = \frac{\rho\left(1+\frac{1}{k}\right)}{2(1-\rho)}\mathcal{E}[\tau] \end{aligned}$$

For M/M/1 we let $k = 1$ and obtain

$$\mathcal{E}[W]_{M/M/1} = \frac{2\rho}{2(1-\rho)}\mathcal{E}[\tau]$$

For M/D/1 $C_X^0 = 0$ so

$$\mathcal{E}[W]_{M/D/1} = \frac{\rho}{2(1-\rho)}\mathcal{E}[\tau]$$

$$\therefore \mathcal{E}[W]_{M/D/1} < \mathcal{E}[W]_{M/E_k/1} \leq \mathcal{E}[W]_{M/M/1}$$

Since $\mathcal{E}[T] = \mathcal{E}[W] + \mathcal{E}[\tau]$ the same ordering applies for total delay.

12.41

$$\begin{aligned} \mathcal{E}[\tau] &= \mathcal{E}[\tau|1]p + \mathcal{E}[\tau|2](1-p) = \frac{1}{\mu_1}p + \frac{1}{\mu_2}(1-p) \\ \mathcal{E}[\tau^2] &= \mathcal{E}[\tau^2|1]p + \mathcal{E}[\tau^2|2](1-p) = \frac{2}{\mu_1^2}p + \frac{2}{\mu_2^2}(1-p) \\ \mathcal{E}[W] &= \frac{\lambda\mathcal{E}[\tau^2]}{2(1-\rho)} = \frac{\lambda/2}{1-\rho}\left[\frac{2}{\mu_1^2}p + \frac{2}{\mu_2^2}(1-p)\right] \\ \mathcal{E}[T] &= \mathcal{E}[W] + \mathcal{E}[\tau] \end{aligned}$$

where $\rho = \lambda\mathcal{E}[\tau]$.

12.42

$$\begin{aligned}
\mathcal{E}[\tau] &= \mathcal{E}[\tau|1]\alpha + \mathcal{E}[\tau|2](1-\alpha) = d\alpha + \frac{1}{\mu}(1-\alpha) \\
\mathcal{E}[\tau^2] &= \mathcal{E}[\tau^2|1]\alpha + \mathcal{E}[\tau^2|2](1-\alpha) = d^2\alpha + \frac{2}{\mu_2^2}(1-\alpha) \\
\mathcal{E}[W] &= \frac{\lambda\mathcal{E}[\tau^2]}{2(1-\rho)} = \frac{\lambda/2}{1-\rho}\left[\alpha d^2 + (1-\alpha)\frac{2}{\mu^2}\right] \\
\mathcal{E}[T] &= \mathcal{E}[W] + \mathcal{E}[\tau]
\end{aligned}$$

where $\rho = \lambda\mathcal{E}[\tau]$.

12.43

$$\begin{aligned}
\tau &= d + \tau_1 \\
\mathcal{E}[\tau] &= d + \mathcal{E}[\tau_1] = d + \frac{1}{\mu} \\
\mathcal{E}[\tau^2] &= d^2 + 2d\mathcal{E}[\tau_1] + \mathcal{E}[\tau_1^2] \\
&= d^2 + \frac{2d}{\mu} + \frac{2}{\mu^2} \\
\mathcal{E}[W] &= \frac{\lambda/2}{1-\rho}\mathcal{E}[\tau^2] = \frac{\lambda/2}{1-\rho}\left[d^2 + \frac{2d}{\mu} + \frac{2}{\mu^2}\right] \\
\mathcal{E}[T] &= \mathcal{E}[W] + \mathcal{E}[\tau]
\end{aligned}$$

12.44

9.37 A message is transmitted until a successful acknowledgement is received:

a) $P[\tau = kd] = (1-p)p^{k-1} \qquad k =, 1, 2, \ldots$

$$\mathcal{E}[\tau] = \frac{d}{1-p} \qquad VAR[\tau] = \frac{d^2p}{(1-p)^2}$$

b) $C_\tau^2 = \dfrac{VAR[\tau]}{\mathcal{E}[\tau]^2} = \dfrac{d^2p}{(1-p)^2}\dfrac{(1-p)^2}{d^2} = p$

$$\begin{aligned}
\mathcal{E}[T] &= \mathcal{E}[\tau] + \mathcal{E}[\tau]\frac{\rho}{2(1-\rho)}(1+C_\tau^2) \qquad \text{where } \rho = \frac{\lambda d}{1-p} \\
&= \frac{2-\lambda d}{2(1-p-\lambda d)}d
\end{aligned}$$

12.47

9.38 a) Let τ = total job time, X = service time, $N(X)$ = # breakdowns during X, R_i = repair times

$$\tau = X + \sum_{i=1}^{N(X)} R_i$$

where $N(X)$ is the total number of times the machine breaks down. To find $C[\tau]$ we use conditional expectation:

$$\begin{aligned}
\mathcal{E}[\tau] &= \mathcal{E}[\mathcal{E}[\tau|X]] \\
\mathcal{E}[\tau|X=t] &= t + \mathcal{E}\left[\sum_{i=1}^{N(t)} R_i\right] = t + \alpha t \mathcal{E}[R] \quad \text{from Eq. 5.13} \\
\Rightarrow \mathcal{E}[\tau] &= \mathcal{E}[X + \alpha X \mathcal{E}[R]] = \underbrace{\mathcal{E}[X] + \alpha \mathcal{E}[X]\mathcal{E}[R]}_{\mathcal{E}[X](1+\alpha\mathcal{E}[R])} = \frac{1}{\mu}\left[1 + \frac{\alpha}{\beta}\right]
\end{aligned}$$

We also use conditional expectation to find $E[\tau^2]$:

$$\begin{aligned}
\mathcal{E}[\tau^2] &= \mathcal{E}[\mathcal{E}[\tau^2|X]] \\
\mathcal{E}[\tau^2|X=t] &= \mathcal{E}\left[\left(t + \sum_{i=1}^{N(t)} R_i\right)^2\right] \\
&= t^2 + 2t\mathcal{E}\left[\sum_{i=1}^{N(t)} R_i\right] + \mathcal{E}\left[\left(\sum_{i=1}^{N(t)} R_i\right)^2\right] \\
&= t^2 + 2t(\alpha t + \mathcal{E}[R]) + \mathcal{E}\left[\left(\sum_{i=1}^{N(t)} R_i\right)^2\right] \\
\mathcal{E}\left[\left(\sum_{i=1}^{N(t)} R_i\right)^2\right] &= \mathcal{E}\left[\mathcal{E}\left[\left(\sum_{i=1}^{N(t)} R_i\right)^2 | N(t)\right]\right] \\
\mathcal{E}\left[\left(\sum_{i=1}^{N(t)} R_i\right)^2 | N(t)=k\right] &= \mathcal{E}\left[\sum_{i=1}^{k}\sum_{j=1}^{k} R_i R_j\right] \\
&= k\mathcal{E}[R^2] + (k^2-k)\mathcal{E}[R]^2 \\
\therefore \mathcal{E}\left[\left(\sum_{i=1}^{N(t)} R_i\right)^2\right] &= \mathcal{E}[N(t)\mathcal{E}[R^2] + [N^2(t) - N(t)]\mathcal{E}[R]^2] \\
&= \mathcal{E}[N(t)]\mathcal{E}[R^2] + (\mathcal{E}[N^2(t)] - \mathcal{E}[N(t)])\mathcal{E}[R]^2
\end{aligned}$$

$$\begin{aligned} &= \alpha t\mathcal{E}[R^2] + (\alpha t + (\alpha t)^2 - \alpha t)\mathcal{E}[R]^2 \\ &= \alpha t\mathcal{E}[R^2] + \alpha^2 t^2 \mathcal{E}[R]^2 \\ \therefore \mathcal{E}[\tau^2|X=t] &= t^2 + 2\alpha t^2 \mathcal{E}[R] + \alpha t\mathcal{E}[R^2] + \alpha^2 t^2 \mathcal{E}[R]^2 \end{aligned}$$

finally

$$\begin{aligned} \mathcal{E}[\tau^2] &= \mathcal{E}[X^2 + 2\alpha X^2 \mathcal{E}[R] + \alpha X\mathcal{E}[R^2] + \alpha^2 X^2 \mathcal{E}[R]^2] \\ &= \mathcal{E}[X^2]\underbrace{[1 + 2\alpha\mathcal{E}[R] + \alpha^2\mathcal{E}[R]^2]}_{(1+\alpha\mathcal{E}[R])^2} + \mathcal{E}[X]\alpha\mathcal{E}[R^2] \end{aligned}$$

$$\begin{aligned} VAR[\tau] &= \mathcal{E}[\tau^2] - \mathcal{E}[\tau]^2 \\ &= \mathcal{E}[X^2](1+\alpha\mathcal{E}[R])^2 + \mathcal{E}[X]\alpha\mathcal{E}[R^2] - \mathcal{E}[X]^2(1+\alpha\mathcal{E}[R])^2 \\ &= VAR[X](1+\alpha\mathcal{E}[R])^2 + \mathcal{E}[X]\alpha\mathcal{E}[R^2] \\ &= \frac{1}{\mu^2}\left(1+\frac{\alpha}{\beta}\right)^2 + \frac{\alpha}{\mu}\frac{2}{\beta^2} \end{aligned}$$

b) The coefficient of variation of τ is:

$$C_\tau^2 = \frac{VAR[\tau]}{\mathcal{E}[\tau]^2} = \frac{\frac{1}{\mu^2}\left(1+\frac{\alpha}{\beta}\right)^2 + \frac{\alpha}{\mu}\frac{2}{\beta^2}}{\frac{1}{\mu^2}\left(1+\frac{\alpha}{\beta}\right)^2} = 1 + \frac{2\alpha}{(\alpha+\beta)^2}$$

Thus the mean delay in the system is

$$\begin{aligned} \mathcal{E}[T] &= \mathcal{E}[\tau] + \mathcal{E}[\tau]\frac{\rho}{2(1-\rho)}(1+C_\tau^2) \\ &= \mathcal{E}[\tau]\left[1 + \frac{\rho}{(1-\rho)}\left(1+\frac{\alpha}{(\alpha+\beta)^2}\right)\right] \end{aligned}$$

where

$$\rho = \lambda\mathcal{E}[\tau] = \frac{\lambda}{\mu}\left[1+\frac{\alpha}{\beta}\right]$$

12.48

9.39 a) The proportion of time that the server works on low priority jobs is

$$\begin{aligned} \rho_2' &= 1-\rho_1 = \lambda_2'\mathcal{E}[\tau_2] \\ \Rightarrow \lambda_2' &= \frac{1-\rho_1}{\mathcal{E}[\tau_2]} = \frac{1-\lambda_1\mathcal{E}[\tau_1]}{\mathcal{E}[\tau_2]} \end{aligned}$$

b) From Eq. (12.105)

$$\begin{aligned} \mathcal{E}[W_1] &= \frac{\lambda_1\mathcal{E}[\tau_1^2]+\lambda_2'\mathcal{E}[\tau_2^2]}{2(1-\rho_1)} \\ &= \frac{\lambda_1\mathcal{E}[\tau_1^2]}{2(1-\rho_1)} + \frac{\lambda_2'\mathcal{E}[\tau_2^2]}{2(1-\rho_1)} \\ &= \frac{\frac{\lambda_1}{2}\mathcal{E}[\tau_1^2]}{1-\lambda_1\mathcal{E}[\tau_1]} + \frac{\mathcal{E}[\tau_2^2]}{2\mathcal{E}[\tau_2]} \end{aligned}$$

since $\rho_1 = \lambda_1\mathcal{E}[\tau_1]$, $1-\rho_1 = \lambda_2'\mathcal{E}[\tau_2]$

12.49

9.40 The server vacations can be viewed as the servicing of a low priority class of fictitious customers whose service times are a vacation time and whose arrival rate saturates the system. The result of Problem 9.39b then implies

$$\mathcal{E}[W] = \frac{\frac{1}{2}\lambda\mathcal{E}[\tau^2]}{1-\lambda\mathcal{E}[\tau]} + \frac{\mathcal{E}[V^2]}{2\mathcal{E}[V]}$$

where W and τ correspond to the real customers.

12.50

9.41 If we suppose that the server in Problem 12.49 takes vacations of fixed duration d, then we have the system described in the problem. Thus

$$\mathcal{E}[W] = \frac{\frac{1}{2}\lambda d^2}{1-\lambda d} + \frac{d^2}{2d} = \frac{\frac{1}{2}\lambda d^2}{1-\lambda d} + \frac{d}{2}$$

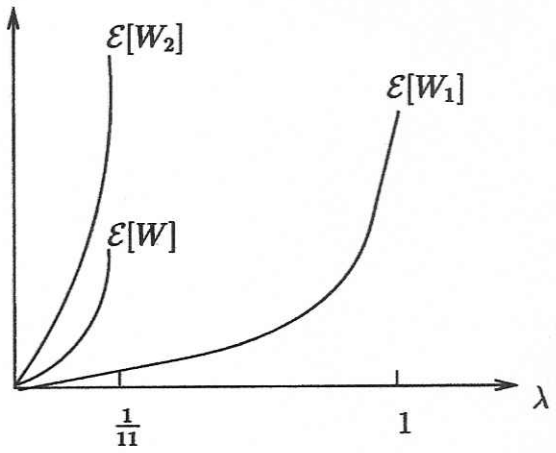
12.51

$$\begin{aligned}\mathcal{E}[\tau^2] &= \mathcal{E}[\tau^2|\text{type 1}]P[\text{type 1}] + \mathcal{E}[\tau^2|\text{type 2}]P[\text{type 2}] \\ &= \frac{2}{\mu_1^2}\frac{\lambda_1}{\lambda_1+\lambda_2} + \frac{2}{\mu_2^2}\frac{\lambda_2}{\lambda_1+\lambda_2} \\ &= 1 + 100 = 101\end{aligned}$$

where $\lambda = \lambda_1 = \lambda_2$ and $\mu_1 = 1$, $\mu_2 = \frac{1}{10}$.

$$\mathcal{E}[W_1] = \frac{\mathcal{E}[R']}{1-\rho_1} = \frac{\frac{2\lambda}{2}\mathcal{E}[\tau^2]}{1-\rho_1} = \frac{101\lambda}{1-\lambda} \quad \text{since } \rho_1 = \lambda/\mu_1 = \lambda$$

$$\begin{aligned}\mathcal{E}[W_2] &= \frac{101\lambda}{(1-\lambda)(1-\lambda-10\lambda)} \quad \text{since } \rho_2 = \lambda/\mu_2 = 10\lambda \\ &= \frac{101\lambda}{(1-\lambda)(1-11\lambda)} \\ \mathcal{E}[W] &= \frac{\lambda_1}{\lambda_1+\lambda_2}\mathcal{E}[W_1] + \frac{\lambda_1}{\lambda_1+\lambda_2}\mathcal{E}[W_2] \\ &= \frac{1}{2}\frac{101\lambda}{1-\lambda}\left[1 + \frac{1}{1-11\lambda}\right]\end{aligned}$$



12.52

9.43 a) The low priority customers are "invisible" to the high priority customers. Thus the mean waiting time and delay of high priority customers is that of a single-class M/G/1 system:

$$\begin{aligned}\mathcal{E}[W_1] &= \frac{\lambda_1 \mathcal{E}[\tau_1^2]}{2(1-\rho_1)} \\ \mathcal{E}[T_1] &= \mathcal{E}[W_1] + \mathcal{E}[\tau]\end{aligned}$$

b) The time required to service all customers found by a low priority arrival is the time required to service all such customers in an ordinary M/G/1 system in which both classes are combined and neither receives priority. The reason for this is that the priority mechanism alters the order in which customers are served but not the rate at which the backlog is reduced. The mean waiting time in such a system is

$$\frac{\lambda \mathcal{E}[\tau^2]}{2(1-\rho)} = \frac{\frac{1}{2}\sum_{j=1}^{2} \lambda_j \mathcal{E}[\tau_j^2]}{(1-\rho_1-\rho_2)}$$

since $\lambda = \lambda_1 + \lambda_2$, $\rho = \rho_1 + \rho_2$

c) The mean time required to service all the high priority customers that arrive while a low priority customer is in the system is

$$\begin{aligned}\mathcal{E}\left[\sum_{i=1}^{N_1(T_2)} \tau_{i_1}\right] &= \mathcal{E}[N_1(T_2)]\mathcal{E}[\tau_1] = \lambda_1 \mathcal{E}[T_2]\mathcal{E}[\tau_1] \\ &= \rho_1 \mathcal{E}[T_2]\end{aligned}$$

d)

$$\begin{aligned}\mathcal{E}[T_2] &= \frac{R_2}{1-\rho_1-\rho_2} + \rho_1 \mathcal{E}[T_2] + \frac{1}{\mu_2} \\ \mathcal{E}[T_2] &= \frac{R_2}{(1-\rho_1)(1-\rho_1-\rho_2)} + \frac{\frac{1}{\mu}}{1-\rho_1} \\ &= \frac{\frac{1}{\mu}(1-\rho_1-\rho_2) + R_2}{(1-\rho_1)(1-\rho_1-\rho_2)}\end{aligned}$$

12.53

9.44 $\lambda = \lambda_1 = \lambda_2 \qquad \mu_1 = 1 \qquad \mu_2 = \frac{1}{10}$

$$\begin{aligned}\mathcal{E}[W_1] &= \frac{\lambda_1 \mathcal{E}[\tau_1^2]}{2(1-\rho_1)} = \frac{\lambda}{1-\lambda} \\ \mathcal{E}[T_1] &= \frac{\lambda}{1-\lambda} + 1 = \frac{1}{1-\lambda} \\ R_2 &= \frac{1}{2}\lambda_1 \frac{2}{\mu_1^2} + \frac{1}{2}\lambda_2 \frac{2}{\mu_2^2} = \lambda + 100\lambda = 101\lambda \\ \mathcal{E}[T_2] &= \frac{101\lambda}{(1-\lambda)(1-11\lambda)} + \frac{10}{1-\lambda} = \frac{10-9\lambda}{(1-\lambda)(1-11\lambda)}\end{aligned}$$

The mean waiting time and delay of class 1 is reduced greatly while those of class 2 are not significantly affected relative to the corresponding values for a non-preemptive priority system.

12.7 M/G/1 Analysis Using Embedded Markov Chains

12.54

$$\rho = \frac{\lambda}{\mu} = \left(\frac{\mu}{2}\right)/\mu = \frac{1}{2}$$

a) For an M/G/1 system we have:

$$G_N(z) = \frac{(1-\rho)(z-1)\hat{\tau}(\lambda(1-z))}{z - \hat{\tau}(\lambda(1-z))}$$

where

$$\begin{aligned}
\hat{\tau}(\lambda(1-z)) &= \frac{4\mu^2}{(s+2\mu)^2}\Big|_{s=\lambda(1-z)} = \frac{4\mu^2}{(\lambda - \lambda z + 2\mu)^2} \\
\Rightarrow G_N(z) &= \frac{\left(1-\frac{1}{2}\right)(z-1)4\mu^2}{z(\lambda - \lambda z + 2\mu) - 4\mu^2} = \frac{8}{z^2 - 9z + 16} \\
&\quad \text{where we used the fact that } \frac{\lambda}{\mu} = \frac{1}{2} \\
&= \frac{8}{(z-z_1)(z-z_2)} \qquad z_1 = \frac{9+\sqrt{17}}{2} \qquad z_2 = \frac{9-\sqrt{17}}{2} \\
&= \frac{8/z_1 z_2}{\left(1-\frac{z}{z_1}\right)\left(1-\frac{z}{z_2}\right)} = \frac{\frac{1}{2}}{\left(1-\frac{1}{z_1}z\right)\left(1-\frac{1}{z_2}z\right)} \\
&= \frac{A}{1-\frac{1}{z_1}z} + \frac{B}{1-\frac{1}{z_2}z} \Rightarrow \begin{matrix} A = \frac{-z_2/2}{z_1 - z_2} \\ B = \frac{z_1/2}{z_1 - z_2} \end{matrix} \quad \text{partial fraction expansion} \\
&= \frac{z_1/2}{(z_1-z_2)\left(1-\frac{1}{z_2}z\right)} = \frac{z_2/2}{(z_1-z_2)\left(1-\frac{z}{z_1}\right)} \\
&= \frac{1}{2(z_1-z_2)}\left[z_1\sum_{j=0}^{\infty}\left(\frac{z}{z_2}\right)^j - z_2\sum_{j=0}^{\infty}\left(\frac{z}{z_1}\right)^j\right] \\
\therefore P[N=j] &= \frac{z_1}{2(z_1-z_2)}\left(\frac{1}{z_2}\right)^j - \frac{z_2}{2(z_1-z_2)}\left(\frac{1}{z_1}\right)^j \quad \text{coefficient of } Z^j \\
P[N=j] &= \frac{9+\sqrt{17}}{4\sqrt{17}}\left(\frac{2}{9-\sqrt{17}}\right)^j - \frac{9-\sqrt{17}}{4\sqrt{17}}\left(\frac{2}{9+\sqrt{17}}\right)^j \\
&= \frac{8}{\sqrt{17}}\left(\frac{2}{9-\sqrt{17}}\right)^{j+1} - \frac{8}{\sqrt{17}}\left(\frac{2}{9+\sqrt{17}}\right)^j \qquad j = 0, 1, \ldots
\end{aligned}$$

b) The Laplace Transform of the waiting time is:

$$\begin{aligned}
\hat{W}(s) &= \frac{(1-\rho)s}{s - \lambda + \lambda\hat{\tau}(s)} = \frac{\frac{1}{2}s}{s - \lambda + \frac{\lambda 4\mu^2}{(s+2\mu)^2}} = \frac{1}{2}\left[\frac{s^2 + 8\lambda s + 16\lambda^2}{s^2 + 7\lambda s + 8\lambda^2}\right] \\
&= \frac{1}{2}\left[1 + \frac{\left(\frac{\sqrt{17}+9}{2\sqrt{17}}\right)\lambda}{s + \left(\frac{7-\sqrt{17}}{2}\right)\lambda} + \frac{\left(\frac{\sqrt{17}-9}{2\sqrt{17}}\right)\lambda}{s + \left(\frac{7+\sqrt{17}}{2}\right)\lambda}\right]
\end{aligned}$$

$$\begin{aligned} f_W(t) &= \mathcal{L}^{-1}[\hat{W}(s)] \\ &= \frac{1}{2}\delta(t) + \frac{1}{2}\left(\frac{\sqrt{17}+9}{2\sqrt{17}}\right)\lambda e^{-\left(\frac{7-\sqrt{17}}{2}\right)\lambda t}u(t) \\ &\quad + \frac{1}{2}\left(\frac{\sqrt{17}-9}{2\sqrt{17}}\right)\lambda e^{-\left(\frac{7+\sqrt{17}}{2}\right)\lambda t}u(t) \end{aligned}$$

The total delay transform is:

$$\begin{aligned} \hat{T}(s) &= \frac{(1-\rho)s\hat{\tau}(s)}{s-\lambda+\lambda\hat{\tau}(s)} = \frac{\frac{1}{2}s\frac{4\mu^2}{(s+2\mu)^2}}{s-\lambda+\lambda\frac{4\mu^2}{(s+2\mu)^2}} \\ &= \frac{8\lambda^2}{s^2+7\lambda s+8\lambda^2} \\ \hat{T}(s) &= \frac{8\lambda}{\sqrt{17}}\left[\frac{1}{s+\left(\frac{7-\sqrt{17}}{2}\right)\lambda} - \frac{1}{s+\left(\frac{7+\sqrt{17}}{2}\right)\lambda}\right] \\ f_T(t) &= \mathcal{L}^{-1}[\hat{T}(s)] = \frac{8\lambda}{\sqrt{17}}\left[e^{-\left(\frac{7-\sqrt{17}}{2}\right)\lambda t} - e^{-\left(\frac{7+\sqrt{17}}{2}\right)\lambda t}\right]u(t) \end{aligned}$$

12.55

~~9.46~~ **a)** $\tau = X + \sum_{i=1}^{N(X)} R_i$ (see solution to 12.47)

$$\begin{aligned} \hat{\tau}(s) &= \mathcal{E}[e^{-s\tau}] = \mathcal{E}[\mathcal{E}[e^{-s\tau}|X]] \\ \mathcal{E}\left[e^{-s\left(X+\sum_{i=1}^{N(x)}R_i\right)}|X=t\right] &= e^{-st}\mathcal{E}\left[e^{-s\sum_{i=1}^{N(t)}R_i}\right] \\ \mathcal{E}\left[e^{-s\sum_{i=1}^{N(t)}R_i}\right] &= \mathcal{E}\left[\mathcal{E}\left[e^{-s\sum_{i=1}^{N}R_i}|N\right]\right] \\ &= \mathcal{E}[\underbrace{\mathcal{E}[e^{-sR}]^N}] \\ &= G_N[\hat{R}(s)] \quad \text{but} \quad G_N(z) = e^{\alpha t(z-1)} \\ &= e^{\alpha t(\hat{R}(s)-1)} \end{aligned}$$

$$\begin{aligned}\hat{\tau}(s) &= \mathcal{E}[e^{-sX}e^{\alpha X(\hat{R}(s)-1)}] = \mathcal{E}[e^{-X(s-\alpha(\hat{R}(s)-1))}] \\ &= \hat{X}(s-\alpha(\hat{R}(s)-1))\end{aligned}$$

But $\hat{R} = \dfrac{\beta}{s+\beta}$ and $\hat{X}(s) = \dfrac{\mu}{s+\mu}$

$$\begin{aligned}\therefore \hat{\tau}(s) &= \frac{\mu}{s-\alpha(\hat{\alpha}(s)-1)+\mu} = \frac{\mu}{s-\alpha\frac{-s}{s+\beta}+\mu} \\ &= \frac{\mu(s+\beta)}{(s+\mu)(s+\beta)+\alpha s} \quad \text{as required.}\end{aligned}$$

b)

$$\begin{aligned}\hat{W}(s) &= \frac{(1-\rho)s}{s-\lambda+\lambda\hat{\tau}(s)} = \frac{(1-\rho)s[(s+\mu)(s+\beta)+\alpha s]}{(s-\lambda)[(s+\mu)(s+\beta)+\alpha s]+\lambda\mu(s+\beta)} \\ &= (1-\rho)s[s^2+(\alpha+\beta+\mu)s+\mu\beta] \\ &= \frac{(1-\rho)(s^2+(\alpha+\beta+\mu)s+\mu\beta)}{s^2+(\alpha+\beta+\mu-\lambda)s-(\alpha+\beta+\mu-\mu\beta-\lambda\mu)} \\ &= (1-\rho)\left[1+\frac{\lambda s+(\alpha+\beta+\mu-\lambda\mu)}{s^2+(\alpha+\beta+\mu-\lambda)s-(\alpha+\beta+\mu-\mu\beta-\lambda\mu)}\right] \\ &\quad \text{where } \lambda_1 \text{ and } \lambda_2 \text{ are roots of denominator} \\ &= (1-\rho)\left[1+\frac{A}{s+\lambda_1}+\frac{B}{s+\lambda_2}\right] \\ f_W(t) &= (1-\rho)\delta(t)+(Ae^{-\lambda_1 t}+Be^{-\lambda_2 t})\mu(t)\end{aligned}$$

where A, B are obtained from a partial fraction expansion

$$\begin{aligned}\hat{T}(s) &= \frac{(1-\rho)s\hat{\tau}(s)}{s-\lambda+\lambda\hat{\tau}(s)} = \frac{(1-\rho)s\mu(s+\beta)}{(s-\lambda)[(s+\mu)(s+\beta)+\alpha s]+\lambda\mu(s+\beta)} \\ &= \frac{(1-\rho)\mu(s+\beta)}{s^2+(\alpha+\beta+\mu-\lambda)s-(\alpha+\beta+\mu-\mu\beta-\lambda\mu)} \\ &= \frac{A'}{s+\lambda_1}+\frac{B'}{s+\lambda_2} \\ f_T(t) &= (A'e^{-\lambda_1 t}+B'e^{-\lambda_2 t})\mu(t)\end{aligned}$$

12.56

9.47 a) $$N_j = N_{j-1} - U(N_{j-1}) + M_j = \begin{cases} N_j - 1 + M_j & N_{j-1} \geq 1 \quad (9.110a) \\ M_j & N_{j-1} = 0 \quad (9.110b) \end{cases}$$ ✓

b)

$$\begin{aligned} \mathcal{E}[N_j] &= \mathcal{E}[N_j] - \mathcal{E}[U(N_{j-1})] + \mathcal{E}[M_j] \\ \Rightarrow \mathcal{E}[M_j] &= \mathcal{E}[U(N_{j-1})] = P[N_{j-1} > 0] \\ \Rightarrow P[N > 0] &= \mathcal{E}[M] = \lambda \mathcal{E}[\tau] \end{aligned}$$

c)

$$\begin{aligned} N_j^2 &= N_{j-1}^2 - 2N_{j-1}U(N_{j-1}) + U(N_{j-1})^2 + M_j^2 \\ &\quad +2(N_{j-1} - U(N_{j-1}))M_j \end{aligned}$$

$N_{j-1}U(N_{j-1}) = N_{j-1}$ and $U(N_{j-1})^2 = U(N_{j-1})$

$$\begin{aligned} 0 = -2\mathcal{E}[N_{j-1}] &+ \mathcal{E}[U(N_{j-1})] + \mathcal{E}[M_j^2] \\ &+ 2\mathcal{E}[N_{j-1}]\mathcal{E}[M_j] - \mathcal{E}[U(N_{j-1})]\mathcal{E}[M_j] \end{aligned}$$

$$\begin{aligned} \mathcal{E}[N_{j-1}] &= \frac{\mathcal{E}[U(N_{j-1})](1 - \mathcal{E}[M_j]) + \mathcal{E}[M_j^2]}{2(1 - \mathcal{E}[M_j])} \\ &= \mathcal{E}[U(N_{j-1})] + \frac{\mathcal{E}[M_j^2]}{2(1 - \mathcal{E}[M_j])} \end{aligned}$$

From part b) $\mathcal{E}(U[N_j]) = \mathcal{E}[M] = \lambda \mathcal{E}[\tau]$

$$\begin{aligned} \mathcal{E}[M^2] &= \mathcal{E}[\mathcal{E}[M^2|\tau]] = \mathcal{E}[X\tau + \lambda^2\tau^2] \\ &= \lambda \mathcal{E}[\tau] + \lambda^2 \mathcal{E}[\tau^2] \end{aligned}$$

Finally

$$\begin{aligned} \mathcal{E}[N] &= \lambda \mathcal{E}[\tau] + \frac{\lambda \mathcal{E}[\tau] + \lambda^2 \mathcal{E}[\tau^2]}{2(1 - \lambda \mathcal{E}[\tau])} \qquad C_\tau^2 = \frac{\lambda^2 \mathcal{E}[\tau^2]}{\lambda^2 \mathcal{E}[\tau]^2} \\ &= \lambda \mathcal{E}[\tau] + \lambda \mathcal{E}[\tau] \frac{(1 + C_\tau^2)}{2(1 - \lambda \mathcal{E}[\tau])} \\ &= \lambda \mathcal{E}[T] \leftarrow \text{as given by 9.94} \quad \surd \end{aligned}$$

12.57

9.48 a)

$$G_N(z) = \frac{(1-\rho)(z-1)\hat{\tau}(\lambda(1-z))}{z-\hat{\tau}(\lambda(1-z))} \qquad \hat{\tau}(s) = e^{-sd}$$

$$= \frac{(1-\rho)(z-1)e^{-\lambda d(1-z)}}{z-e^{-\lambda d(1-z)}} = \frac{(1-\rho)(1-z)}{1-ze^{\rho(1-z)}} \qquad \text{where } \rho = \lambda d$$

b)

$$\frac{1}{1-e^{\rho(1-z)}z} = \sum_{k=0}^{\infty} e^{\rho(1-z)k}z^k = \sum_{k=0}^{\infty} e^{-\rho zk}e^{\rho k}z^k$$

$$= \sum_{k=0}^{\infty}\sum_{l=0}^{\infty}\frac{(-\rho kz)^l}{l!}e^{\rho k}z^k$$

$$\therefore G_N(z) = (1-\rho)(1-z)\sum_{k=0}^{\infty}\sum_{l=0}^{\infty}\frac{(-\rho k)^l}{l!}e^{\rho k}z^{l+k}$$

$$= (1-\rho)\sum_{k=0}^{\infty}\sum_{l=0}^{\infty}\frac{(-\rho k)^l}{l!}e^{\rho k}(z^{l+k}-z^{l+k+1})$$

$$= \sum_{k'=0}^{\infty} P[N=k']z^{k'}$$

where

$$P[N=k'] = (1-\rho)\left\{\sum_{l,k:l+k=k'}\frac{(-\rho k)^l e^{\rho k}}{l!} - \sum_{l,k:l+k+1=k'}\frac{(-\rho k)^l e^{\rho k}}{l!}\right\}$$

$$P[N=k'] = (1-\rho)\sum_{j=0}^{k'}\left[\frac{(-j\rho)^{k'-j}e^{j\rho}}{(k'-j)!} - \frac{(-j\rho)^{k'-j-1}e^{j\rho}}{(k'-j-1)!}\right]$$

$$= (1-\rho)\sum_{j=0}^{k'}\frac{(-j\rho)^{k'-j-1}(-j\rho-k'+j)}{(k'-j)!}e^{j\rho} \qquad \checkmark$$

12.58

$$\hat{W}(s) = \frac{(1-\rho)s}{s-\lambda+\lambda\hat{\tau}(s)} = \frac{1-\rho}{1-\lambda\frac{1-\hat{\tau}(s)}{s}} = \frac{1-\rho}{1-\rho\frac{1-\hat{\tau}(s)}{s\mathcal{E}[\tau]}}$$

$$= \frac{1-\rho}{1-\rho\hat{R}(s)} \quad \text{where } \hat{R}(s) = \frac{1-\hat{\tau}(s)}{s\mathcal{E}[\tau]}$$

But

$$f_R(t) = \mathcal{L}^{-1}\left[\frac{1-\hat{\tau}(s)}{s\mathcal{E}(\tau)}\right] = \frac{1}{\mathcal{E}[\tau]}[1-F_\tau(x)]$$

which is pdf of residual service time as given by Eqn. 12.87

$$\therefore \hat{W}(s) = (1-\rho)\sum_{k=0}^{\infty}(\rho\hat{R}(s))^k$$

$$= \sum_{k=0}^{\infty}(1-\rho)\rho^k\hat{R}^k(s)$$

and

$$f_W(t) = \sum_{k=0}^{\infty}(1-\rho)\rho^k f^{(k)}(x)$$

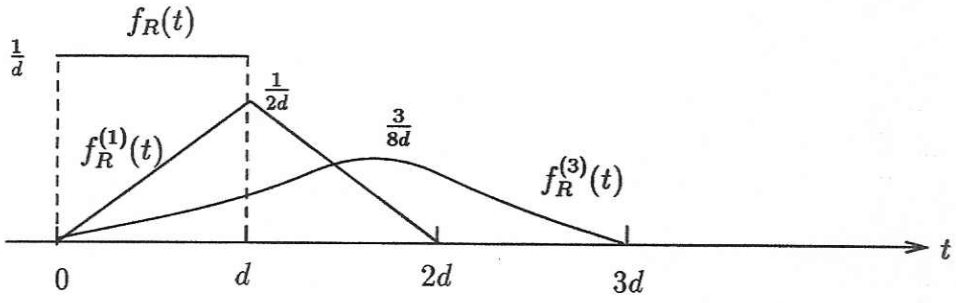
where

$$f^{(k)}(x) = \mathcal{L}^{-1}[\hat{R}^k(s)]$$

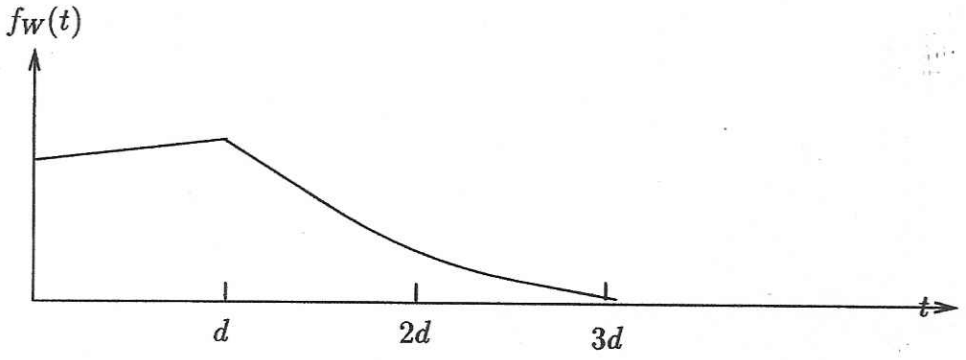
12.59

9.50 For M/D/1 $\hat{\tau}(s) = e^{-sd}$ and

$$\begin{aligned}\hat{R}(s) &= \frac{1-e^{-sd}}{sd} \\ \Rightarrow f_R(t) &= \frac{1-\mu(t-d)}{d} = \begin{cases} 1 & 0 \le t \le d \\ 0 & ew \end{cases}\end{aligned}$$



$$\begin{aligned}\therefore f_W(t) &\approx (1-\rho)f_R(t) + (1-\rho)\rho f_R^{(1)}(t) + (1-\rho)\rho^2 f_R^{(2)}(t) \\ &= \frac{1}{2}f_R(t) + \frac{1}{4}f_R^{(1)}(t) + \frac{1}{8}f_R^{(2)}(t)\end{aligned}$$



12.8 Burke's Theorem: Departures From M/M/*c* Systems

12.60

9.51 a) If a departure leaves the system nonempty, then another customer commences service immediately. Thus the time until the next departure is an exponential random variable with mean $1/\mu$.

b) If a departure leaves the system empty, then the time until the next departure is equal to the sum of an exponential interarrival time (of mean $1/\lambda$) followed by an exponential service time (of mean $1/\mu$).

c) The Laplace transform of the interdeparture time is

$$\frac{\mu}{s+\mu} \quad \text{when a departure leaves system nonempty}$$

$$\frac{\lambda}{s+\lambda}\frac{\mu}{s+\mu} \quad \text{when a departure leaves system empty}$$

$$\therefore \mathcal{E}[e^{-sT_d}] = \frac{\mu}{s+\mu}\underbrace{\rho}_{\substack{\text{prob. system}\\ \text{left nonempty}}} + \frac{\lambda}{s+\lambda}\frac{\mu}{s+\mu}\underbrace{(1-\rho)}_{\substack{\text{prob. system}\\ \text{left empty}}}$$

$$= \frac{\lambda}{s+\mu} + \frac{\lambda(\mu-\lambda)}{(s+\lambda)(s+\mu)} = \frac{\lambda(s+\lambda)+\lambda\mu-\lambda^2}{(s+\lambda)(s+\mu)}$$

$$= \frac{\lambda}{s+\lambda} \Rightarrow T_d \text{ exponential with mean } 1/\lambda$$

12.61

9.52 Claim:

$$P[N_1 = n, N_2 = m] = (1-\rho_1)\rho_1^n(1-\rho_2)\rho_2^m \qquad \begin{matrix} n, m \geq 0 \\ \rho_i = \lambda/\mu_i \end{matrix}$$

Eq. 9.135a

$$\begin{aligned} \lambda P[N_1 = 0, N_2 = 0] &= \lambda(1-\rho_1)(1-\rho_2) \\ &= \mu_2(1-\rho_1)(1-\rho_2)\rho_2 = \mu_2 P[N_1 = 0, N_2 = 1] \quad \surd \end{aligned}$$

Eq. 9.135b

$$\begin{aligned} (\lambda+\mu_1)P[N_1 = 0, N_2 = 0] &= (\lambda+\mu_1)(1-\rho_1)\rho_1^n(1-\rho_2) \\ &= \lambda(1-\rho_1)\rho_1^n(1-\rho_2) + \mu_1(1-\rho_1)\rho_1^n(1-\rho_2) \\ &= \mu_2\rho_2(1-\rho_1)\rho_1^n(1-\rho_2) + \lambda(1-\rho_1)\rho_1^{n-1}(1-\rho_2) \\ &= \mu_2 P[N_1 = n, N_2 = 1] + \lambda P[N_1 = n-1, N_2 = 0] \quad \surd \end{aligned}$$

Eqn. 9.135c

$$\begin{aligned} (\lambda+\mu_2)P[N_1 = 0, N_2 = m] &= (\lambda+\mu_2)(1-\rho_1)(1-\rho_2)\rho_2^m \\ &= \mu_2(1-\rho_1)(1-\rho_2)\rho_2^{m+1} + \mu_1(1-\rho_1)\rho_1(1-\rho_2)\rho_2^{m-1} \\ &= \mu_2 P[N_1 = 0, N_2 = m+1] + \mu_1 P[N_1 = 1, N_2 = m-1] \quad \surd \end{aligned}$$

Eqn. 9.135d

$$\begin{aligned} (\lambda+\mu_1+\mu_2)P[N_1 = n, N_2 = m] &= \lambda(1-\rho_1)\rho_1^n(1-\rho_2)\rho_2^m + \mu_1(1-\rho_1)\rho_1^n(1-\rho_2)\rho_2^m \\ &\quad +\mu_2(1-\rho_1)\rho_1^n(1-\rho_2)\rho_2^m \\ &= \mu_2(1-\rho_1)\rho_1^n(1-\rho_2)\rho_2^{m+1} + \lambda(1-\rho_1)\rho_1^{n-1}(1-\rho_2)\rho_2^m \\ &\quad +\mu_1(1-\rho_1)\rho_1^{n+1}(1-\rho_2)\rho_2^{m-1} \\ &= \mu_2 P[N_1 = n, N_2 = m+1] + \lambda P[N_1 = n-1, N_2 = m] \\ &\quad +\mu_1 P[N_1 = n+1, N_2 = m-1] \quad \surd \end{aligned}$$

12.62

9.53 The arrival process at queue #3 is the merge of two independent Poisson processes with combined rate

$$\lambda_1 + \frac{1}{2}\lambda_2$$

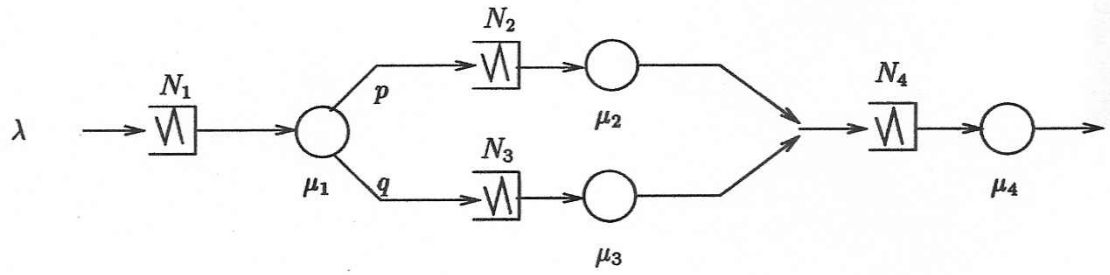
The state of queue #3 at time t is independent of those at queues #1 and #2 at time t:

$$P[N_1(t) = i, N_2(t) = j, N_3(t) = k] = (1-\rho_1)\rho_1^2(1-\rho_2)\rho_2^j(1-\rho_3)\rho_3^k$$

where

$$\begin{aligned} \rho_1 &= \frac{\lambda_1}{\mu_1} < 1 \\ \rho_2 &= \frac{\lambda_2}{\mu_2} < 1 \\ \text{and } \rho_3 &= \frac{(\lambda_1 + \frac{1}{2}\lambda_2)}{\mu_3} < 1 \end{aligned}$$

12.63



Claim:

$$\begin{aligned} P[N_1 = k, N_2 = l, N_3 = m, N_4 = n] &= (1-\rho)\rho_1^k(1-\rho_2)\rho_2^l(1-\rho_3)\rho_3^m(1-\rho_4)\rho_4^n \\ &\triangleq A\rho_1^k\rho_2^l\rho_3^m\rho_4^n \end{aligned}$$

where

$$\rho_1 = \frac{\lambda}{\mu_1} \quad \rho_2 = \frac{p\lambda}{\mu_2} \quad \rho_3 = \frac{q\lambda}{\mu_3} \quad \rho_4 = \frac{\lambda}{\mu_4}$$

Let $\underline{e}_i$ be the ith unit vector.

Let $\underline{s} = (klmn)$ where $k, l, m, n > 0$. The balance equation for this state is

$$\begin{aligned} (\lambda+\mu_1+\mu_2+\mu_3+\mu_4)P[\underline{s}] &= \lambda P[\underline{s}-\underline{e}_1] + p\mu_1 P[\underline{s}+\underline{e}_1-\underline{e}_2] \\ &\quad +q\mu_1 P[\underline{s}+\underline{e}_1-\underline{e}_2] + \mu P[\underline{s}+\underline{e}_2-\underline{e}_4] \\ &\quad +\mu_3 P[\underline{s}+\underline{e}_3-\underline{e}_4] + \mu_4 P[\underline{s}+\underline{e}_4] \\ &= \lambda P[\underline{s}]\rho_1^{-1} + p\mu_1 P[\underline{s}]\rho_1\rho_2^{-1} + q\mu_1 P[\underline{s}]\rho_1\rho_3^{-1} \\ &\quad +\mu_2 P[\underline{s}]\rho_2\rho_4^{-1} + \mu_3 P[\underline{s}]\rho_3\rho_4^{-1} | \mu_4 P[\underline{s}]\rho_4 \\ &= \mu_1 P[\underline{s}] + \mu_2 P[\underline{s}] + \mu_3 P[\underline{s}] + \mu_4 P[\underline{s}] \\ &\quad +\lambda P[\underline{s}] \quad \surd \end{aligned}$$

$\therefore P[\underline{s}] = A\rho_1^k\rho_2^l\rho_3^m\rho_4^n$ satisfies this balance equation.

There are 15 other special cases of boundary balance equations. These are shown to be satisfied by $P[\underline{s}]$ in similar fashion.

12.9 Networks of Queues: Jackson's Theorem

12.64

9.55 $\lambda_1 = \lambda \qquad \lambda_2 = \frac{1}{2}\lambda + \frac{1}{2}\lambda_3 \qquad \lambda_3 = \lambda_2 + \frac{1}{2}\lambda$

$$\Rightarrow \quad \lambda_1 = \lambda \qquad \lambda_2 = \frac{3}{2}\lambda \qquad \lambda_3 = 2\lambda$$

$$\Rightarrow \quad \rho_1 = \frac{\lambda}{\mu_1} \qquad \rho_2 = \frac{3\lambda}{2\mu_2} \qquad \rho_3 = \frac{2\lambda}{\mu_3}$$

Assuming $\rho_i < 1$, $i = 1, 2, 3$, then

$$P[N_1 = k, N_2 = l, N_3 = m] = (1-\rho_1)\rho_1^k(1-\rho_2)\rho_2^l(1-\rho_3)\rho_3^m \qquad k, l, m \geq 0$$

12.65

9.56 $I = 3$

$$\left.\begin{array}{rcl} \pi_0 &=& p\pi_0 + \pi_1 + \pi_2 \\ \pi_1 &=& \frac{1}{2}(1-p)\pi_0 \\ \pi_2 &=& \frac{1}{2}(1-p)\pi_0 \end{array}\right\} \begin{array}{l} \pi_0 = \frac{1}{2-p} \\ \pi_1 = \pi_2 = \frac{1-p}{2(2-p)} \end{array}$$

a) Then

$$\lambda_0 = \lambda(3)\pi_0 = \frac{\lambda(3)}{2-p} \qquad \rho_0 = \frac{\lambda_0}{\mu}$$

$$\lambda_1 = \lambda(3)\pi_1 = \frac{\lambda(3)(1-p)}{2(2-p)} \qquad \rho_1 = \frac{\lambda_1}{\mu_1}$$

$$\lambda_2 = \lambda_1 \qquad \rho_2 = \frac{\lambda_1}{\mu_2}$$

$$\begin{aligned} S(3) &= (1-\rho_0)(1-\rho_1)(1-\rho_2)[\rho_0^3 + \rho_1^3 + \rho_2^3 + \rho_0\rho_1^2 + \rho_0\rho_2^2 \\ &\quad + \rho_1\rho_2^2 + \rho_1\rho_0^2 + \rho_2\rho_0^2 + \rho_2\rho_1^2 + \rho_0\rho_1\rho_2] \\ &= (1-\rho_0)(1-\rho_1)(1-\rho_2)[(\rho_0^2 + \rho_1^2 + \rho_2^2)(\rho_0 + \rho_1 + \rho_2) + \rho_0\rho_1\rho_2] \end{aligned}$$

$$\therefore P[N_0 = i, N_1 = j, N_2 = 3-i-j] = \frac{\rho_0^i \rho_1^j \rho_2^{3-i-j}}{(\rho_0^2 + \rho_1^2 + \rho_2^2)(\rho_0 + \rho_1 + \rho_2) + \rho_1\rho_2\rho_3}$$

$$0 \leq i, j \quad \text{and} \quad i + j \leq 3$$

b) The program completion rate is

$$p\mu[1 - P[N_0 = 0]] = p\mu\frac{\rho_0^3 + \rho_0^2\rho_1 + \rho_0^2\rho_2 + \rho_0\rho_2^2 + \rho_0\rho_1^2 + \rho_0\rho_1\rho_2}{(\rho_0^2 + \rho_1^2 + \rho_2^2)(\rho_0 + \rho_1 + \rho_2) + \rho_1\rho_2\rho_3}$$

12.10 Simulation and Data Analysis of Queueing Systems

12.66

From Eq. (12.56) we have:

$$\pi_0 = \frac{1}{2-p} \qquad \pi_1 = \pi_2 = \frac{1-p}{2(2-p)}$$

We need to find $p\lambda_0(3) = p\pi_0\lambda(3)$

$I = 1$

$$\mathcal{E}[T_0(1)] = \frac{1}{\mu} \qquad \mathcal{E}[T_1(1)] = \frac{1}{\mu_1} \qquad \mathcal{E}[T_2(1)] = \frac{1}{\mu_2}$$

$$\lambda(1) = 1\left[\frac{\frac{1}{\mu}}{2-p} + \left(\frac{1}{\mu_1}+\frac{1}{\mu_2}\right)\frac{1-p}{2(2-p)}\right]^{-1}$$

$$\mathcal{E}[N_0(1)] = \frac{\frac{1}{\mu}}{\frac{1}{\mu}+\frac{1-p}{2}\left(\frac{1}{\mu_1}+\frac{1}{\mu_2}\right)} \triangleq \frac{a}{a+b+c}$$

$$\mathcal{E}[N_1(1)] = \frac{\frac{1-p}{2}\frac{1}{\mu_1}}{\frac{1}{\mu}+\frac{1-p}{2}\left(\frac{1}{\mu_1}+\frac{1}{\mu_2}\right)} \triangleq \frac{b}{a+b+c}$$

$$\mathcal{E}[N_2(1)] = \frac{\frac{1-p}{2}\frac{1}{\mu_2}}{\frac{1}{\mu}+\frac{1-p}{2}\left(\frac{1}{\mu_1}+\frac{1}{\mu_2}\right)} \triangleq \frac{c}{a+b+c}$$

where

$$a \triangleq \frac{1}{\mu} \qquad b = \frac{1-p}{2}\frac{1}{\mu_1} \qquad c = \frac{1-p}{2}\frac{1}{\mu_2}$$

$I = 2$

$$\mathcal{E}[T_0(2)] = \frac{1}{\mu}\left[\frac{2a+b+c}{a+b+c}\right]$$

$$\mathcal{E}[T_1(2)] = \frac{1}{\mu_1}\left[\frac{a+2b+c}{a+b+c}\right]$$

$$\mathcal{E}[T_2(2)] = \frac{1}{\mu_2}\left[\frac{a+b+2c}{a+b+c}\right]$$

$$\lambda(2) = 2\left[\frac{1}{\frac{1}{2-p}\mathcal{E}[T_0(2)] + \frac{1-p}{2}\frac{1}{2-p}\mathcal{E}[T_1(2)] + \frac{1-p}{2}\frac{1}{2-p}\mathcal{E}[T_2(2)]}\right]$$

$$= 2\left[\frac{(2-p)(a+b+c)}{\underbrace{\frac{1}{\mu}}_{a}(2a+b+c)+\underbrace{\frac{1-p}{2}\frac{1}{\mu_1}}_{b}(a+2b+c)+\underbrace{\frac{1-p}{2}\frac{1}{\mu_2}}_{c}(a+b+2c)}\right]$$

$$= 2\frac{(2-p)(a+b+c)}{2a^2+2b^2+2c^2+2ab+2ac+2bc}$$

$$= \frac{(2-p)(a+b+c)}{a^2+b^2+c^2+ab+ac+bc}$$

$$\mathcal{E}[N_0(2)] = \lambda(2)\pi_0\mathcal{E}[T_0(2)] = \lambda(2)\frac{1}{2-p}\frac{1}{\mu}\left[\frac{2a+b+c}{a+b+c}\right]$$

$$= \frac{1}{\mu}\frac{2a+b+c}{a^2+b^2+c^2+ab+ac+bc} = \frac{2a^2+ab+ac}{a^2+b^2+c^2+ab+ac+bc}$$

$$\mathcal{E}[N_1(2)] = \frac{\overbrace{\frac{1}{\mu_1}\frac{1-p}{2}}^{b}(a+2b+c)}{a^2+b^2+c^2+ab+ac+bc} = \frac{ab+2b^2+bc}{a^2+b^2+c^2+ab+ac+bc}$$

$$\mathcal{E}[N_2(2)] = \frac{c(a+b+2c)}{a^2+b^2+c^2+ab+ac+bc} = \frac{ac+bc+2c^2}{a^2+b^2+c^2+ab+ac+bc}$$

$I = 3$

$$\mathcal{E}[T_0(3)] = \frac{1}{\mu}[1+\mathcal{E}[N_0(2)]] = \frac{3a^2+b^2+c^2+2ab+2ac+bc}{a^2+b^2+c^2+ab+ac+bc}$$

$$\mathcal{E}[T_1(3)] = \frac{1}{\mu_1}\left[\frac{2ab+3b^2+2bc+a^2+c^2+ac}{a^2+b^2+c^2+ab+ac+bc}\right]$$

$$\mathcal{E}[T_2(3)] = \frac{1}{\mu_2}\left[\frac{2ac+2bc+3c^2+a^2+b^2+c^2+ab}{a^2+b^2+c^2+ab+ac+bc}\right]$$

$$\lambda(3) = 3\frac{1}{\frac{1}{2-p}\mathcal{E}[T_0(3)]+\frac{1-p}{2}\frac{1}{2-p}\mathcal{E}[T_1(3)]+\frac{1-p}{2}\frac{1}{2-p}\mathcal{E}[T_2(3)]}$$

Program completion rate is

$$p\lambda_0 = px_0\lambda(3)$$

$$= \frac{p}{2-p}\lambda(3)$$

$$
\begin{aligned}
&= \frac{3p}{\mathcal{E}[T_0(3)] + \frac{1-p}{2}\mathcal{E}[T_1(3)] + \frac{1-p}{2}\mathcal{E}[T_2(3)]} \\
&= [3p(a^2+b^2+c^2+ab+ac+bc)]/[a(3a^2+b^2+c^2+2ab+2ac+bc) \\
&\quad +b(2ab+3b^2+2bc+a^2+c^2+ac)+c(2ac+2bc+3c^2+a^2+b^2+ab)] \\
&= \frac{p\mu\overbrace{\left(\frac{1}{\mu}\right)}^{a}(a^2+b^2+c^2+ab+ac+bc)}{a^3+b^3+c^3+ab^2+ac^2+a^2b+a^2c+b^2c+bc^2+abc}
\end{aligned}
$$

We will multiply the numerator and denominator above by $\left(\frac{\lambda(3)}{2-p}\right)^3$ but first note that

$$
\begin{aligned}
\frac{\lambda(3)a}{2-p} &= \frac{\lambda(3)}{\mu(2-p)} = \lambda_0(3)\frac{1}{\mu} = \rho_0 \\
\frac{\lambda(3)b}{2-p} &= \frac{\lambda(3)(1-p)}{2(2-p)\mu_1} = \rho_1 \qquad \frac{\lambda(3)c}{2-p} = \rho_2
\end{aligned}
$$

$\therefore$ Completion Rate

$$= p\mu\frac{\rho_0^3+\rho_0^2\rho_1+\rho_0^2\rho_2+\rho_0\rho_1^2+\rho_0\rho_1\rho_2}{(\rho_0^2+\rho_1^2+\rho_2^2)(\rho_0+\rho_1+\rho_2)+\rho_1\rho_2\rho_3} \qquad \surd$$

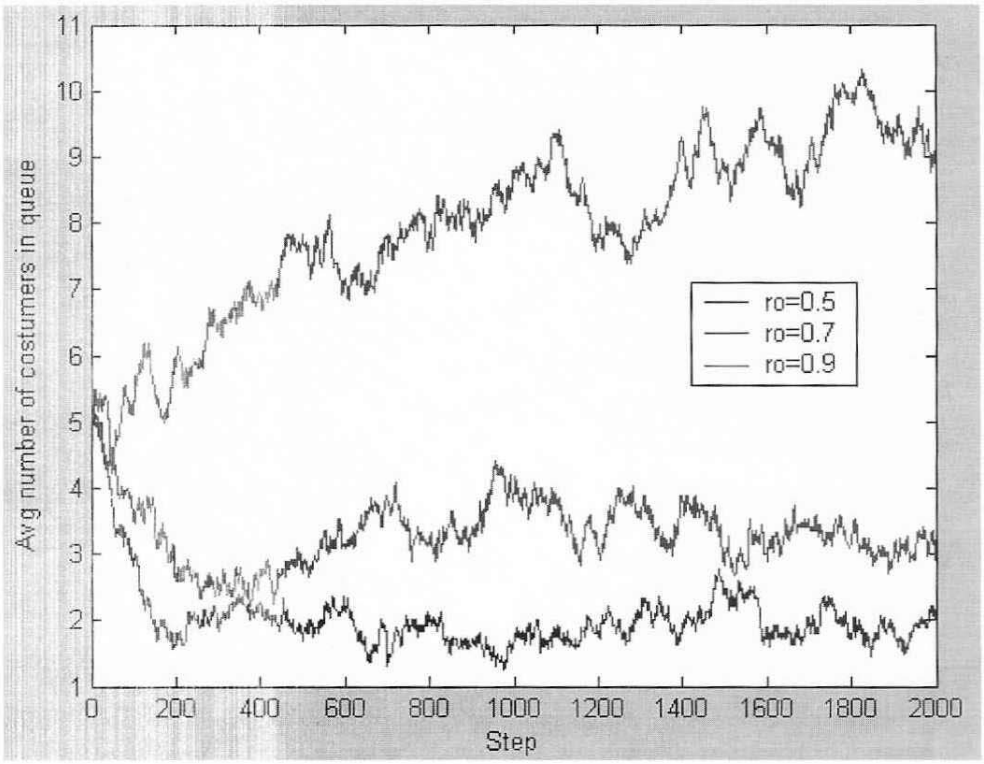
12.67

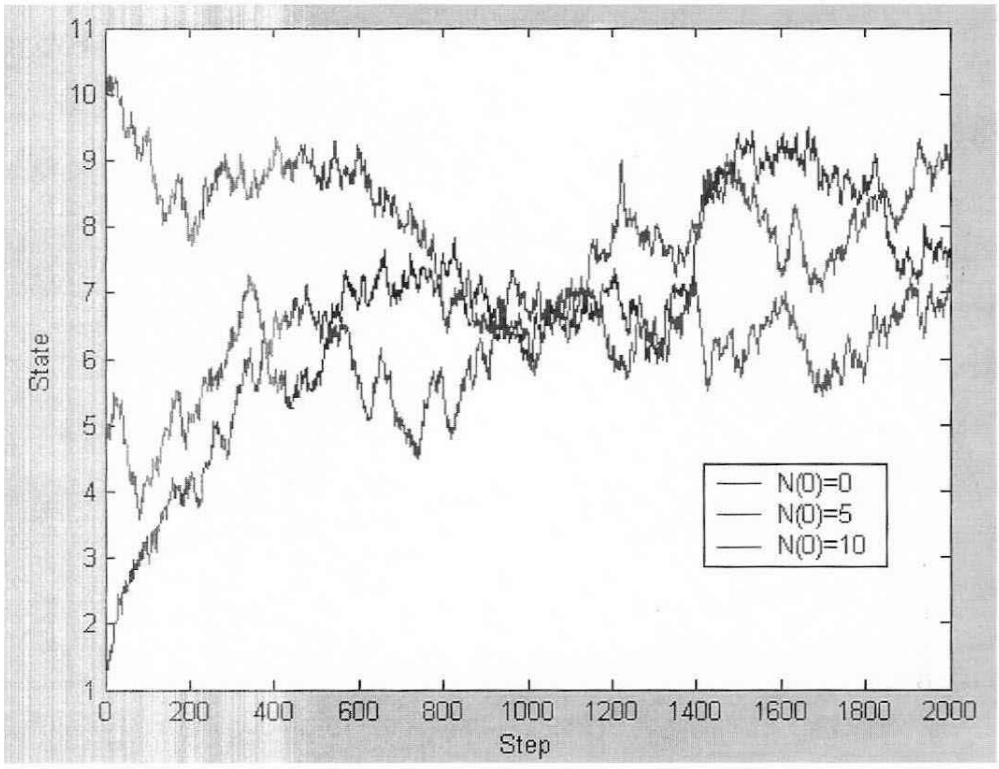
```
Nmax=50;
P=zeros(Nmax+1,3);
mu=1;
lambda=.9;
delta=.1;
a=delta*lambda;
b=delta*mu;
P(1,:)=[0,1-a,a];
r=[(1-a)*b,a*b+(1-a)*(1-b),(1-b)*a];
for n=2:Nmax;
   P(n,:)=r;
end
P(Nmax+1,:)=[(1-a)*b,1-(1-a)*b,0];
IC=zeros(Nmax+1,1);
IC(1,1)=1;
L=2000;
avg_seq=zeros(L,1);
avg_cor=zeros(L,1);
for j=1:25
   seq=queueState(Nmax,P,IC,L);
   cor_seq=autocorr(seq,L);
   for l=1:L
      avg_seq(l)=(avg_seq(l)*(j-1)+seq(l))/j;
      avg_cor(l)=(avg_cor(l)*(j-1)+cor_seq(l))/j;
   end
end

plot(avg_seq);

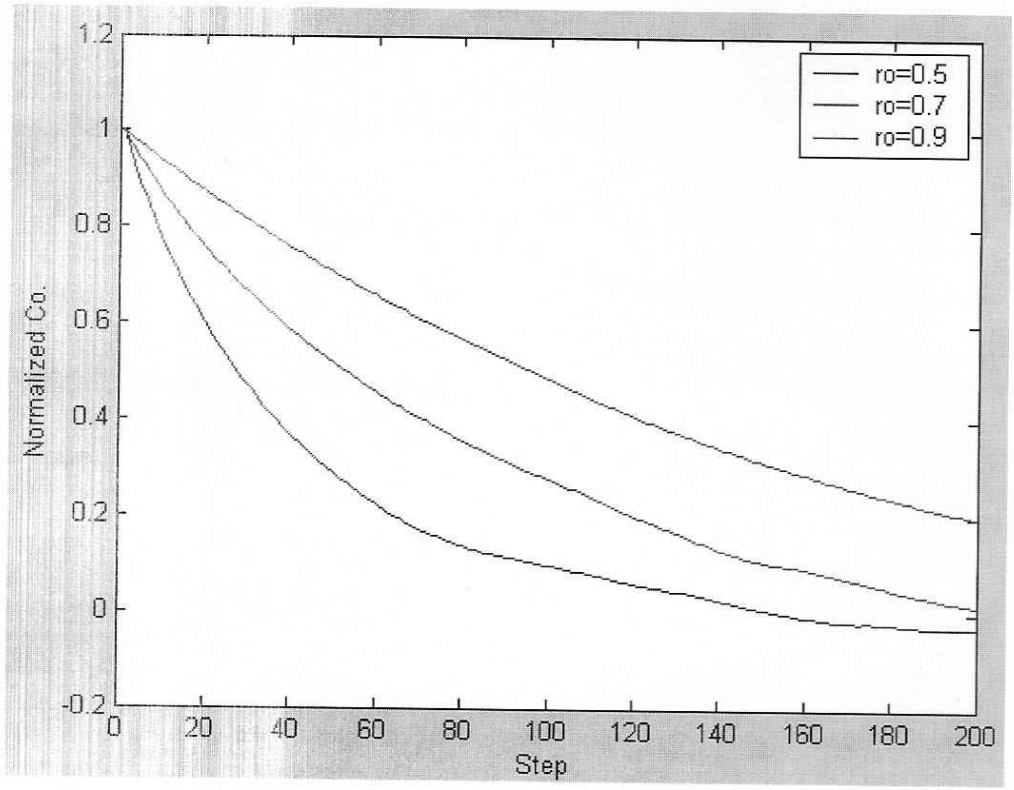


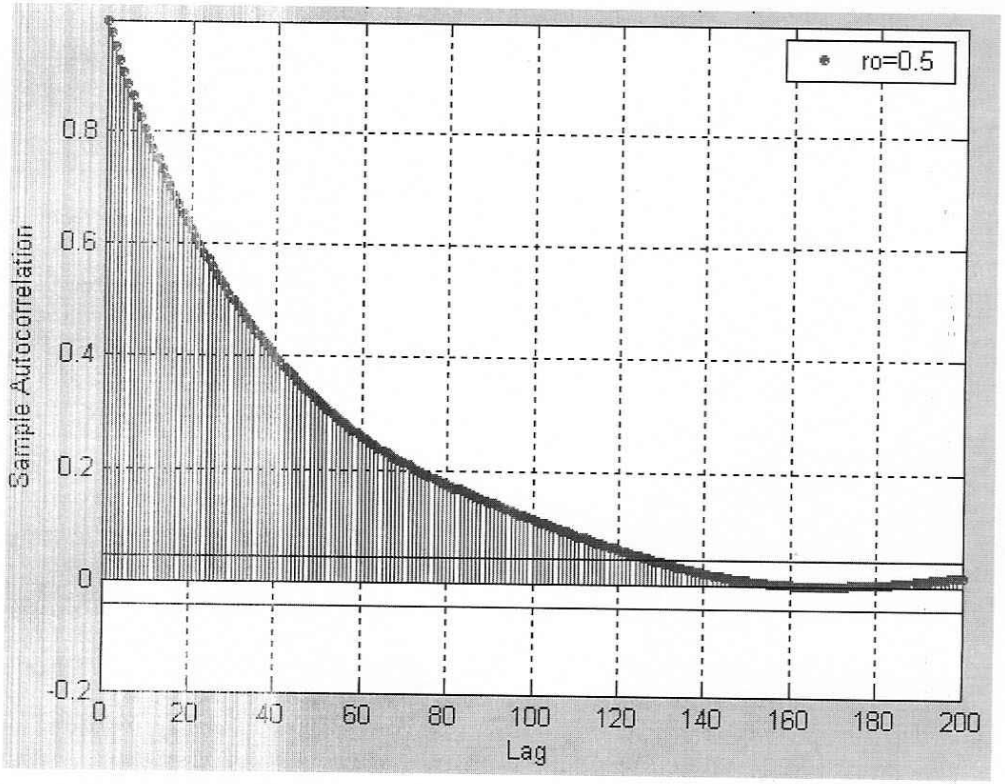
function stseq = queue_state(Nmax,P,IC,L)
stseq=zeros(1,L);
s=[1:Nmax+1];
step=[-1,0,1];
%Initst= floor(1000*rand);
Initst=ceil(10*rand);
stseq(1)=Initst;
for n=2:L+1
   k=rand;
   if(k<P(stseq(n-1),1))
      nxt=step(1);
   elseif (k<(P(stseq(n-1)+1,1)+P(stseq(n-1)+1,2)))
         nxt=step(2);
   else
      nxt=step(3);
   end
   nextst=stseq(n-1)+nxt;
   stseq(n)=nextst;
end
```

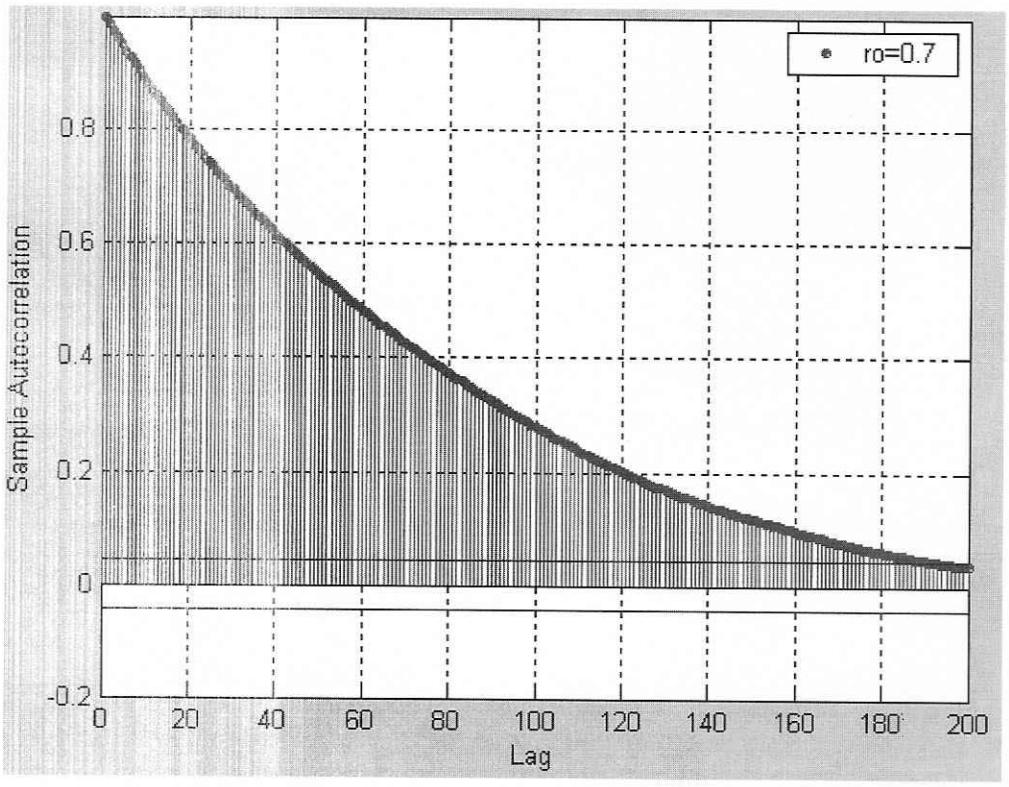

Avg number of costumers in queue
Step
ro=0.5
ro=0.7
ro=0.9

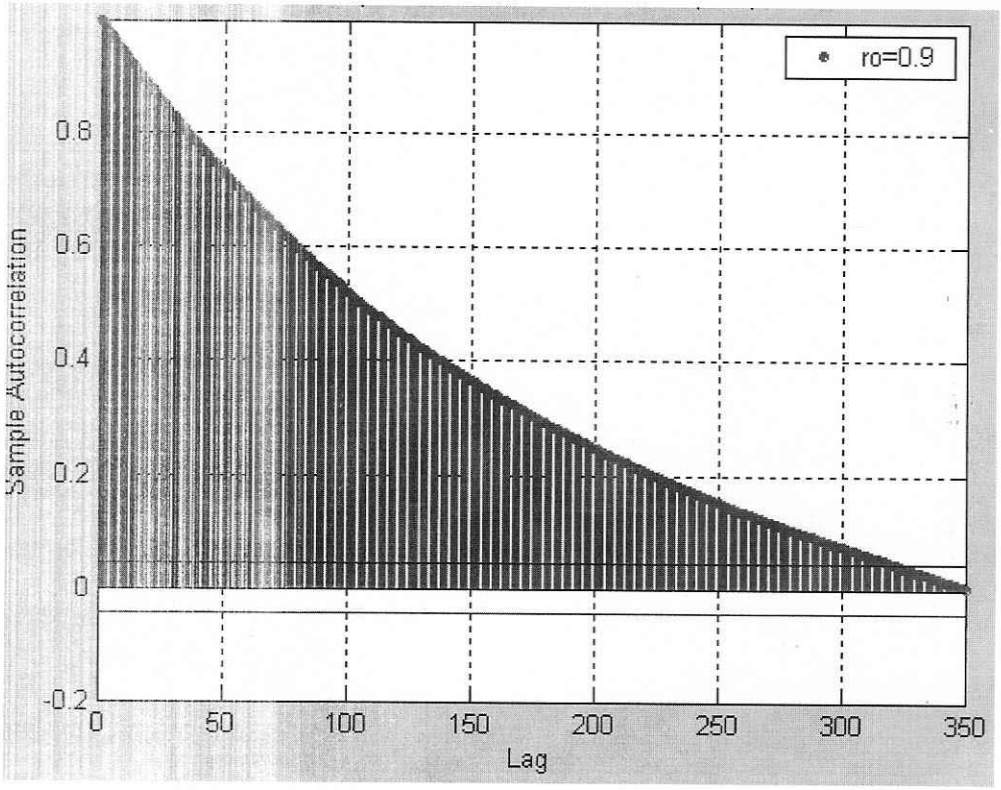

State
Step
N(0)=0
N(0)=5
N(0)=10

12.68


ro=0.5
ro=0.7
ro=0.9
Normalized Co.
Step
1.2
1
0.8
0.6
0.4
0.2
0
-0.2
0
20
40
60
80
100
120
140
160
180
200


ro=0.5
Sample Autocorrelation
Lag
0.8
0.6
0.4
0.2
0
-0.2
0
20
40
60
80
100
120
140
160
180
200


ro=0.7
Sample Autocorrelation
0.8
0.6
0.4
0.2
0
-0.2
0
20
40
60
80
100
120
140
160
180
200
Lag


ro=0.9
Sample Autocorrelation
0.8
0.6
0.4
0.2
0
-0.2
0
50
100
150
200
250
300
350
Lag

12.70

```
Nmax=50;
P=zeros(Nmax+1,3);
mu=1;
lambda=.5;
delta=.1;
a=delta*lambda;
b=delta*mu;
P(1,:)=[0,1-a,a];
r=[(1-a)*b,a*b+(1-a)*(1-b),(1-b)*a];
for n=2:Nmax;
   P(n,:)=r;
end
P(Nmax+1,:)=[(1-a)*b,1-(1-a)*b,0];
IC=zeros(Nmax+1,1);
IC(1,1)=1;
L=5000;
n=50;
avg_seq=zeros(L,1);
avg_cor=zeros(L,1);
avg=zeros(n,6);
conf=zeros(n,2);
avgseq=zeros(n,1);
prcnt=zeros(n,1);
z=1.68;  %90% confidence interval

for j=1:n
   seq=queueState(Nmax,P,IC,L);
   avgseq(j)=sum(seq(1:L))/L;
   conf(j,1)=avgseq(j)-z*sqrt(var(seq))/sqrt(n);
   conf(j,2)=avgseq(j)+z*sqrt(var(seq))/sqrt(n);
   for l=1:L
      avg_seq(l)=(avg_seq(l)*(j-1)+seq(l))/j;
   end
   prcnt(j)=accuracy(avg_seq,conf(j,1), conf(j,2),L);
   for i=1:6
      ii=200*i+1;
      avg(j,i)=sum(seq(ii:ii+800))/800;
   end
end

avg1=zeros(6,1);
for i=1:6
   ii=200*i+1;
   avg1(i)=sum(avg_seq(ii:ii+800))/800;
end
plot(avg_seq);



function acc = accuracy(seq,a,b,L)
cnt=0;
for i=1:L
   a1=seq(i)-a;
   a2=seq(i)-b;
   if a1>=0 && a2<=0
      cnt=cnt+1;
   end
end
acc=cnt/L;
```

Each column shows mean for one batch is 50 simulations (ro=0.5):

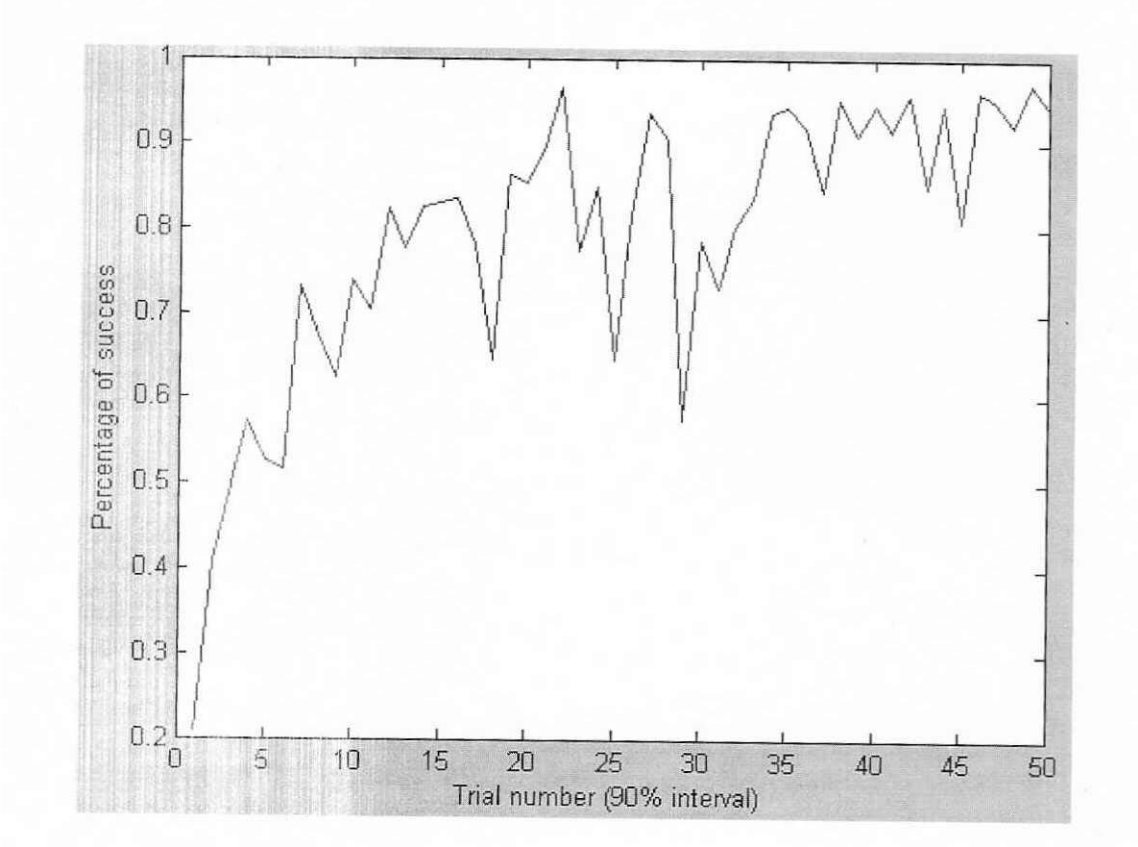
avg =

1.8275	1.7862	2.0550	2.1800	2.0513	1.8300
1.9150	1.8663	1.7900	1.2825	1.2788	2.1550
1.7038	1.6450	2.2388	2.3138	2.4337	2.2788
1.7938	1.8900	1.9163	1.6563	1.5562	1.7650
2.2212	2.3112	2.1363	1.5375	1.8975	2.0825
1.7763	1.8975	1.9175	1.8313	1.4763	1.4950
1.9288	1.9238	1.6275	1.5137	1.6863	1.7250
1.5900	1.7000	1.5337	1.5225	1.6000	1.5738
2.2425	2.3075	2.3438	1.5213	1.5362	1.5225
2.0013	1.9500	1.9788	1.5862	1.6450	1.7900
1.7637	1.5700	1.7200	1.9212	1.9225	1.8925
1.4688	1.6863	2.4663	2.3500	2.3863	2.2037
2.3537	2.3100	2.3737	2.3950	2.2212	2.0850
2.1050	2.2837	2.0313	2.2763	2.1250	1.9288
1.9412	1.8713	2.2588	2.1175	2.1913	1.8825
2.7375	2.8725	2.8363	3.2287	2.3912	3.0638
1.8162	1.5662	1.6438	1.6600	1.5738	2.1425
3.3963	3.3363	2.7062	1.7825	1.6888	1.4475
2.2500	2.1975	2.2763	1.6912	1.4800	1.4400
1.7388	1.9087	2.0275	1.9863	1.8713	2.2925
1.5900	1.6087	2.4137	2.3262	2.4550	2.3912
2.1500	1.7475	1.4238	1.4150	1.5313	1.6100
1.6612	1.6838	1.4775	1.4288	1.4663	1.4325
1.5488	1.6200	1.5800	1.6650	1.6950	1.8438
1.6587	1.6688	1.7175	1.7563	1.6187	1.5425
1.5425	1.5337	1.5900	1.9538	1.8375	2.0838
1.5350	1.7900	1.8587	2.8700	3.3163	3.2437
1.6175	1.5438	1.5413	1.6087	1.5388	2.0675
1.9925	2.0125	1.9813	2.0412	1.7962	1.7313
2.2825	1.8875	1.5700	1.4837	1.5788	1.5538
2.0187	1.8925	1.5950	1.3175	1.3987	1.3875
1.6538	1.6775	2.0675	1.8963	1.9688	1.9800
1.7588	1.8275	1.8150	1.6463	1.8663	1.7913
1.8125	1.7950	1.9737	1.8400	1.7825	2.0100
2.3287	2.4613	2.3712	1.7413	1.9125	1.6550
1.7038	1.5538	1.7950	1.8575	1.7962	1.9163
2.5675	2.4562	1.8725	1.9763	1.7237	1.7575
1.9038	1.7363	1.8075	1.7537	1.6638	1.4188
1.7163	1.4400	1.4000	1.4950	1.4775	1.6825
2.1450	2.2175	2.1338	2.2525	1.9913	2.0713
1.8538	1.9825	1.9475	1.8162	1.9525	1.8825
2.4000	1.8975	1.9512	1.7988	1.5950	1.5575
1.9188	2.1400	2.8988	2.4925	2.5750	2.2687
1.6663	1.8438	1.7962	1.7563	1.5037	1.4988
3.2550	1.8038	1.5300	2.0300	1.9675	2.4712
2.4362	2.6200	2.6675	1.8900	1.7875	2.2050
2.2050	1.9087	1.8550	1.6475	1.5263	1.6538
2.0888	2.3625	2.2363	2.1200	1.6575	1.5100
1.6400	1.8963	2.0950	2.0137	1.9150	1.5300
1.8188	1.8750	2.0425	2.4188	2.3963	2.4562

conf =

1.6179	2.1865
1.4681	2.0367
1.5845	2.1531
1.6285	2.1971
1.3779	1.9465
1.3155	1.8841
2.0663	2.6349
1.8899	2.4585
1.6341	2.2027
1.5071	2.0757
1.5911	2.1597
1.6489	2.2175
1.5311	2.0997
1.5765	2.1451
1.5279	2.0965
1.5345	2.1031
1.8395	2.4081
1.8585	2.4271
1.5859	2.1545
1.7831	2.3517
1.6665	2.2351
1.6427	2.2113
1.6305	2.1991
1.8207	2.3893
1.7083	2.2769
1.5471	2.1157
1.4711	2.0397
1.6845	2.2531
1.5713	2.1399
1.4545	2.0231
1.5913	2.1599
1.8241	2.3927
1.6721	2.2407
1.6101	2.1787
1.4299	1.9985
1.6529	2.2215
1.4947	2.0633
1.6019	2.1705
1.5661	2.1347
1.6317	2.2003
1.9669	2.5355
1.4301	1.9987
1.7863	2.3549
2.0861	2.6547
1.9275	2.4961
1.3875	1.9561
1.5473	2.1159
1.5055	2.0741
1.6113	2.1799
1.3331	1.9017

The percentage of times that the real value of the state is in the confidence interval is shown below:



12.71

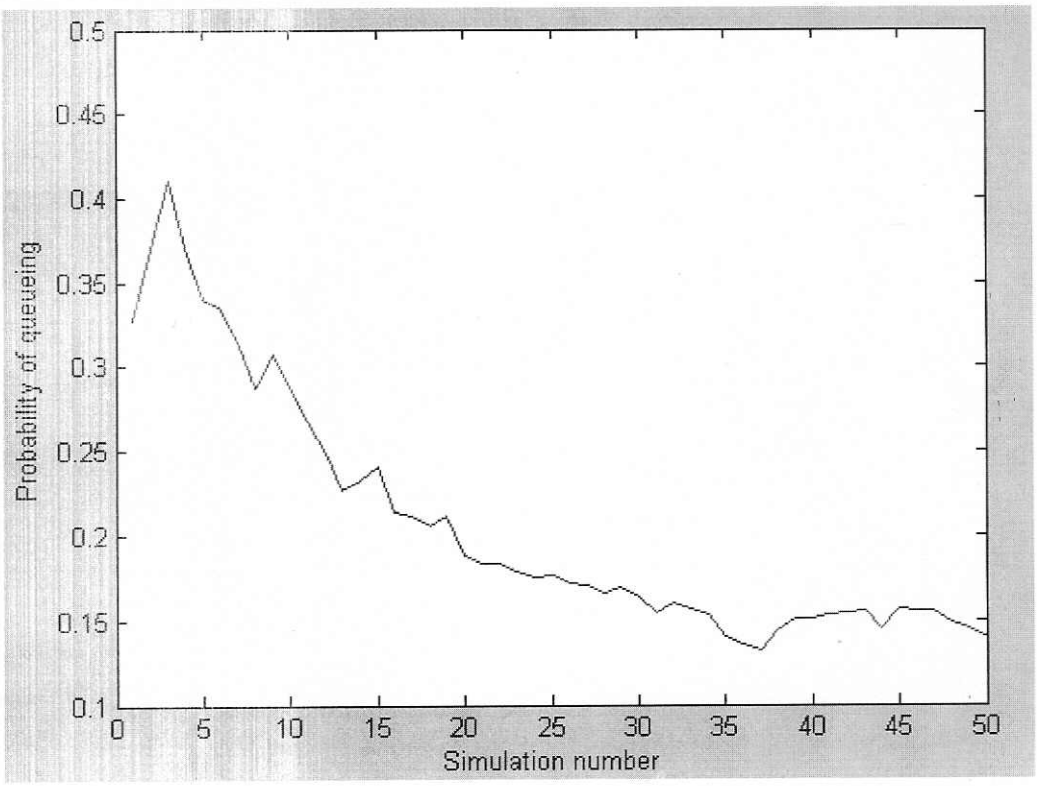
```
Nmax=50;
P=zeros(Nmax+1,3);
mu=1;
lambda=2;
delta=.1;
c=3;
a=delta*lambda;
b=delta*mu*c;
b2=delta*mu*2;
P(1,:)=[0,1-a,a];
P(2,:)=[(1-a)*b2,a*b2+(1-a)*(1-b2),(1-b2)*a];
r=[(1-a)*b,a*b+(1-a)*(1-b),(1-b)*a];
for n=3:Nmax;
   P(n,:)=r;
end
P(Nmax+1,:)=[(1-a)*b,1-(1-a)*b,0];
IC=zeros(Nmax+1,1);
IC(1,1)=1;
L=5000;
n=50;
avg_seq=zeros(L,1);
avg_cor=zeros(L,1);
avg=zeros(n,6);
conf=zeros(n,2);
avgseq=zeros(n,1);
prcnt=zeros(n,1);
prq=zeros(n,1);
z=1.68;  %90% confidence interval

for j=1:n
   seq=queueState(Nmax,P,IC,L);
   avgseq(j)=sum(seq(1:L))/L;
   conf(j,1)=avgseq(j)-z*sqrt(var(seq))/sqrt(n);
   conf(j,2)=avgseq(j)+z*sqrt(var(seq))/sqrt(n);
   for l=1:L
      avg_seq(l)=(avg_seq(l)*(j-1)+seq(l))/j;
   end
   prcnt(j)=accuracy(avg_seq,conf(j,1), conf(j,2),L);
   prq(j)=prQue(avg_seq,c,L);
   for i=1:6
      ii=200*i+1;
      avg(j,i)=sum(seq(ii:ii+800))/800;
   end
end
```

```
avg1=zeros(6,1);
for i=1:6
   ii=200*i+1;
   avg1(i)=sum(avg_seq(ii:ii+800))/800;
end
plot(seq);


function acc = prQue(seq,c,L)
cnt=0;
for i=1:L
   if seq(i)>c
      cnt=cnt+1;
   end
end
acc=cnt/L;
```

Probability of being queued before getting the service:



12.72

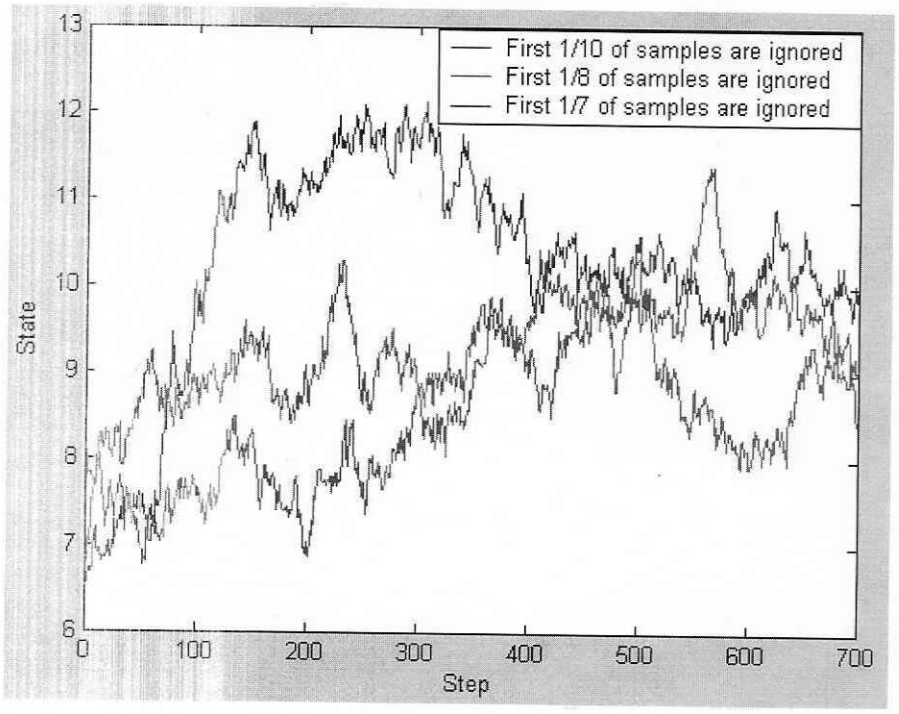
```
Nmax=50;
mu=1;
lambda=.9;
G=zeros(Nmax,2);
cnt=zeros(Nmax,1);
for i=1:Nmax
   G(i,:)=[mu,lambda];
end
G(1,1)=0;
%G(Nmax,2)=0;
IC=zeros(Nmax+1,1);
IC(1,1)=1;
T=100;
[stseq,OccTime,n]= contTm(Nmax,G,IC,T);
%pmf
l=length(stseq);
for i=1:l
   idx=stseq(i);
   cnt(idx)=cnt(idx)+1;
end

for i=1:Nmax
   pmf(i)=cnt(i)/l;
end

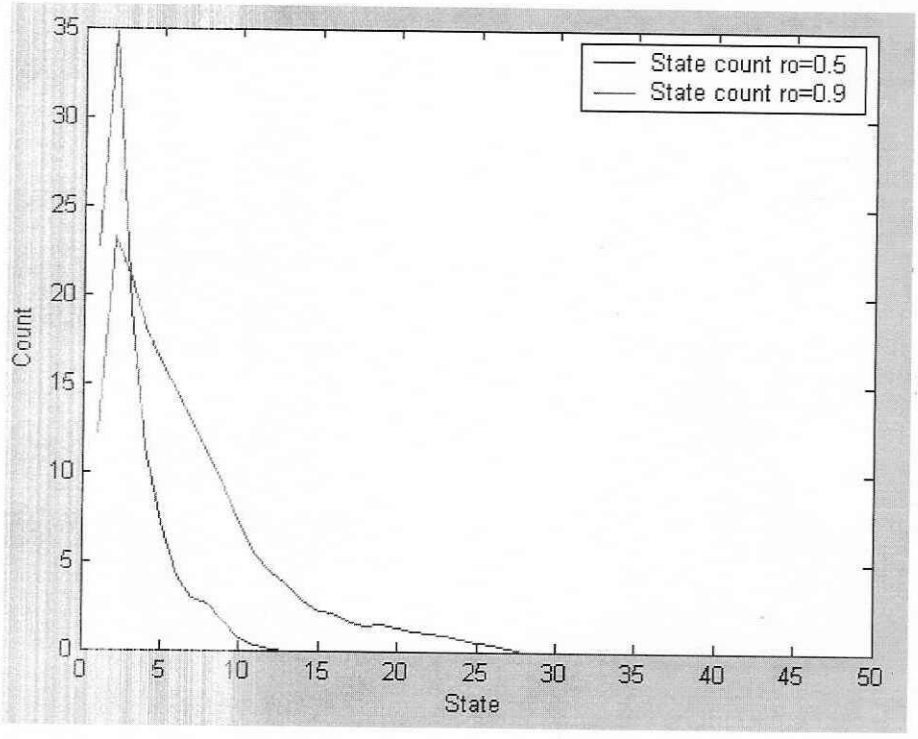

function [stseq,OccTime,n]= contTime(Nmax,G,IC,T)

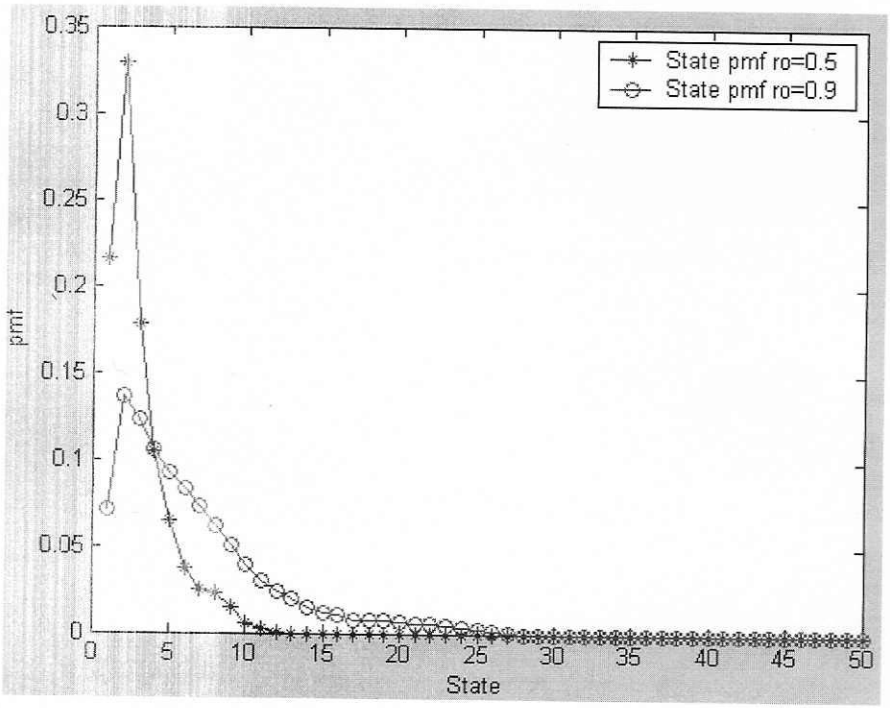
Taggr=-1;
L=round(T*(G(Nmax-1,1)+G(Nmax-1,2)));
%stseq=zeros(1,L);
%OccTime=zeros(1,L+1);
%Q=zeros(Nmax,2);
Q=G/(G(Nmax-1,1)+G(Nmax-1,2));
s=[1:Nmax+1];
step=[-1,1];
Initst= ceil(10*rand);
stseq(1)=Initst;
%Initst=1;
n=1;
OccTime(n)=randraw('exp',G(stseq(n),1)+G(stseq(n),2));
Taggr=OccTime(n);
while(Taggr<T)
   n=n+1;
   Q(stseq(n-1),:)=[G(stseq(n-1),1),G(stseq(n-1),2)]/(G(stseq(n-1),1)+G(stseq(n-1),2));
   nxt=dscRnd(1,Q(stseq(n-1),:),step);
   nextst=stseq(n-1)+nxt;
   stseq(n)=nextst;
   OccTime(n)=randraw('exp',G(stseq(n),1)+G(stseq(n),2));
   Taggr=Taggr+OccTime(n);
end
```

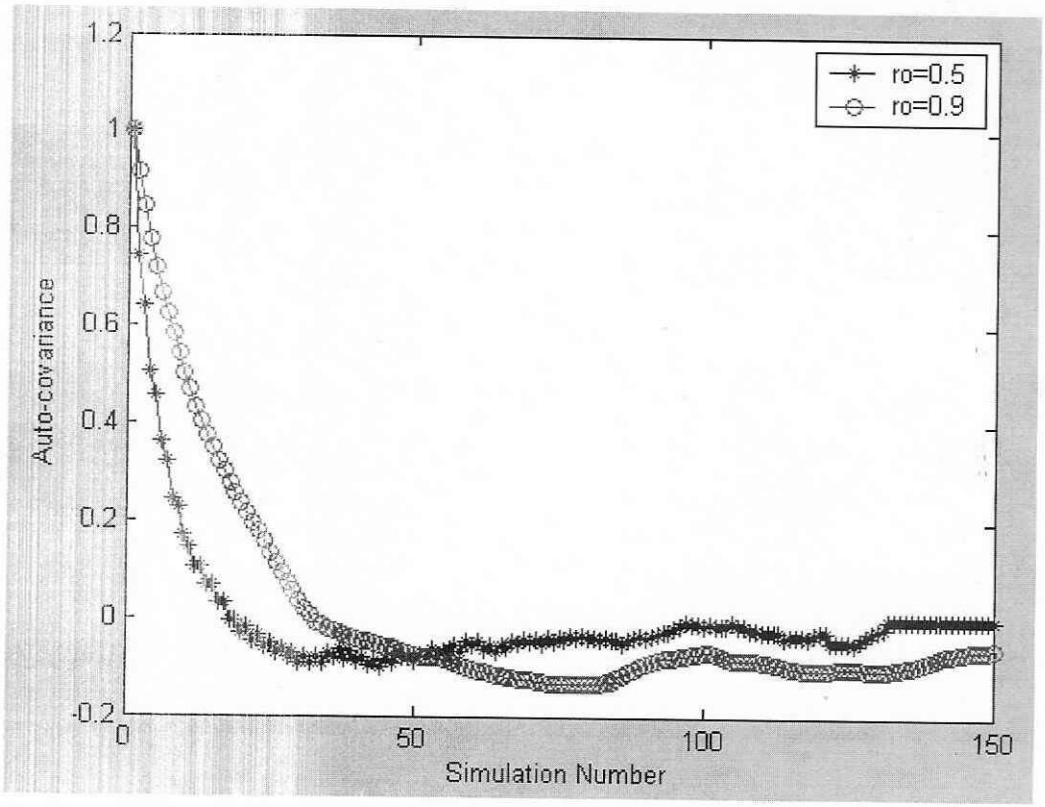
This diagram compares three different warm-up periods:



The diagram of stay in each one of the states and the pmf's:




State pmf ro=0.5
State pmf ro=0.9
pmf
State
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
0
5
10
15
20
25
30
35
40
45
50


ro=0.5
ro=0.9
Auto-covariance
Simulation Number
1.2
1
0.8
0.6
0.4
0.2
0
-0.2
0
50
100
150

12.73

```
% Prepare Transition Probability Matrix
ro=.7;
s=1000;
cnt=zeros(s+1,1);
P=zeros(s,s);
for j=1:s
   P(1,j)=exp(-ro)*ro^j/factorial(j);
   P(2,j)=exp(-ro)*ro^j/factorial(j);
end

for i=3:s
   for j=i-1:s
      P(i,j)=exp(-ro)*ro^(j-i+2)/factorial(j-i+2);
   end
end

L=s;
stseq=zeros(1,L+1);
step=1:s;
Initst= ceil(10*rand);
%Initst=1;
stseq(1)=Initst;
for n=2:L+1
   stseq(n)=dscRnd(1, P(stseq(n)+1,:), step);;
end

l=length(stseq);
for i=1:l
   idx=stseq(i);
   cnt(idx+1)=cnt(idx+1)+1;
end

for i=1:s+1
   pmf(i)=cnt(i)/s;
end

plot(stseq);

function [sample] = dscRnd(np, pdf, v)

 %if (sum(pdf) ~= 1)
 %  error('Probabilities does not sum up to 1');
 %end

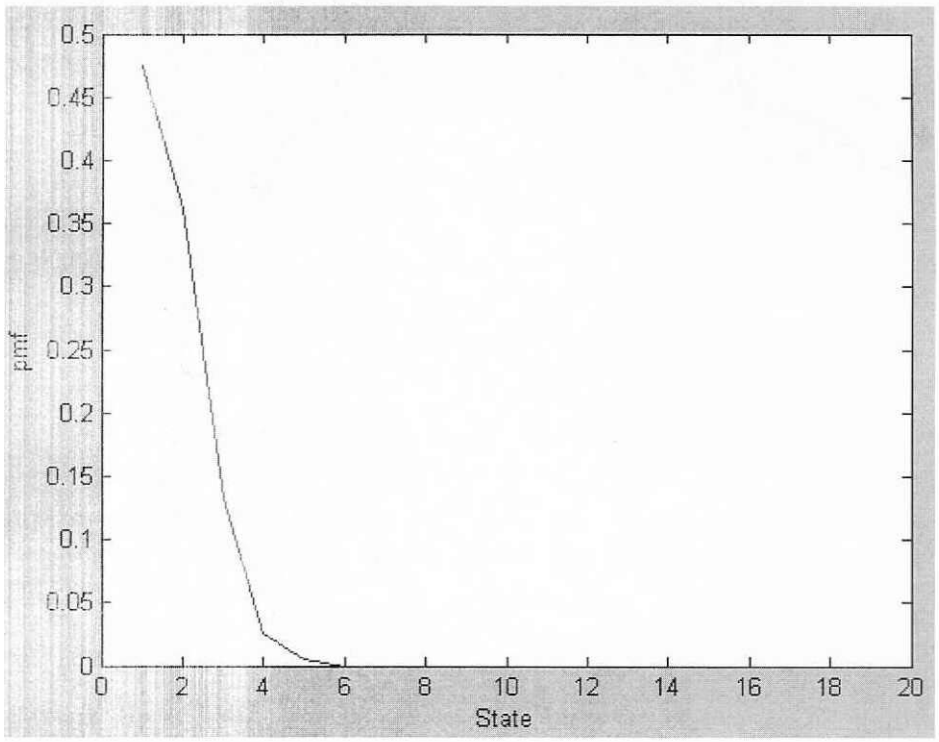
 n = length(pdf);

 if (nargin==2)
   v = [1:n];
 end

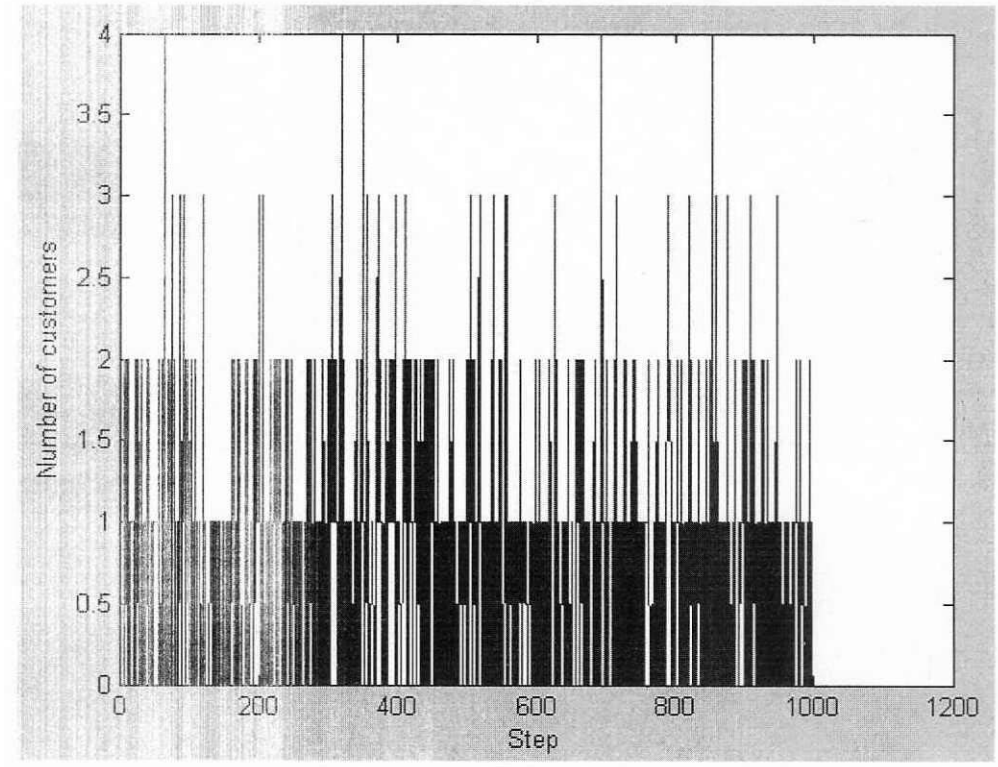
 cumprob = [0 cumsum(pdf)];

 runi = rand(1, np); % random uniform sample

 sample = zeros(1, np);
 for j=1:n
   ind = find((runi>cumprob(j)) & (runi<=cumprob(j+1)));
   sample(ind) = v(j);
 end
```



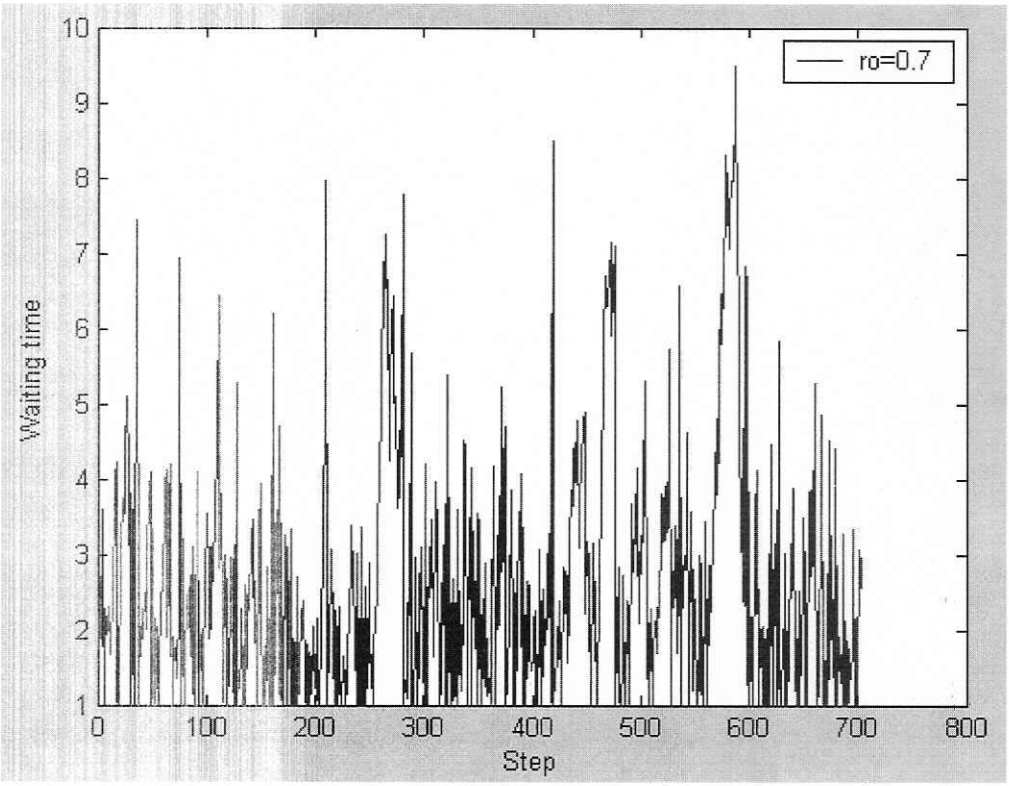
Number of customers in each step:

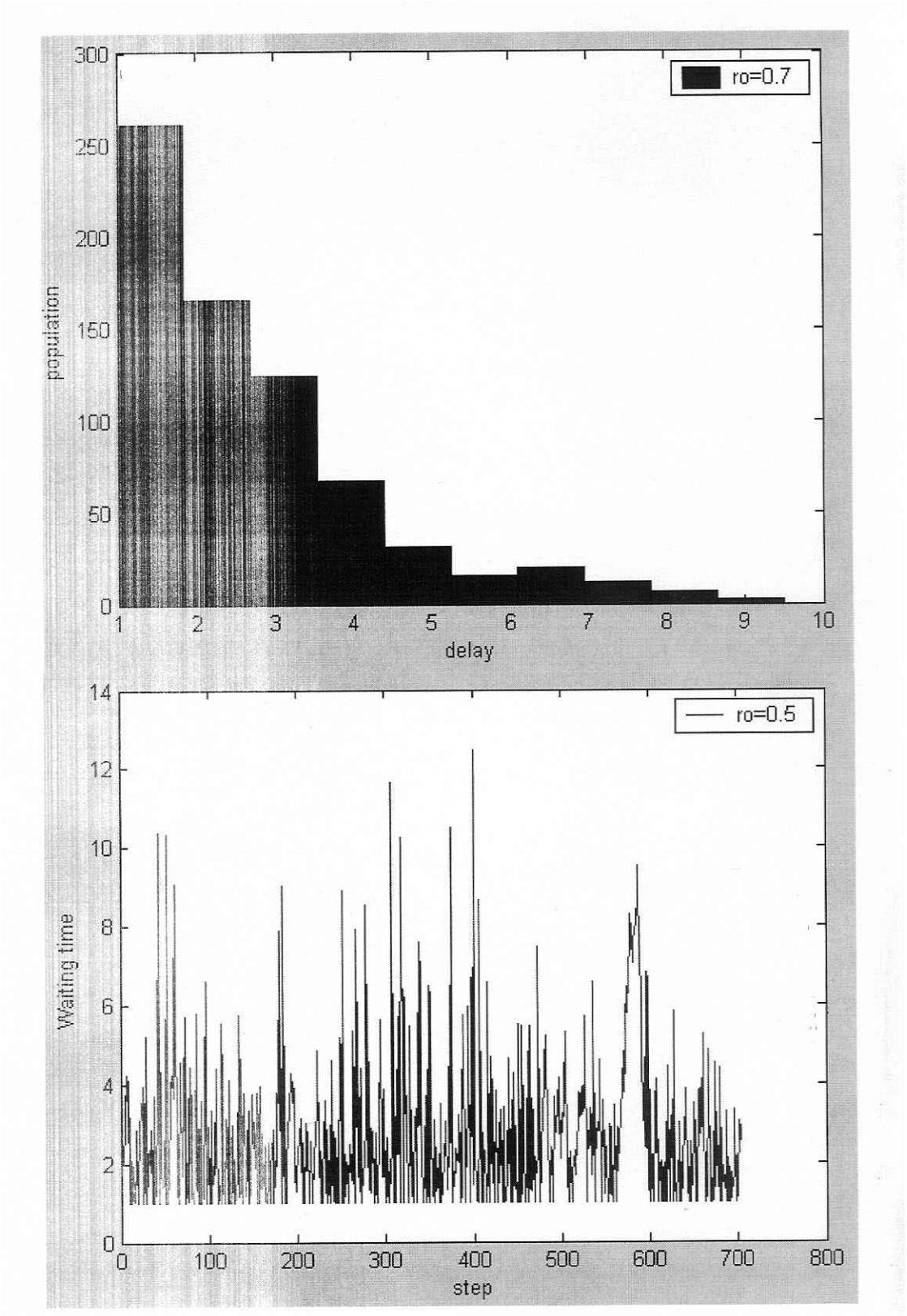


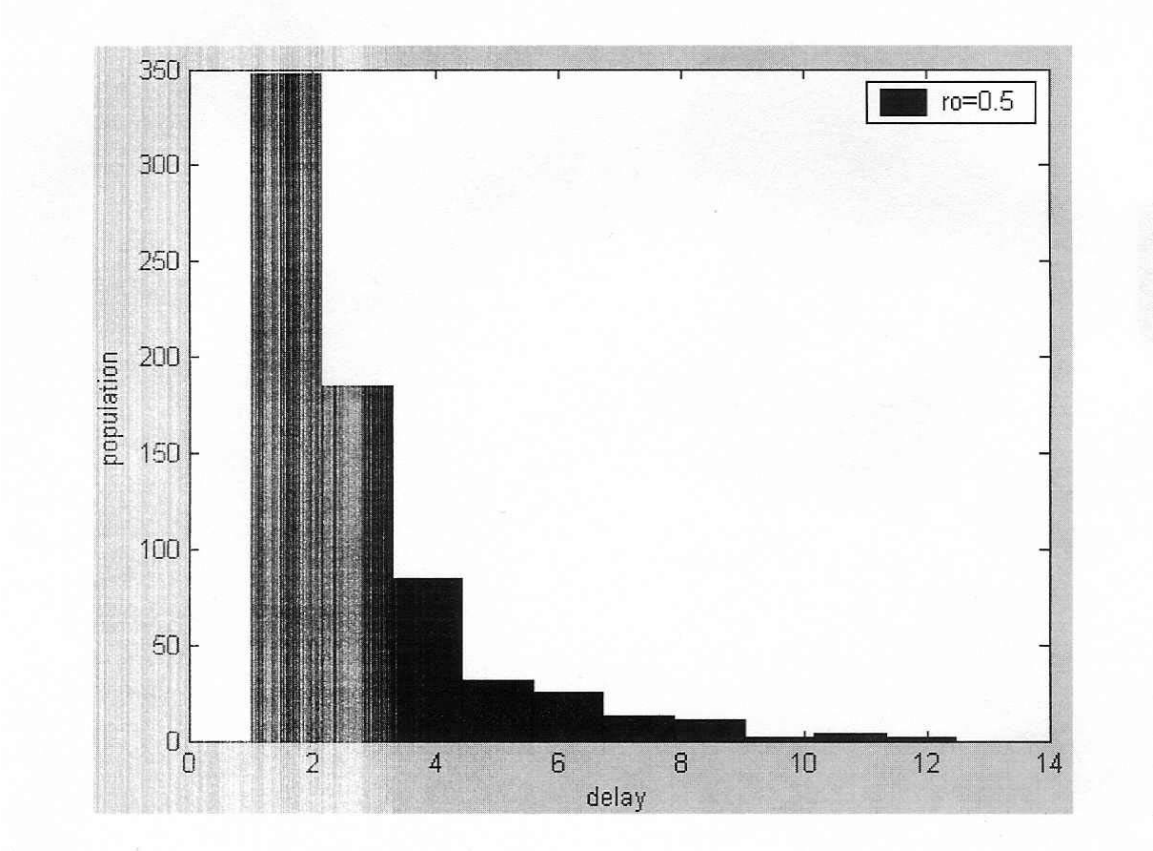
12.74

```
T=1000;
lambda=0.5;
arrtime=-log(rand)/lambda;  % Poisson arrivals
i=1;
while (min(arrtime(i,:))<=T)
   arrtime = [arrtime; arrtime(i, :)-log(rand)/lambda];
   i=i+1;
end
n=length(arrtime);          % arrival times t_1,...t_n
w=ones(1,n+1);
for j=2:n-1
   w(j+1)=max(0, w(j)+1-(arrtime(j+1)-arrtime(j)));
   TT(j)=w(j)+arrtime(j)-arrtime(j-1);
end

plot(TT);
```



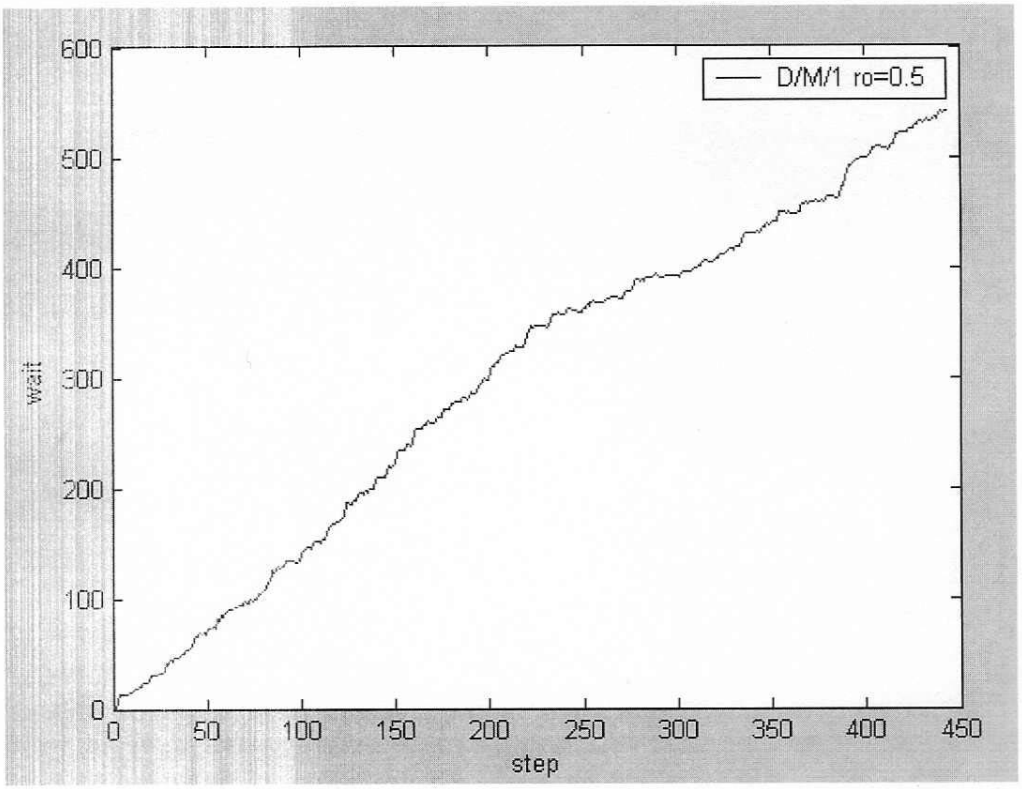

ro=0.7
population
delay
ro=0.5
Waiting time
step

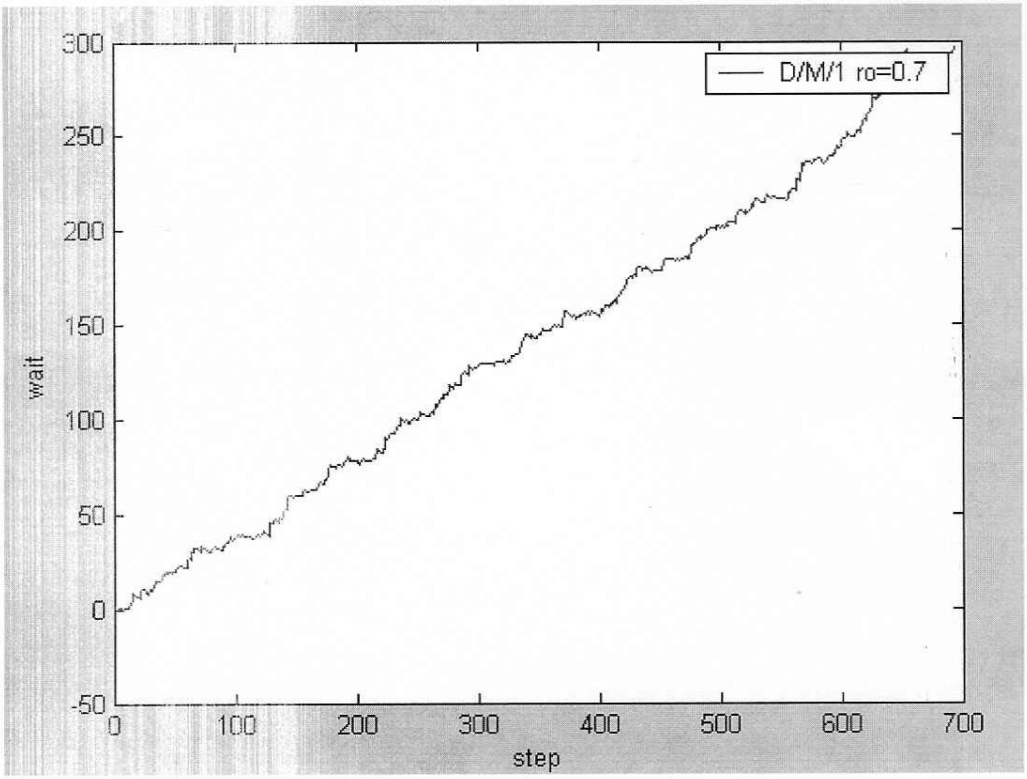


12.75

```
T=1000;
lambda=0.5;
srvtime=-log(rand)/lambda;  % Poisson arrivals
i=1;
while (min(srvtime(i,:))<=T)
   srvtime = [srvtime; srvtime(i, :)-log(rand)/lambda];
   i=i+1;
end
n=length(srvtime);          % arrival times t_1,...t_n
w=ones(1,n+1);
for j=2:n-1
   w(j+1)=max(0, w(j)+(srvtime(j+1)-srvtime(j))-1);
   TT(j)=w(j)-1;
end

plot(TT(1:n-1));
```

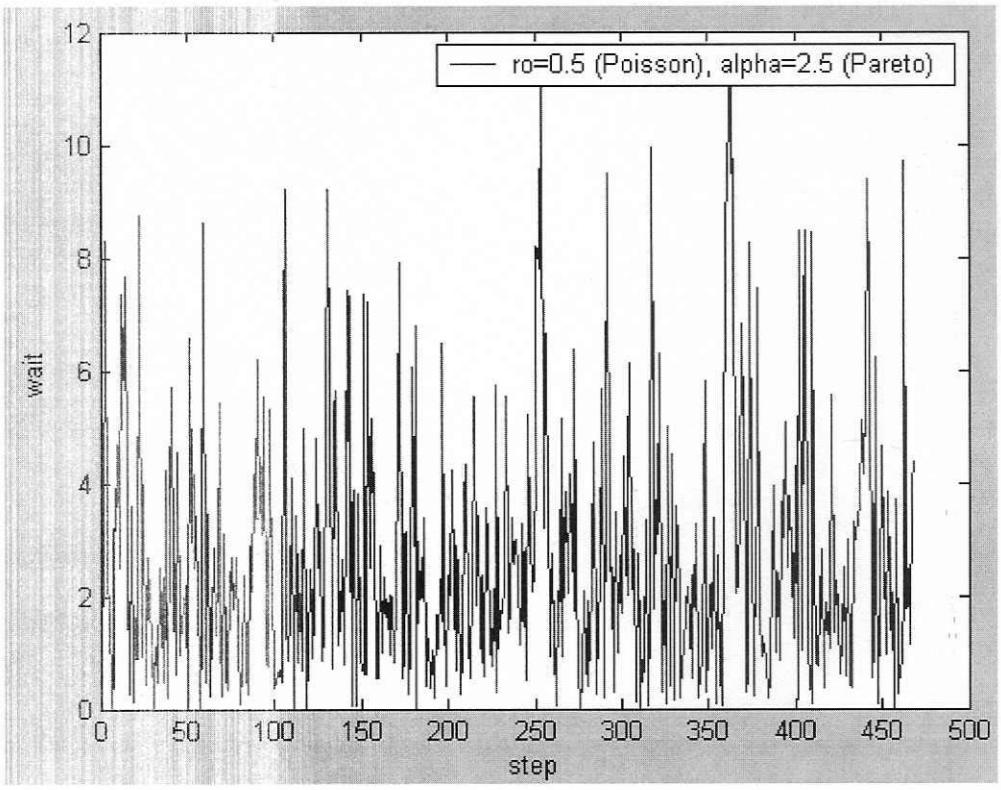

D/M/1 ro=0.5
wait
step
0
50
100
150
200
250
300
350
400
450
100
200
300
400
500
600

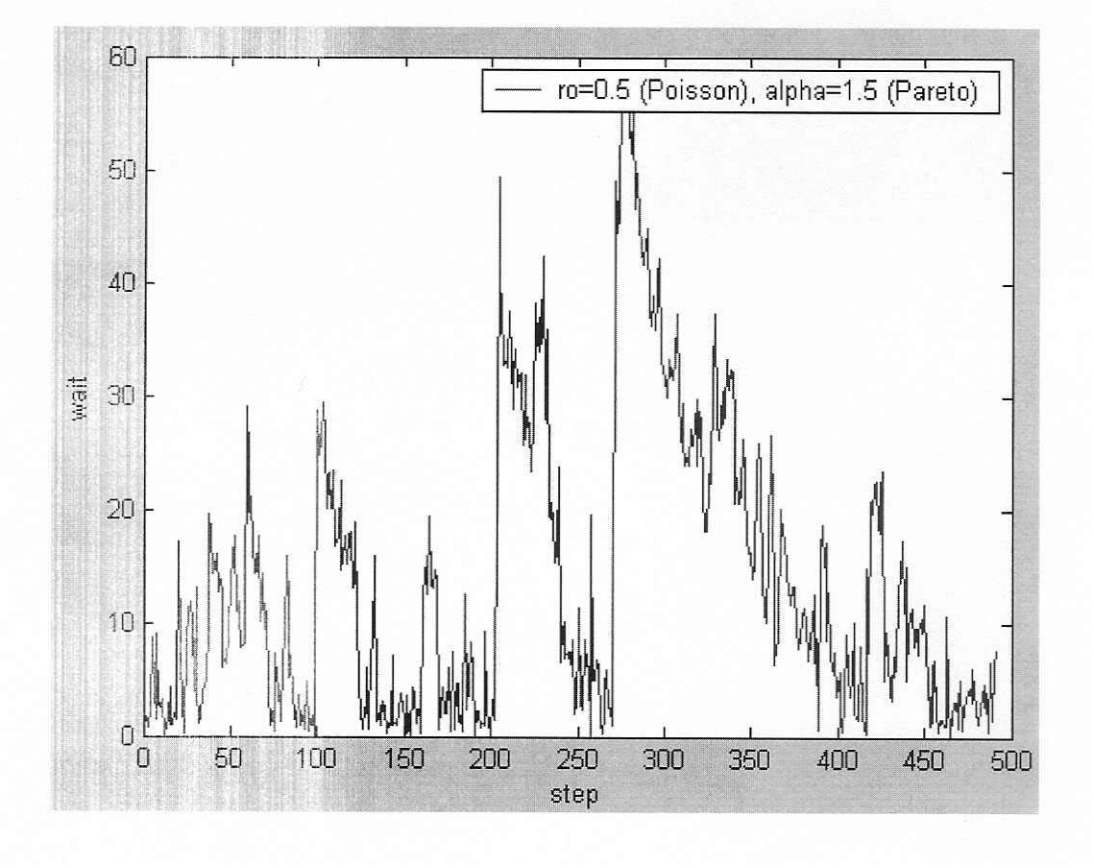

D/M/1 ro=0.7
wait
step
0
100
200
300
400
500
600
700
-50
50
100
150
200
250
300

12.76

```
T=1000;
lambda=0.5;
alpha=1.5;
arrtime=-log(rand)/lambda;  % Poisson arrivals
srvtime=(1-rand).^(-1/alpha)-1; % Pareto
i=1;
while (min(arrtime(i,:))<=T)
   arrtime = [arrtime; arrtime(i, :)-log(rand)/lambda];
   srvtime = [srvtime; srvtime(i, :)+(1-rand).^(-1/alpha)-1];
   i=i+1;
end
n=length(arrtime);          % arrival times t_1,...t_n
w=ones(1,n+1);
for j=2:n-1
   w(j+1)=max(0, w(j)+(srvtime(j)-srvtime(j-1))-(arrtime(j+1)-arrtime(j)));
   TT(j)=w(j)+(arrtime(j+1)-arrtime(j));
end

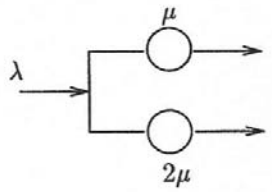
plot(TT(1:n-1));
```




ro=0.5 (Poisson), alpha=1.5 (Pareto)
wait
step
0
10
20
30
40
50
60
0
50
100
150
200
250
300
350
400
450
500

Problems Requiring Cumulative Knowledge

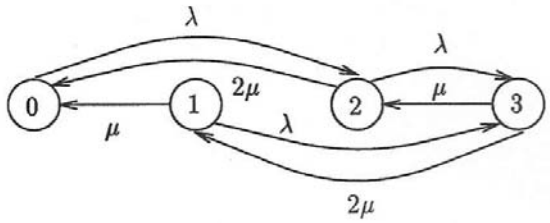
12.80



a) There are four possible cases: both servers are idle, the slower server is idle, the faster server is idle or both servers are busy. If we define each of these cases as a state, then $X(t)$ will be a four-state continuous-time Markov chain.

b) In this case, the transition rate diagram will be as follows:

State 1: Slower server is busy & faster server is idle.
State 2: Faster server is busy & slower server is idle.



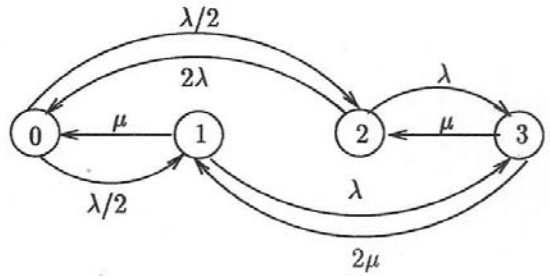
Writing global balance equations, we will have:

$$\begin{aligned} P_0\lambda &= \mu P_1 + 2\mu P_2 \\ 2\mu P_3 &= (\mu+\lambda)P_1 \\ (\lambda+2\mu)P_2 &= \mu P_3 + \lambda P_0 \\ 2\mu P_3 &= \lambda P_2 + \lambda P_1 \end{aligned}$$

Incorporating the fact that $\sum_i P_i = 1$ and having $\rho = \frac{\lambda}{\mu}$, the following is obtained:

$$\begin{aligned} P_0 &= \frac{2\rho+3}{\rho^3+2\rho^2+3.5\rho+3} \\ P_1 &= \frac{\rho^2}{\rho^3+2\rho^2+3.5\rho+3} \\ P_2 &= \frac{0.5\rho^2+1.5\rho}{\rho^3+2\rho^2+3.5\rho+3} \\ P_3 &= \frac{\rho^3+0.5\rho^2}{\rho^3+2\rho^2+3.5\rho+3} \end{aligned}$$

c) The Markov chain is as follows:



$$\begin{aligned}
\rho P_0 &= \mu P_1 + 2\mu P_2 \\
2\mu P_3 + \frac{\lambda}{2} P_0 &= (\lambda + \mu) P_1 \\
(\lambda + 2\mu) P_2 &= \mu P_3 + \frac{\lambda}{2} P_0 \\
3\mu P_3 &= \lambda P_2 + \lambda P_1
\end{aligned}$$

$$\Rightarrow \quad P_0 = \frac{4}{\rho^2 + 3\rho + 4}$$

$$P_1 = \frac{2\rho}{\rho^2 + 3\rho + 4}$$

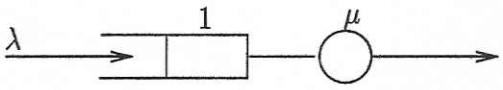
$$P_2 = \frac{\rho}{\rho^2 + 3\rho + 4}$$

$$P_3 = \frac{\rho^2}{\rho^2 + 3\rho + 4}$$

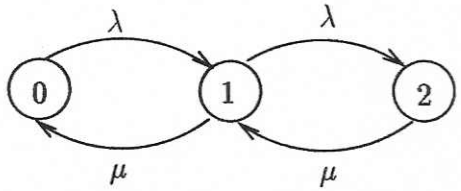
It is expected that the average waiting time in the case of part (b) will be less than that of part (c).

12.81

~~9.59~~ M/M/1/2



This is a three-state continuous-time Markov chain:



The associated Chapman-Kolmogorov: $P_i' = \sum_j \Pi_{ji} P_j$

$$\Rightarrow \quad \begin{aligned} P_0' &= -\lambda P_0 + \mu P_1 \\ P_1' &= \lambda P_0 + \mu P_2 - (\mu+\lambda)P_1 \\ P_2' &= \lambda P_1 - \mu P_2 \end{aligned}$$

This set of differential equations can be solved much easier using Laplace transform:

$$\begin{cases} sP_0(s) - P_0(0) = -\lambda P_0(s) + \mu P_1(s) \\ sP_1(s) - P_0(0) = \lambda P_0(s) + \mu P_2(s) - (\mu+\lambda)P_1(s) \\ sP_2(s) - P_2(0) = \lambda P_1(s) - \mu P_2(s) \end{cases}$$

$P_0(s)$, $P_1(s)$, and $P_2(s)$ are obtained by solving this set of linear equations from which the transient pmfs, $P_i(t) = P[N(t) = i]$, are obtained.

a) Suppose that the system is empty at $t = 0 \Rightarrow P_0(0) = 1, P_1(0) = P_2(0) = 0$

$$\Rightarrow \quad P_1(s) = \frac{\lambda(s+\mu)}{s(s^2+2(\mu+\lambda)s+\mu\lambda+\lambda^2+\mu^2)} = \frac{\lambda(s+\mu)}{F(s)}$$

$$P_0(0) = \frac{\mu\lambda(s+\mu)}{s(s+\mu)(s^2+2(\mu+\lambda)s+\mu\lambda+\lambda^2+\mu^2)} + \frac{1}{s+\lambda}$$

$$P_2(s) = \frac{\lambda^2}{s(s^2+2(\mu+\lambda)s+\mu\lambda+\lambda^2+\mu^2)}$$

$\Rightarrow$ By partial fraction expansion, we will have:

$$P_i(s) = \frac{A_i}{s} + \frac{B_i}{s+\lambda} + \frac{C_i}{s+a_1} + \frac{D_i}{s+a_2}$$

where $a_1, a_2 = -(\mu+\lambda) \pm \sqrt{\mu\lambda}$.
Applying inverse Laplace transform, we obtain:

$$P_i(t) = A_i + B_i e^{-\lambda t} + C_i e^{-a_1 t} + D_i e^{-a_2 t}$$

Finding the factors $A_i - D_i$ is very easy because $P_i(s)$ have only simple 1st order roots.

b) If the system is full at $t = 0 \Rightarrow P_2(0) = 1, P_0(0) = P_1(0) = 0$

$$\Rightarrow \quad P_1(s) = \frac{\mu(s+\lambda)}{F(s)}$$

$$P_0(s) = \frac{\mu^2}{F(s)}$$

$$P_2(s) = \frac{\mu\lambda(s+\lambda)}{(s+\mu)F(s)} + \frac{1}{s+\mu}$$

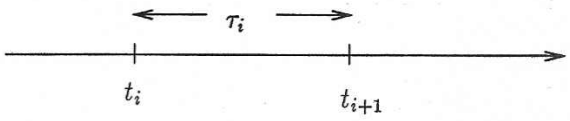
The expansion of these fractions will be as follows:

$$P_i(s) = \frac{A_i}{s} + \frac{B_i}{s+\mu} + \frac{C_i}{s+a_1} + \frac{P_i}{s+a_2}$$

$$\Rightarrow \quad P_i(t) = A_i + B_i e^{-\mu t} + C_i e^{-a_i t} + D_i e^{-a_2 t}$$

12.82

9.60 a) Let's divide the time axis into time intervals or cycles τ_i which are the times when customers arrive to an empty system.



We have $t_j = \tau_0 + \tau_1 + \tau_2 + ... + \tau_i$.

If the τ_i's are identically distributed independent random variables, then they form a renewal process. In an M/G/1, each time a customer arrives to an empty system restarts a process which is independent of all past events. In addition, all cycles have the same statistics, i.e. they constitute a repeated trial.

b) Let $N(t)$ be the number of customers in the system.

$$P[N(t) = j] = \lim_{t\to\infty} \frac{\text{Total time that there are 'j' customers in the system}}{t}$$

To find this, let's associate a cost, C_i, to each renewal interval τ_i. C_i is equal to amount of time that there are 'j' customers in the system in time τ_i. Also, $C(t) = \sum_i C_i$.

$$P[N(t) = j] = \lim_{t\to\infty} \frac{C(t)}{t}$$

According to the renewal theorem

$$\lim_{t\to\infty} \frac{C(t)}{t} = \frac{E(C)}{E[\tau]}$$

Therefore,

$$P[N(t) = j] = \frac{E(C)}{E(\tau)} \approx \frac{\frac{1}{n}\sum_{j=1}^{n} C_j}{\frac{1}{n}\sum_{j=1}^{n} \tau_j}$$

c) We can find confidence intervals for the numerator and denominator of the above expression using the methods from section 8.4.

12.83

9.61 a)
$$\begin{aligned} P[N_1(t) = i, N_2(t) = j] &= P[N_1(t) = i, N(t) = i + j] \\ &= P[N_1(t) = i/N(t) = i + j]P[N(t) = i + j] \end{aligned}$$

If we know that $N(t) = i + j$, then each of the $i + j$ arrivals picks its arrival time according to a uniform distribution in the interval $(0, t)$ independently of the other arrivals

Therefore the probability that an arrival for $N(t)$ is also an arrival for $N_1(t)$ is

$$\int_0^t Pr[\text{arrival for } N_1(t)/\text{arrival time is } t]\frac{1}{t}dt = \frac{1}{t}\int_0^t p(t)dt \triangleq p$$

Since the $i + j$ arrivals are independent, we have

$$P[N_1(t) = i|N(t) = i + j] = \binom{i+j}{i} p^i(1-p)^j$$

Finally, we have

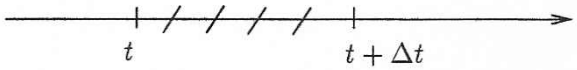
$$\begin{aligned} P[N_1(t)=i, N_2(t)=j] &= \binom{i+j}{i} p^i(1-p)^j \frac{(\lambda t)^{i+j}}{(i+j)!} e^{-\lambda t} \\ &= \frac{p^i(1-p)^j}{i!j!}(\lambda t)^{i+j} e^{-\lambda t} \\ &= \frac{(\lambda p)^i}{i!}\frac{(\lambda(1-p))^j}{j!} e^{-\lambda t} e^{-\lambda(1-p)t} \\ &= P[N_1(t)=i]P[N_2(t)=j] \,. \end{aligned}$$

b) In order for $N_1(t)$ and $N_2(t)$ to be independent random processes we require that

$$P[N_1(t_1)=i, N_2(t_2)=j] = P[N_1(t_1)=i]P[N_2(t_2)=j]$$

for any i, j and any choices of times t_1 and t_2.

Consider a very small time interval at time t of duration Δt:



Consider the evolution of the joint process $(N_1(t), N_2(t))$

$$\begin{aligned} P[N_1(t+\Delta t)=i, N_2(t+\Delta t)=j] &= P[N_1(t)=i-1, \text{ a type 1 arrival in } \Delta t] \\ &\quad + P[N_2(t)=j-1, \text{ a type 2 arrival in } \Delta t] \\ &\quad + P[N_1(t)=i, N_2(t)=j, \text{ no arrival in } \Delta t] \\ &\quad + O(\Delta t) \\ &= P[N_1(t)=i-1]\lambda p(t)\Delta t \\ &\quad + P[N_2(t)=j-1]\lambda(1-p(t))\Delta t \\ &\quad + P[N_1(t)=i, N_2(t)=j](1-\lambda\Delta t) \\ &\quad + O(\Delta t) \end{aligned}$$

Now consider the individual evolution of the two processes:

$$\begin{aligned} P[N_1(t+\Delta t)=i] &= P[N_1(t)=i-1]\lambda p(t)\Delta t \\ &\quad + P[N_1(t)=i](1-\lambda p(t)\Delta t) + O(\Delta t) \\ P[N_2(t+\Delta t)=j] &= P[N_2(t)=j-1]\lambda(1-p(t))\Delta t \\ &\quad + P[N_2(t)=j](1-\lambda(1-p(t))\Delta t) + O(\Delta t) \end{aligned}$$

If we multiply these two equations we obtain:

$$\begin{aligned} P[N_1(t+\Delta t)=i]P[N_2(t+\Delta t)=j] &= P[N_1(t)=i-1]P[N_2(t)=j]\lambda p(t)\Delta t \\ &\quad + P[N_1(t)=i]P[N_2(t)=j-1]\lambda(1-p(t))\Delta t \\ &\quad + P[N_1(t)=i]P[N_2(t)=j](1-\lambda\Delta t) + O(\Delta t) \end{aligned}$$

By comparing this equation to that of the joint process, we see that we will end up with independent random processes.

12.84

9.62 a) Suppose a customer arrived at time $t_1 < t$, then the customer has completed its service time by time t with probability

$$P[X < t - t_1] = F_X(t - t_1) \triangleq p(t_1)$$

Therefore customers that arrived in the interval $(0, t)$ have probability of completing service:

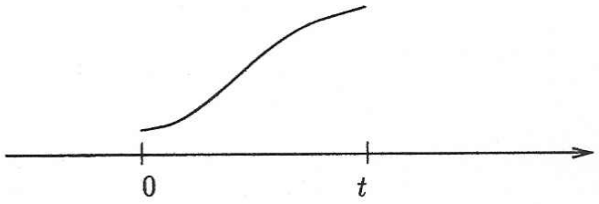
$$p = \frac{1}{t}\int_0^t F_X(t - t_1)dt_1$$

Thus

$$P[N_1(t) = i, N_2(t) = j] = \frac{(\lambda tp)^i}{i!}e^{-\lambda tp}\frac{(\lambda t(1-p))^j}{j!}e^{-\lambda t(1-p)}$$

Note that

$$\begin{aligned}\lambda tp &= \lambda\int_0^t F_X(t - t_1)dt_1 \\ \lambda t(1-p) &= \lambda t\left[\frac{1}{t}\int_0^t (1 - F_X(t - t_1))dt_1\right] \\ &= \lambda\int_0^t (1 - F_X(t - t_1))dt_1\end{aligned}$$



b) As $t \to \infty$

$$\lambda t(1-p) \to \lambda\int_0^t (1 - F_X(t))dt = \lambda E[X]$$

$$\therefore P[N_2(t) = j] = \frac{(\lambda E[X])^j}{j!}e^{-\lambda F[X]} \quad \text{Poisson RV!}$$

c) Little's formula

$$E[N] = \lambda E[X] \qquad \surd$$